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(12) **United States Patent**
Velderman et al.

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(45) **Date of Patent:** **Sep. 9, 2025**

(54) **POWER TOOL**

(71) Applicant: **Black & Decker Inc.**, New Britain, CT (US)

(72) Inventors: **Matthew J. Velderman**, Baltimore, MD (US); **Andrew E. Seman, Jr.**, Pylesville, MD (US); **Daniel J. White**, Baltimore, MD (US); **Paul Becke**, Stewartstown, PA (US); **William A. Rigdon**, Baltimore, MD (US)

(73) Assignee: **BLACK & DECKER INC.**, New Britain, CT (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/646,102**

(22) Filed: **Apr. 25, 2024**

(65) **Prior Publication Data**

US 2024/0275243 A1 Aug. 15, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/411,727, filed on Aug. 25, 2021, now Pat. No. 11,984,771, which is a (Continued)

(51) **Int. Cl.**
H02K 11/215 (2016.01)
B23B 45/02 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H02K 11/215** (2016.01); **B23B 45/02** (2013.01); **B24B 23/02** (2013.01); **B25F 5/008** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC H02K 11/215; H02K 5/207; H02K 9/227; H02K 5/10; H02K 7/116

(Continued)

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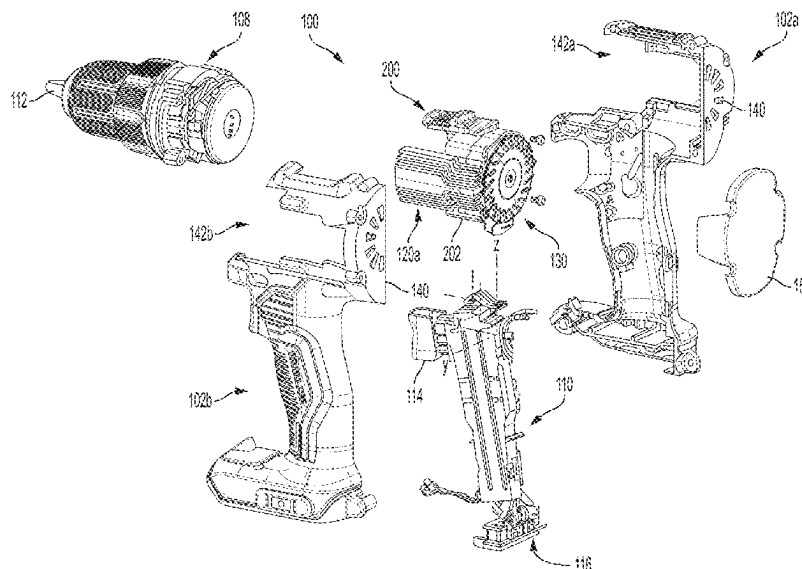
Machine Translation JP2017226025 (Year: 2017).*
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Primary Examiner — Jose A Gonzalez Quinones
(74) *Attorney, Agent, or Firm* — Amir R. Rohani

(57) **ABSTRACT**

A power tool is provided comprising: a motor, an output shaft driven by the motor along a longitudinal axis, a fan rotationally coupled to the motor to generate an airflow, and a housing comprising two calm shells mated together to support the motor and forming a back wall behind the motor along the longitudinal axis. The rear wall includes a plurality of air intakes. An end cap is attached to the housing adjacent the rear wall with an air gap therebetween. The airflow generated by the fan enters through the air gap in a substantially radial direction, then passes through the plurality of air intakes to enter the housing in thermal contact with the motor.

20 Claims, 65 Drawing Sheets



Related U.S. Application Data			
continuation of application No. 16/281,475, filed on Feb. 21, 2019, now Pat. No. 11,139,722.			
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(52)	U.S. Cl. CPC B26B 15/00 (2013.01); H02K 5/10 (2013.01); H02K 5/15 (2013.01); H02K 5/161 (2013.01); H02K 5/18 (2013.01); H02K 5/207 (2021.01); H02K 7/083 (2013.01); H02K 7/116 (2013.01); H02K 7/145 (2013.01); H02K 9/06 (2013.01); H02K 9/227 (2021.01); H02K 21/16 (2013.01)	FOREIGN PATENT DOCUMENTS	
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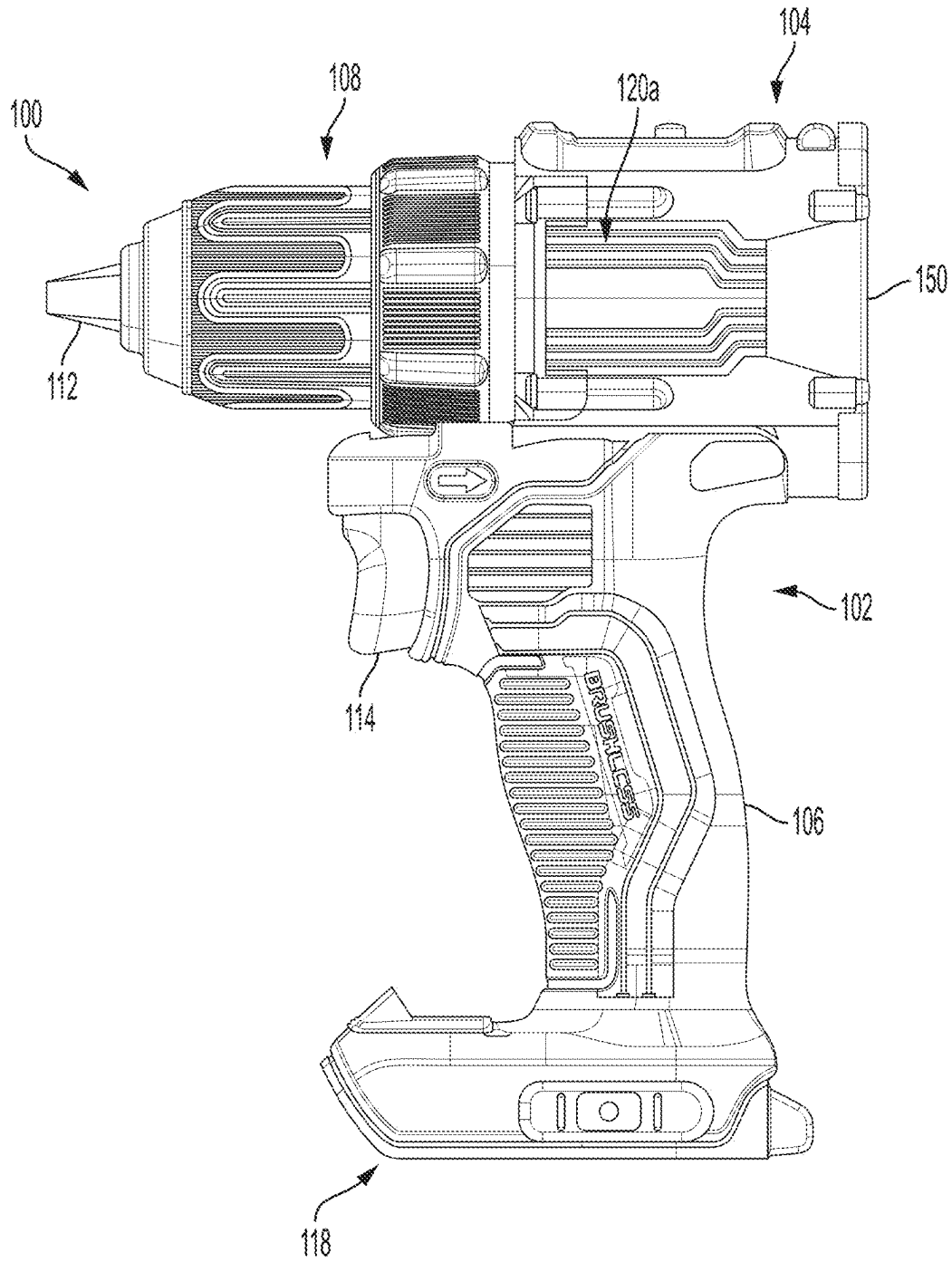
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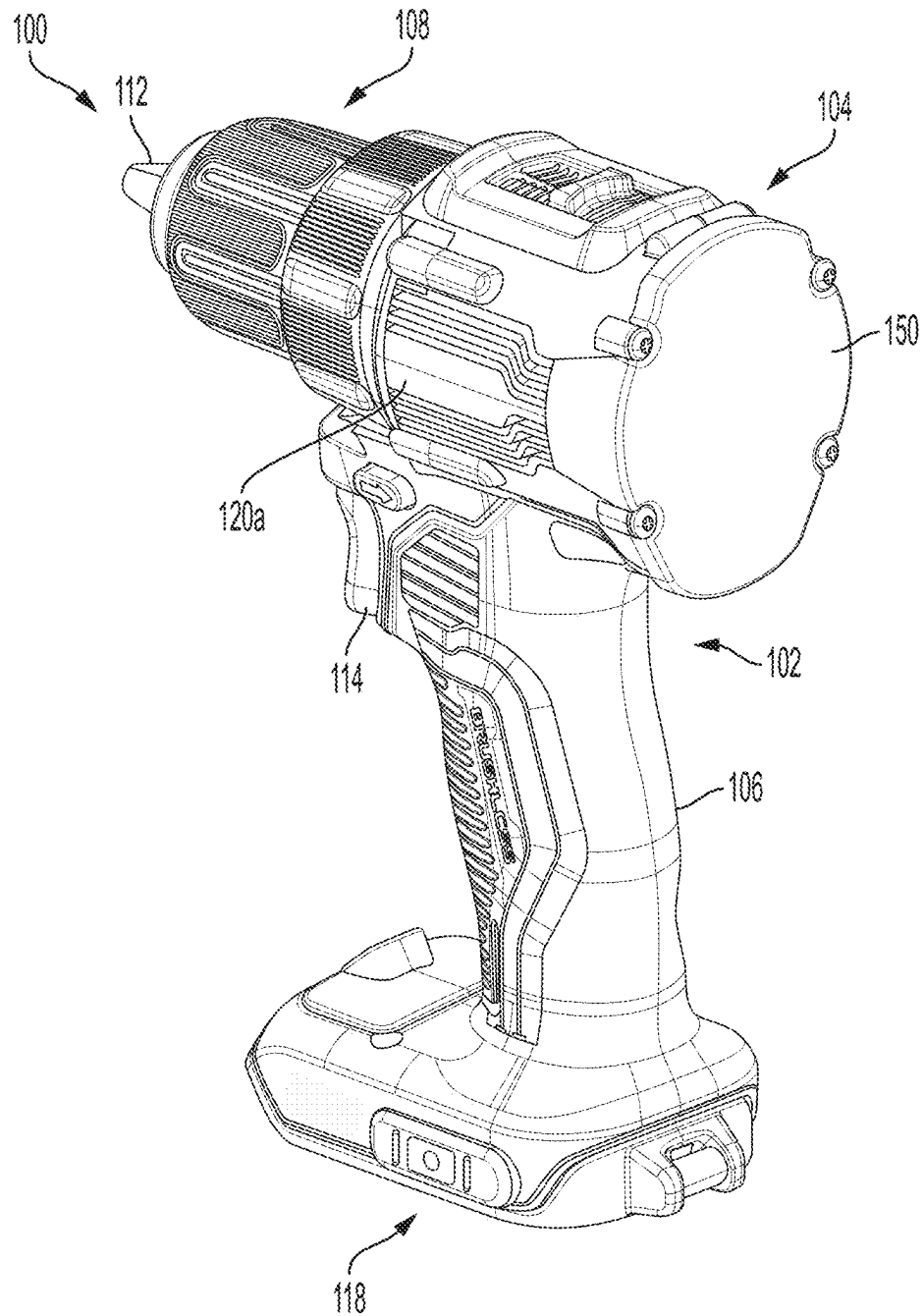
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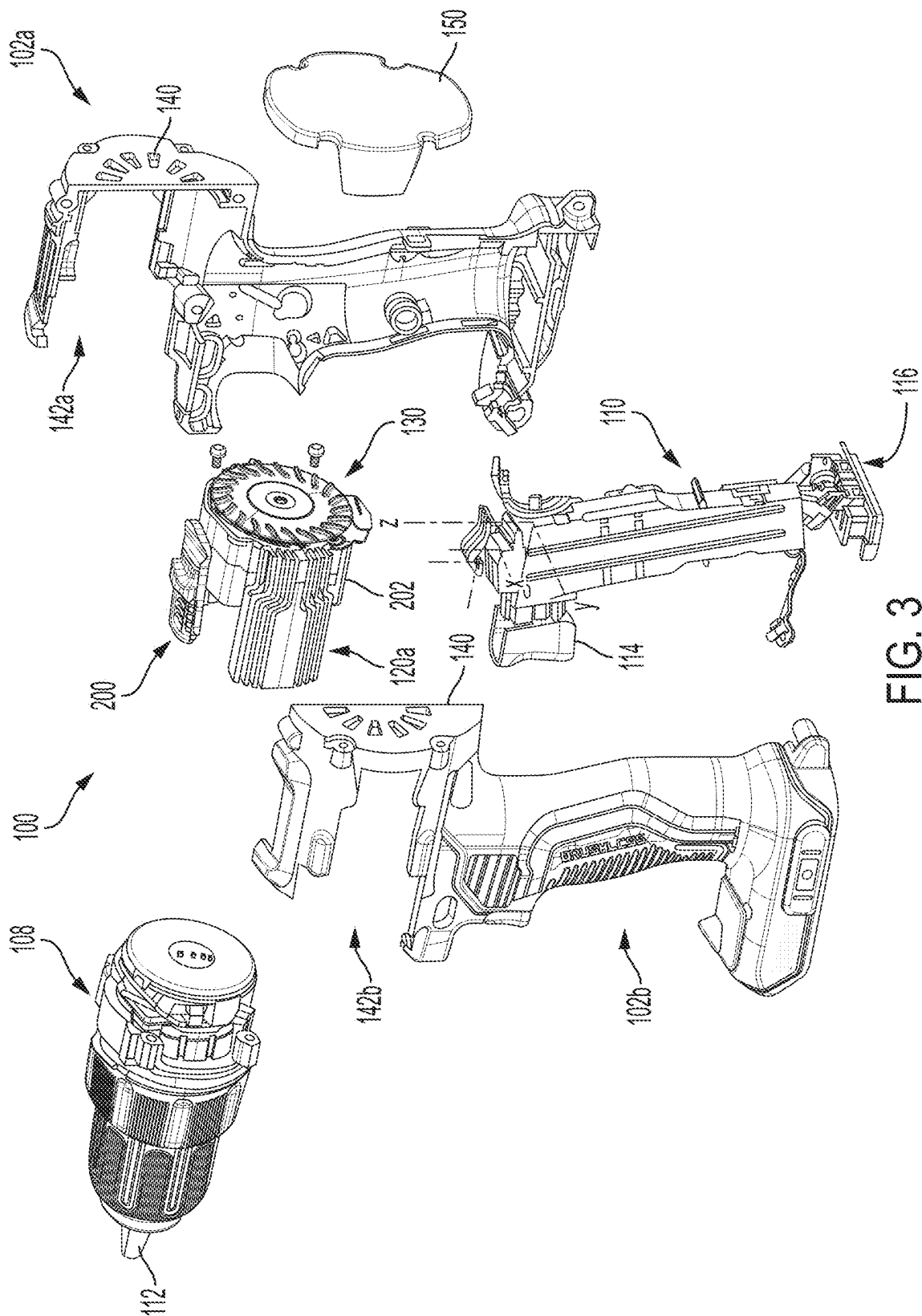
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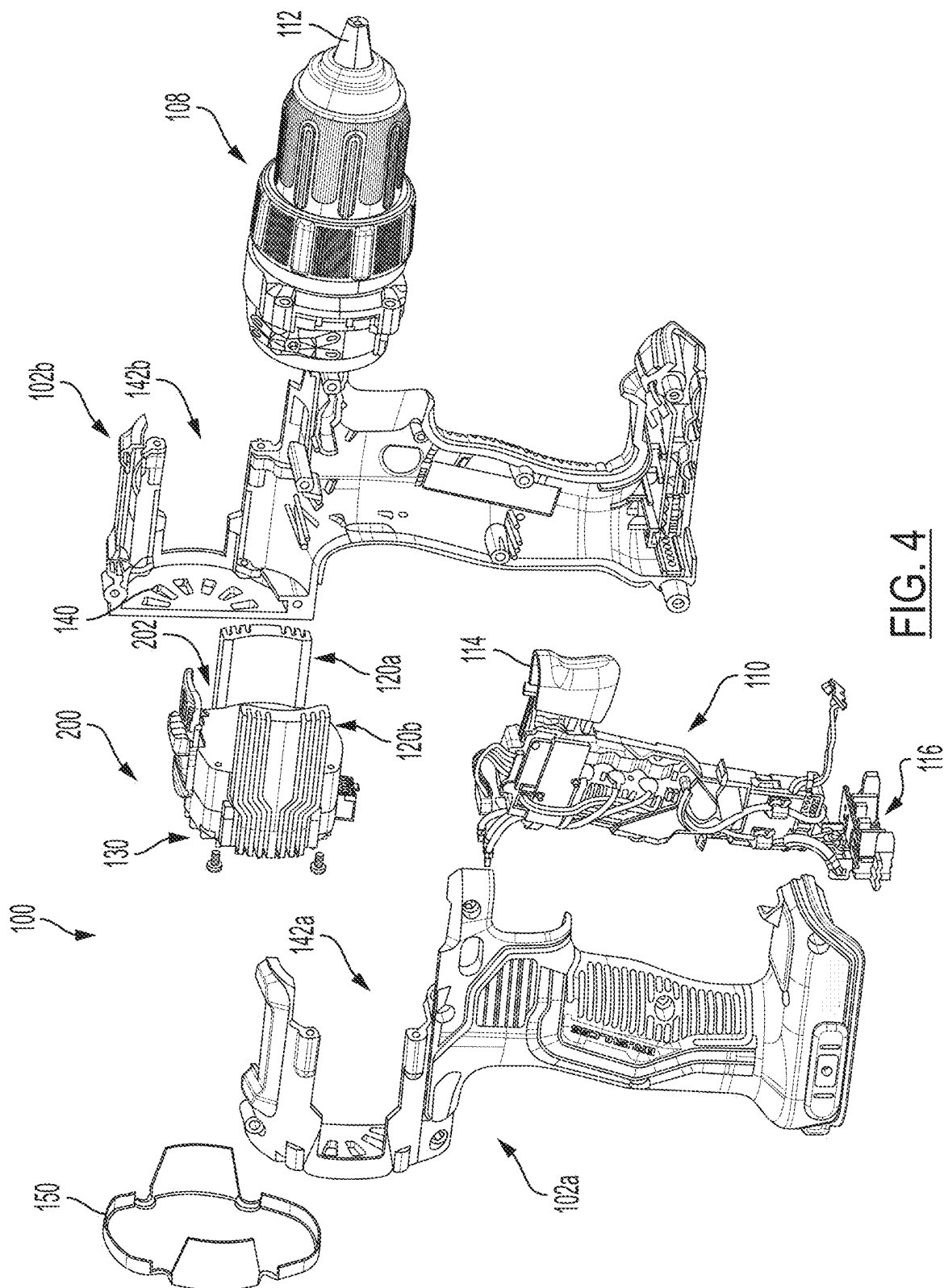
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FIG. 1

FIG. 2





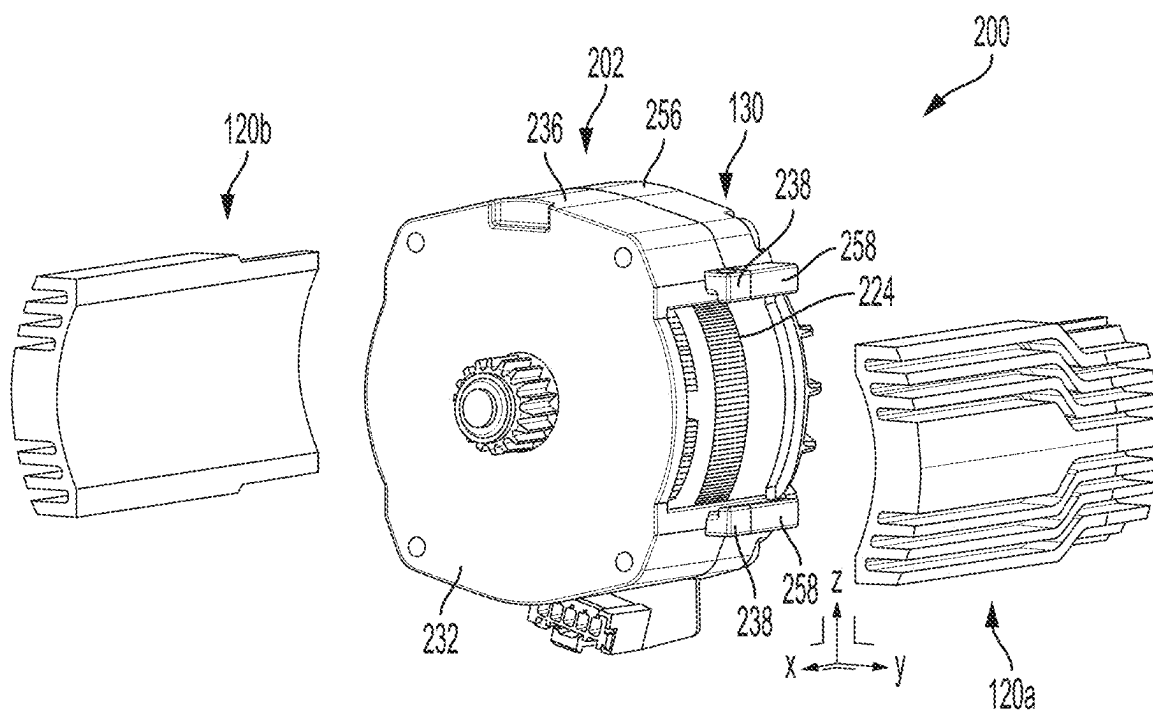


FIG. 5

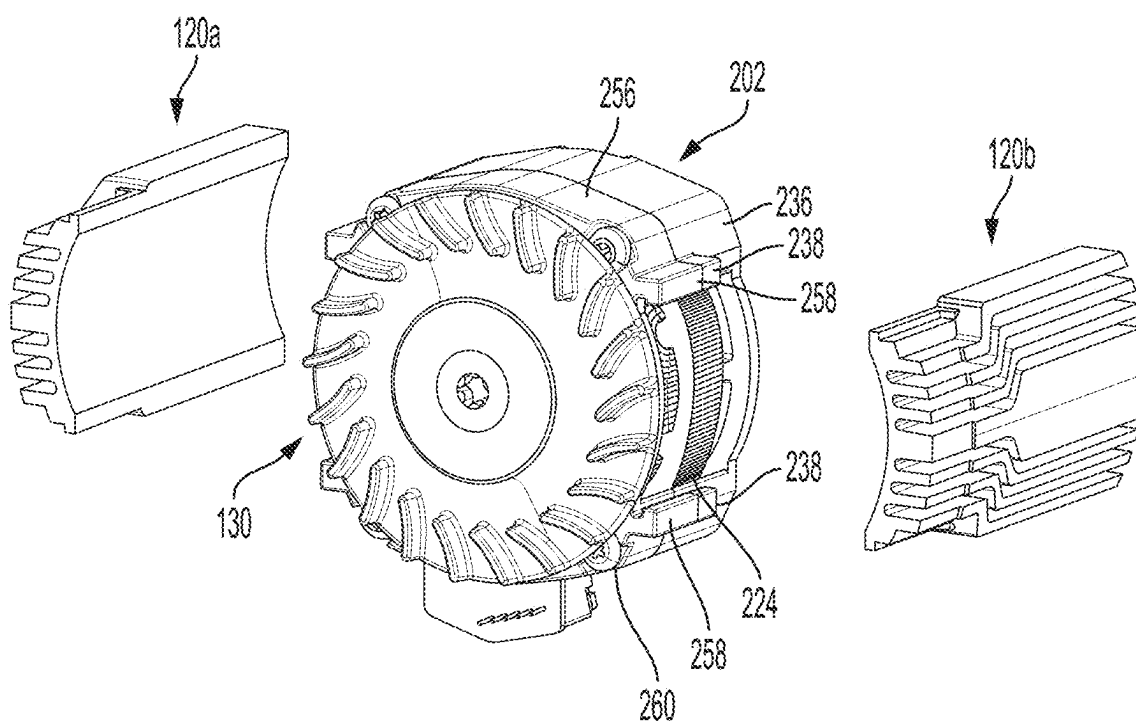


FIG. 6

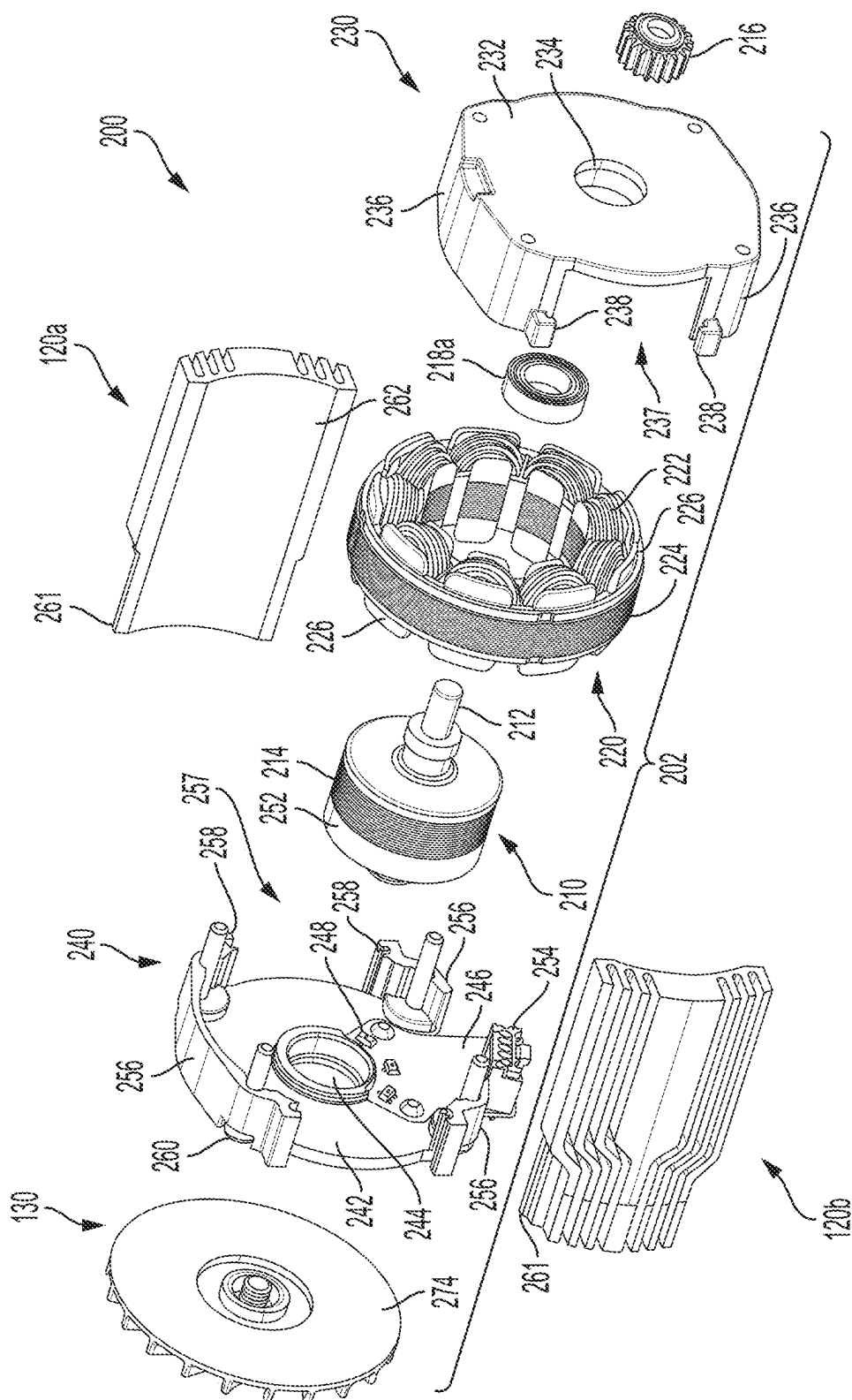


FIG. 7

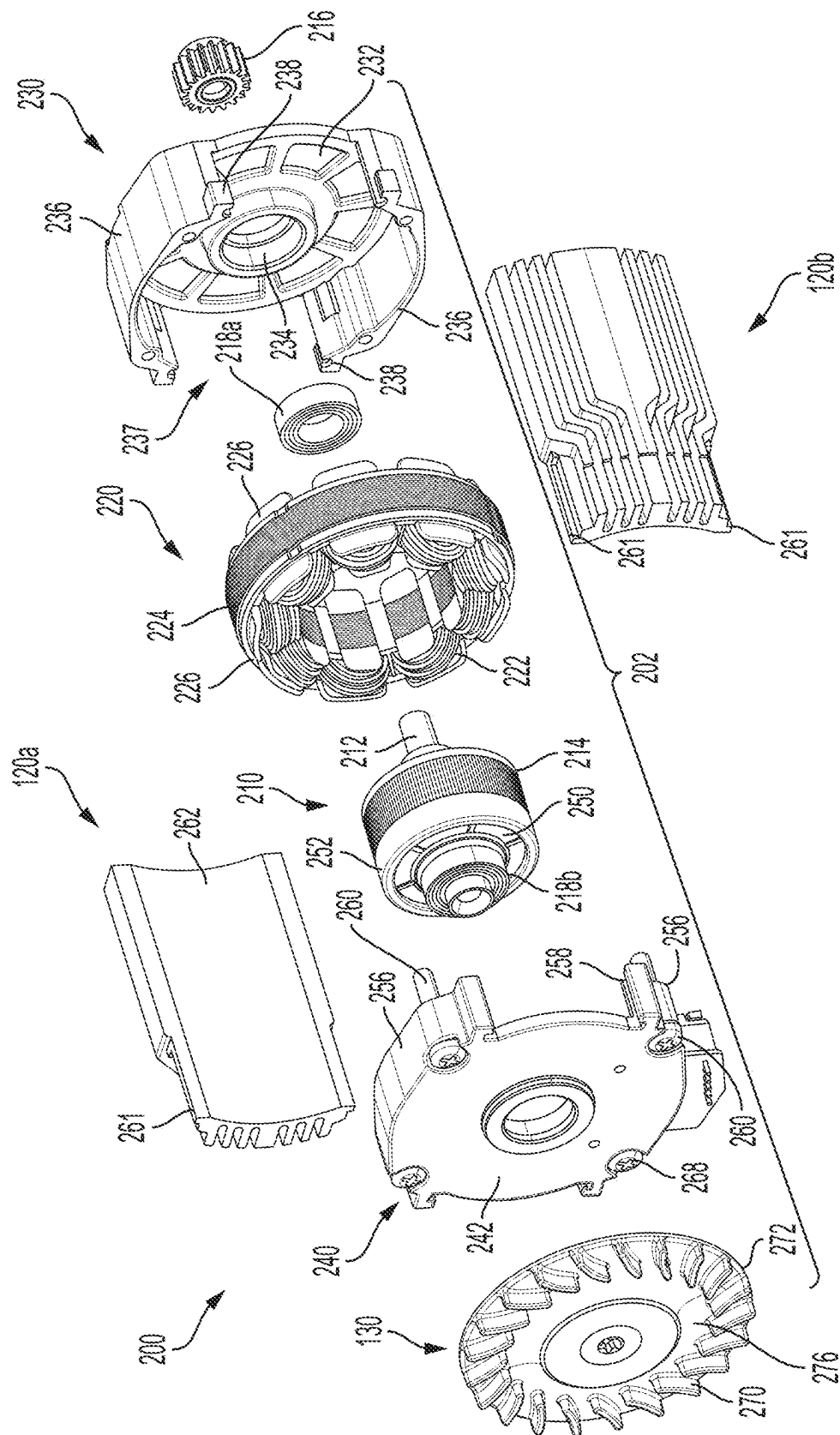


FIG. 8

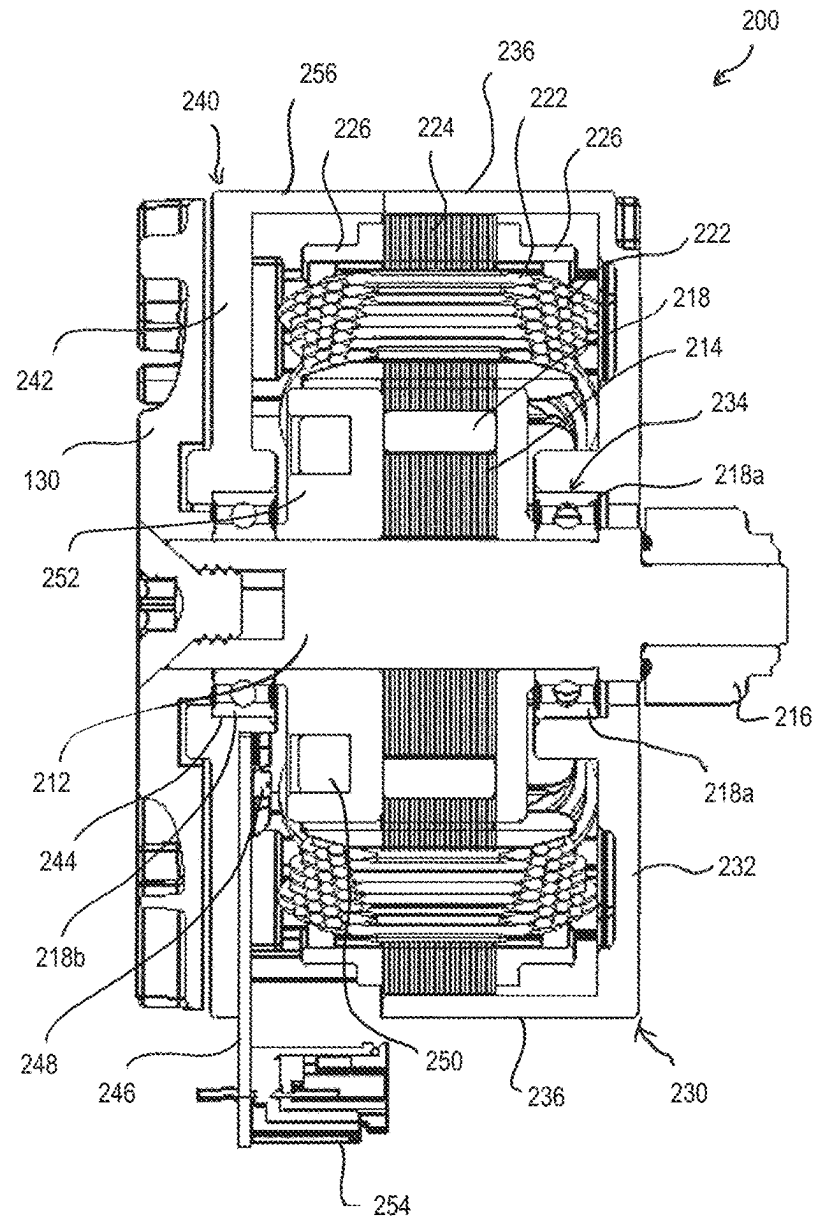


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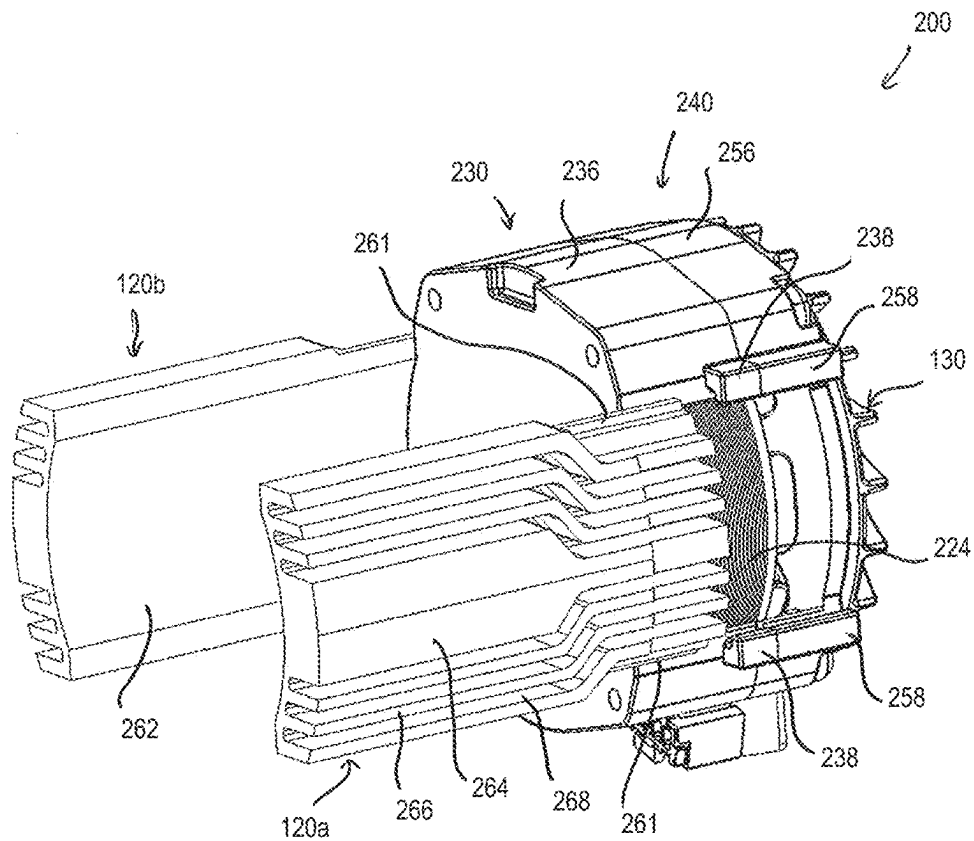


FIG. 10

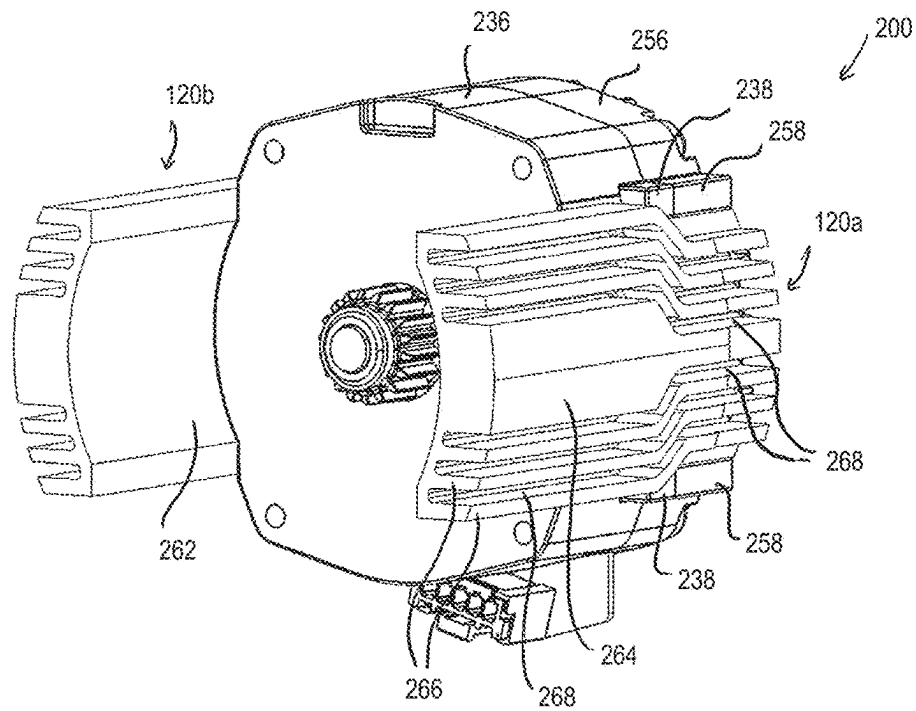


FIG. 11

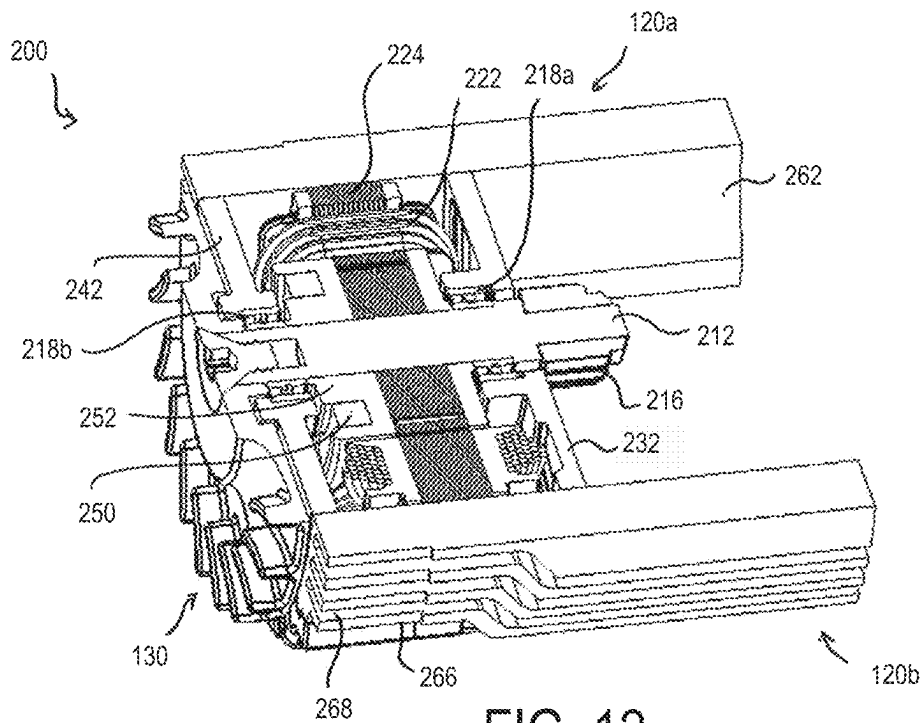


FIG. 12

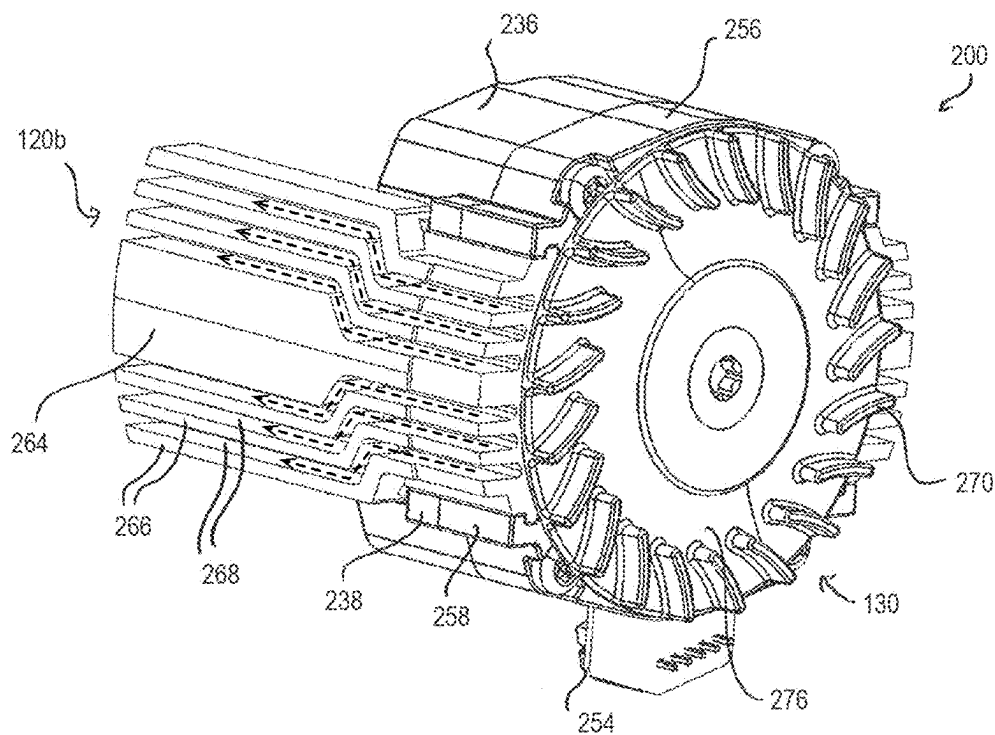


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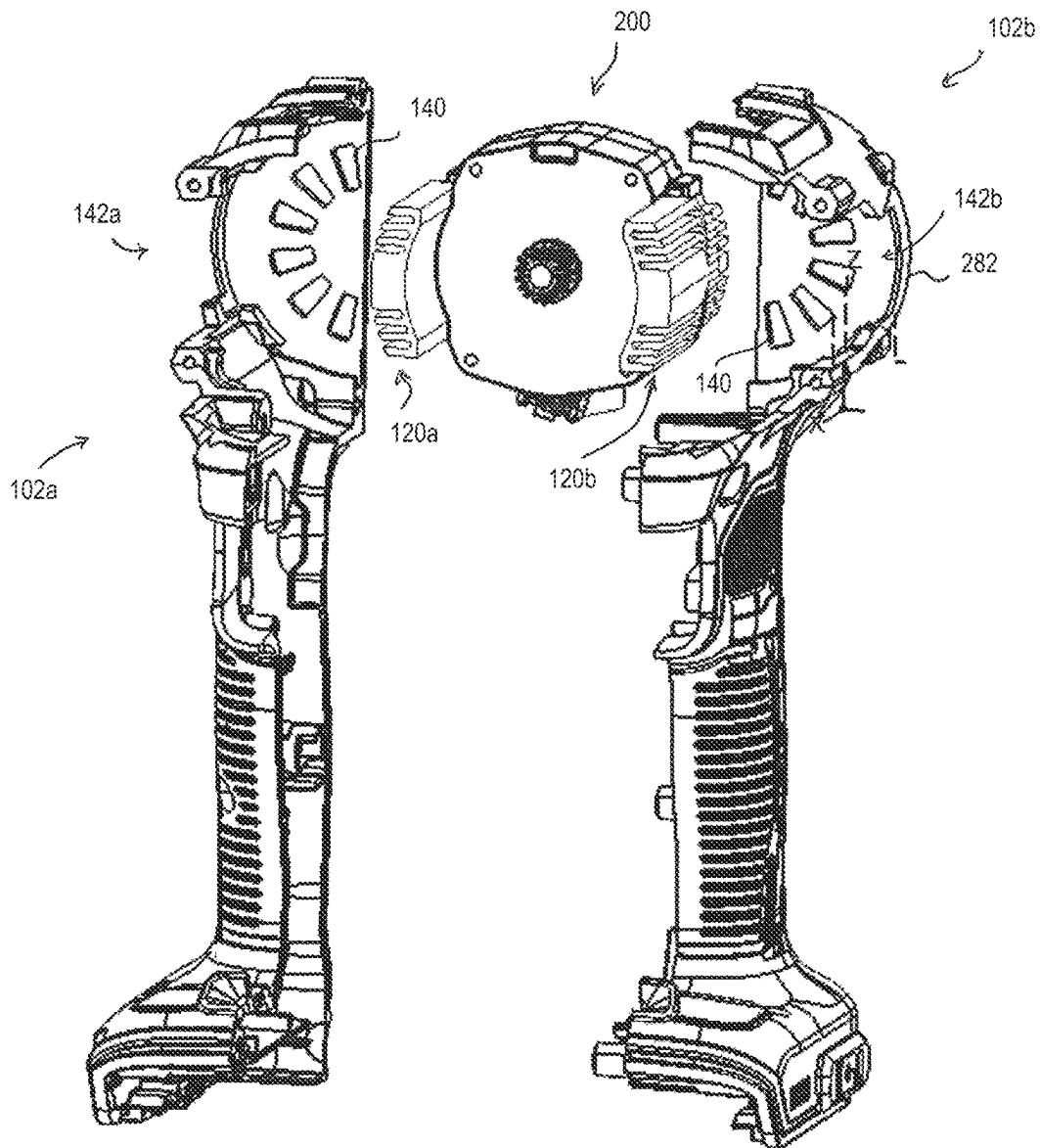


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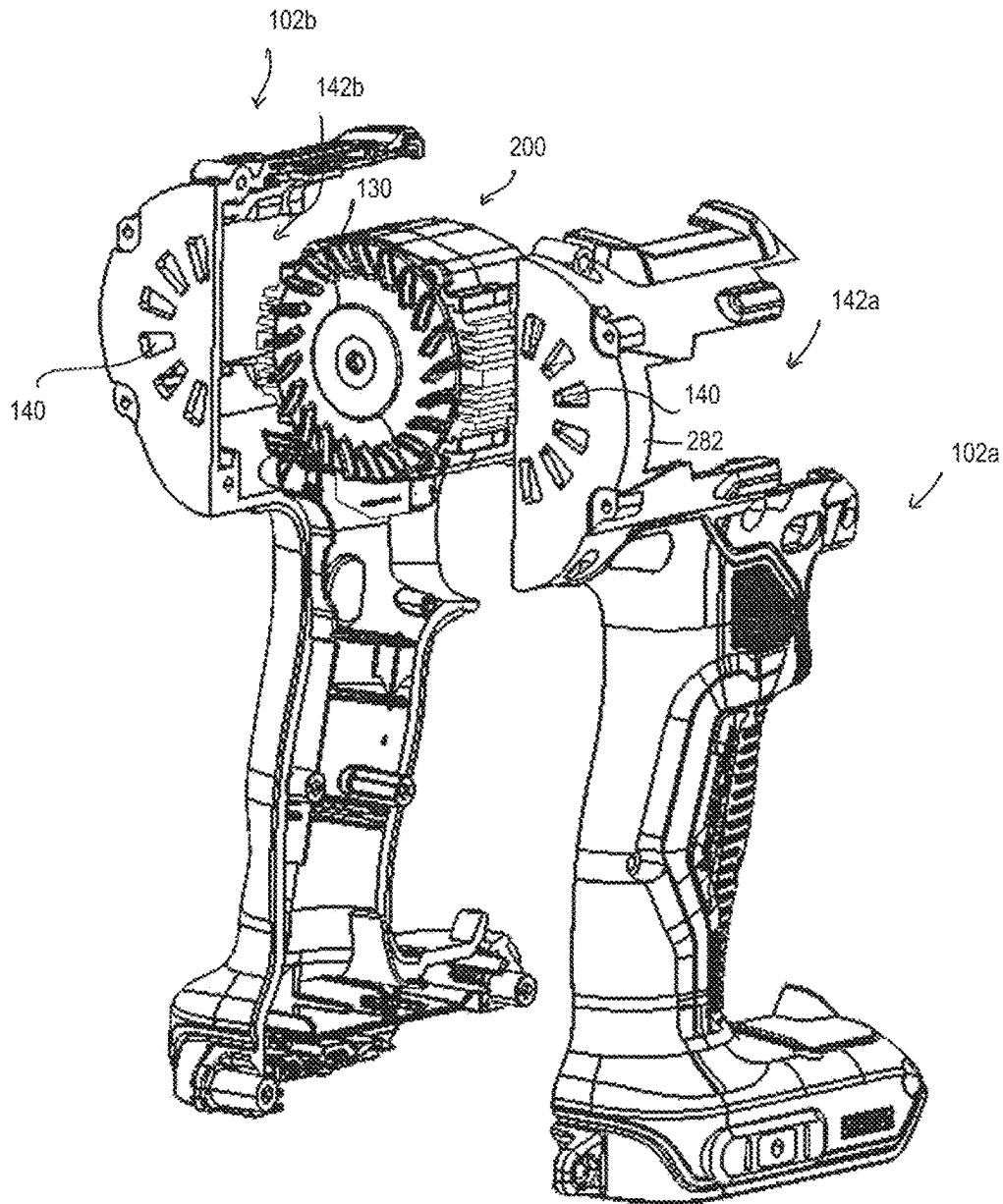


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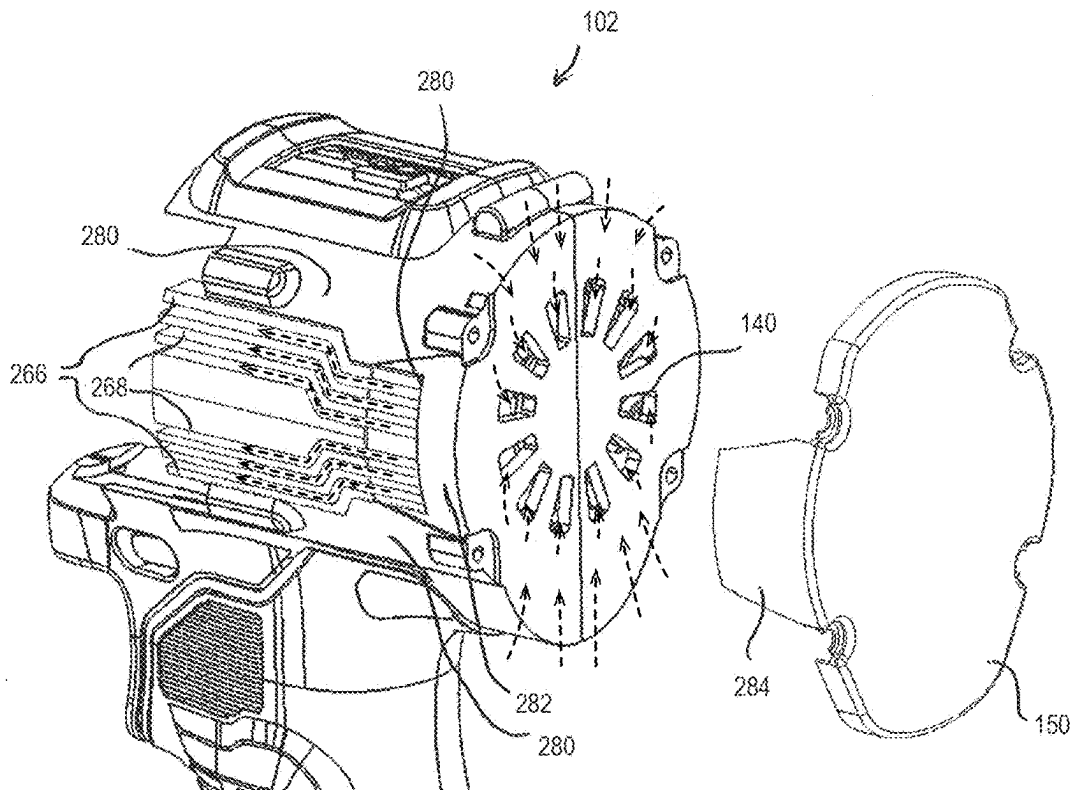


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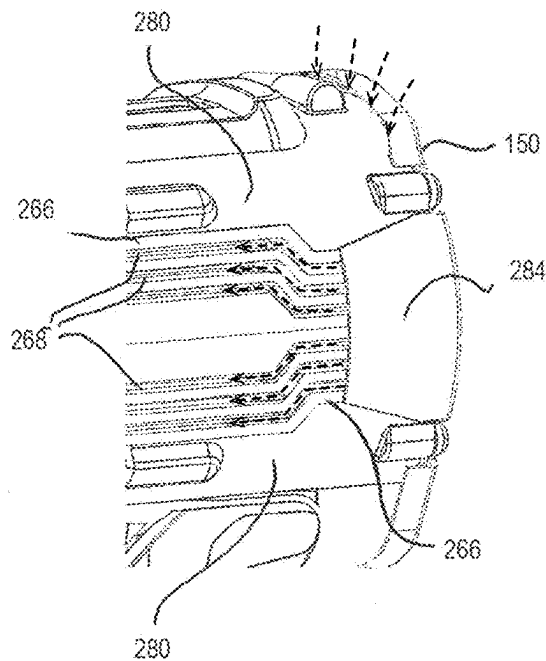


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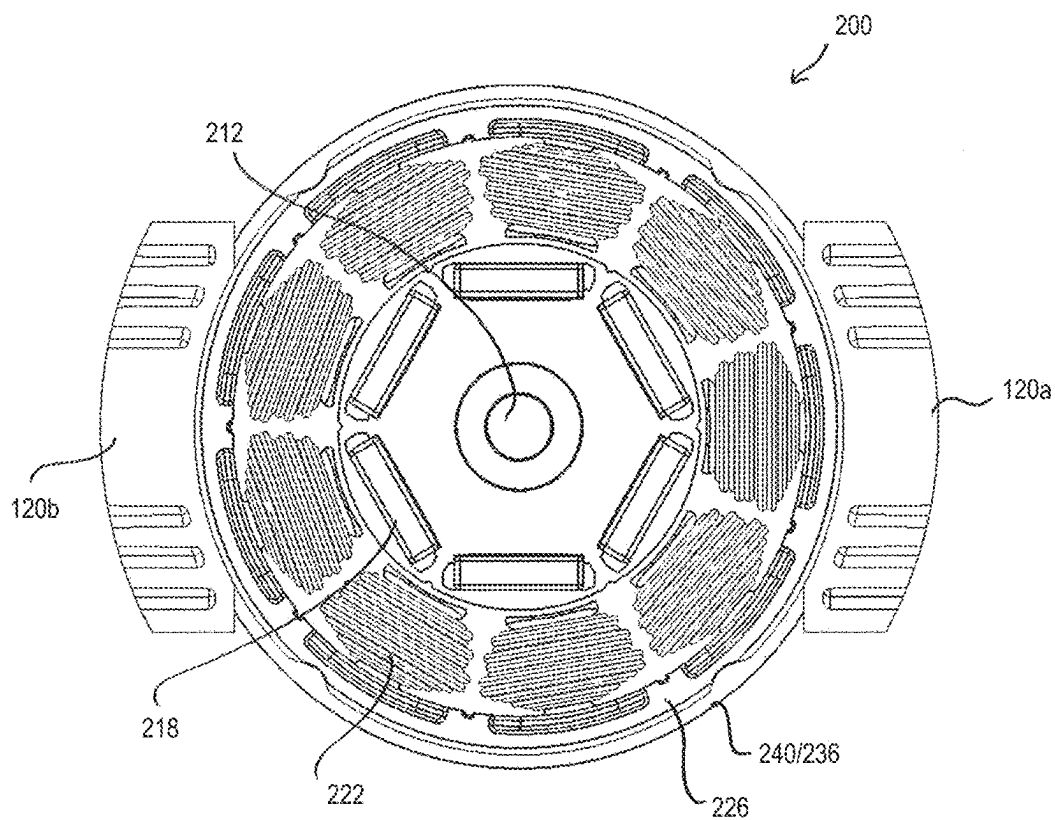


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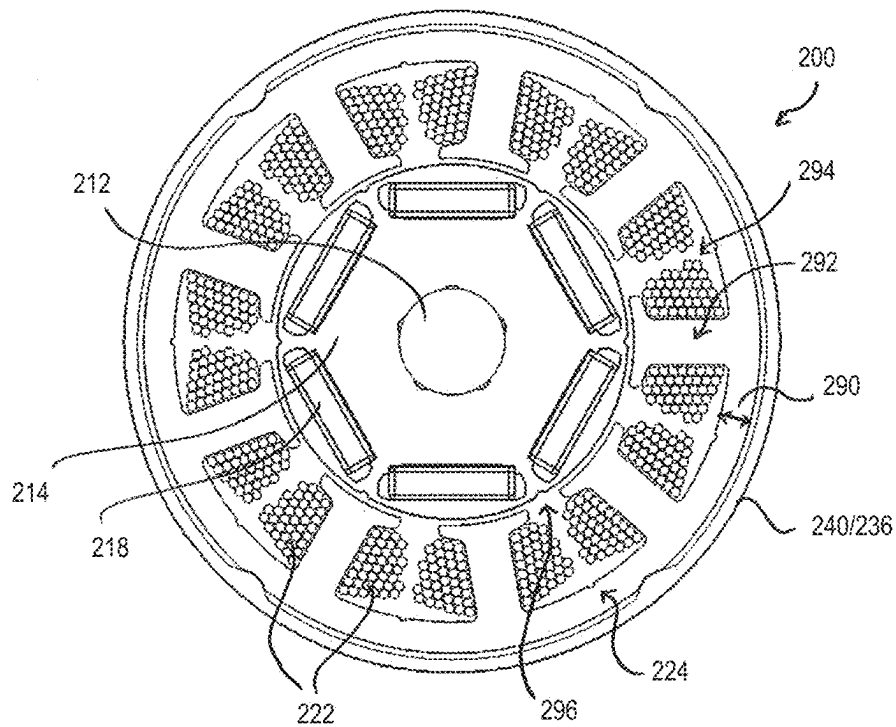


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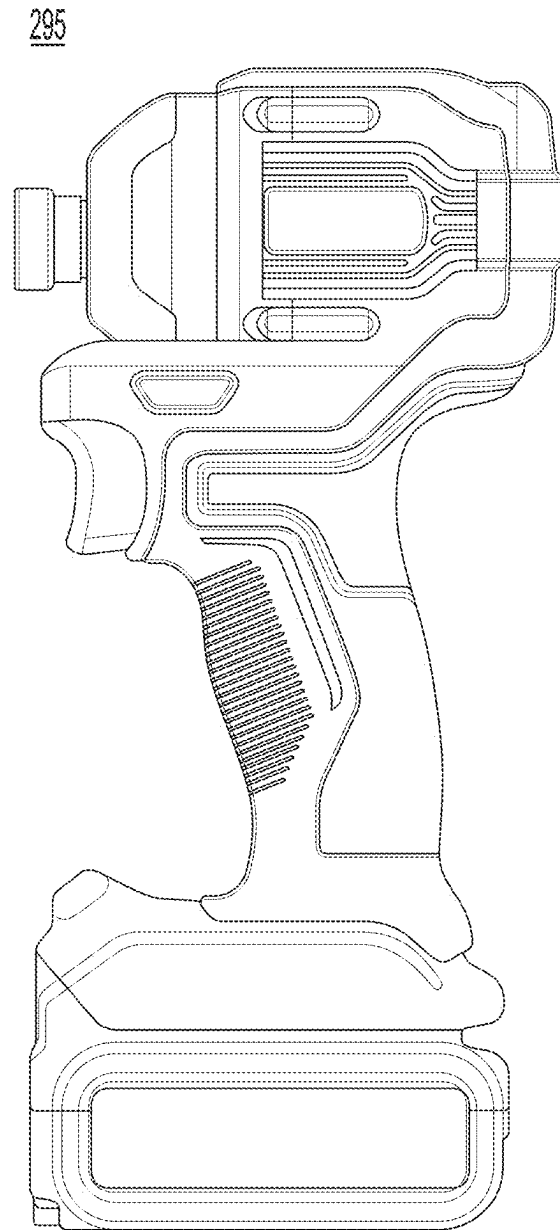


FIG. 20

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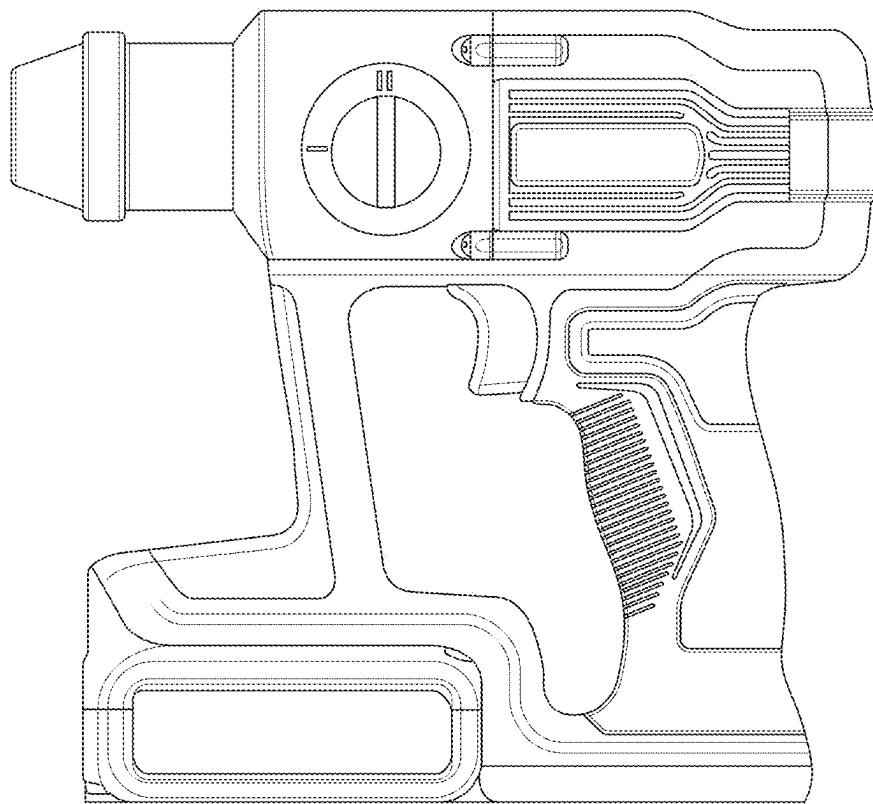


FIG. 21

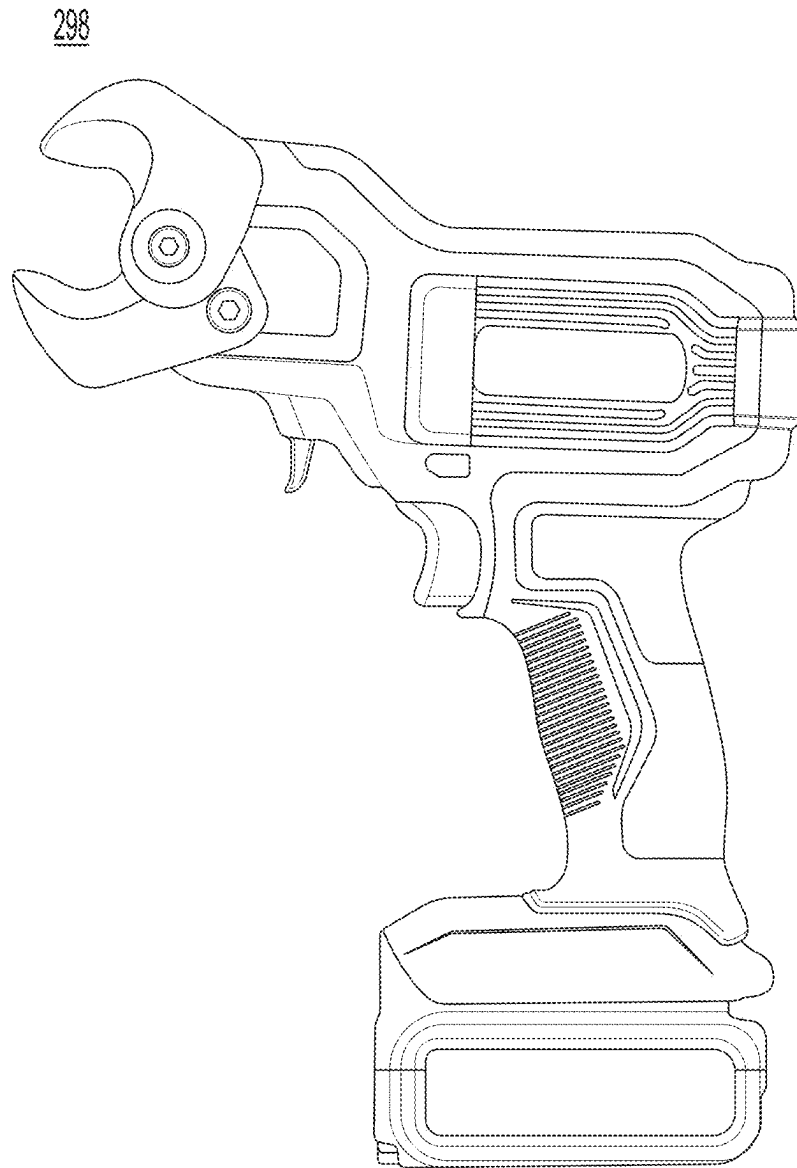


FIG. 22

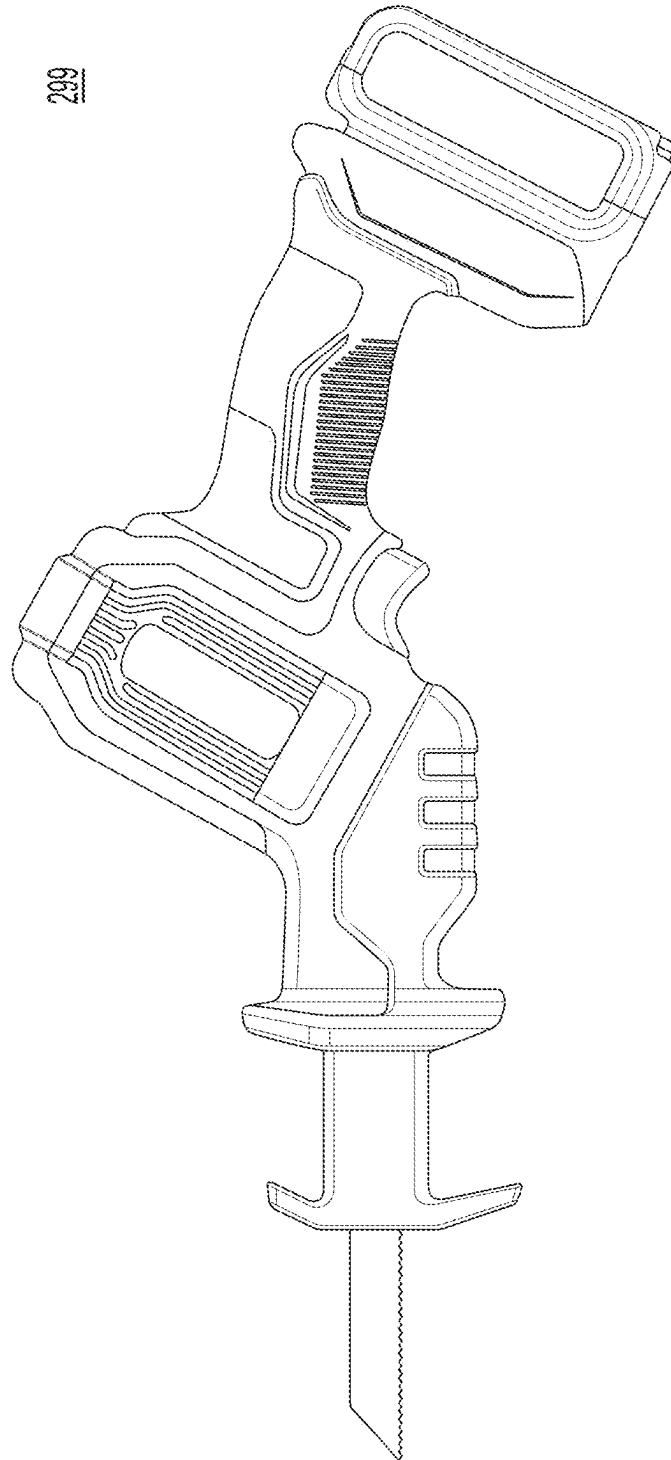


FIG. 23

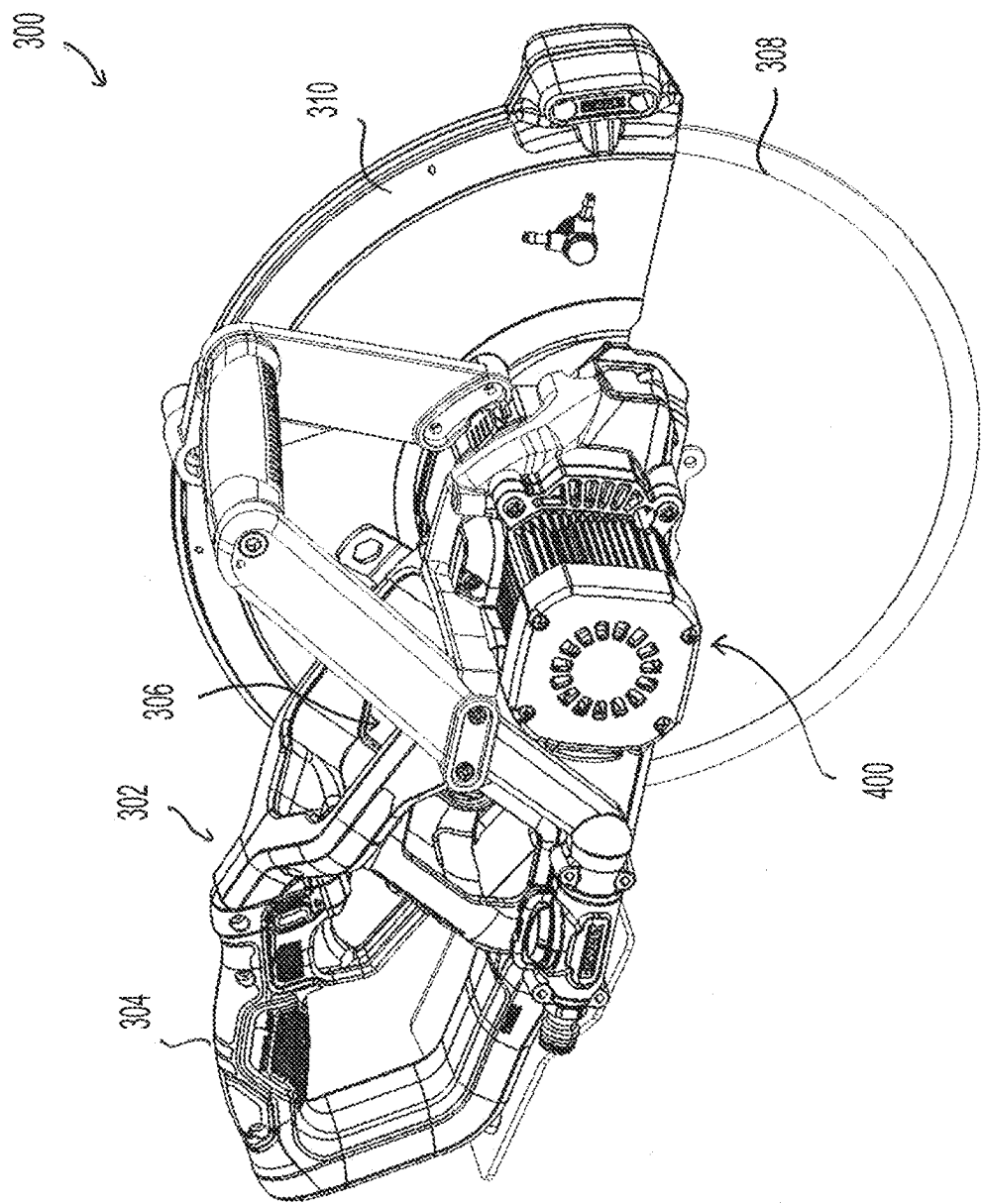


FIG. 24

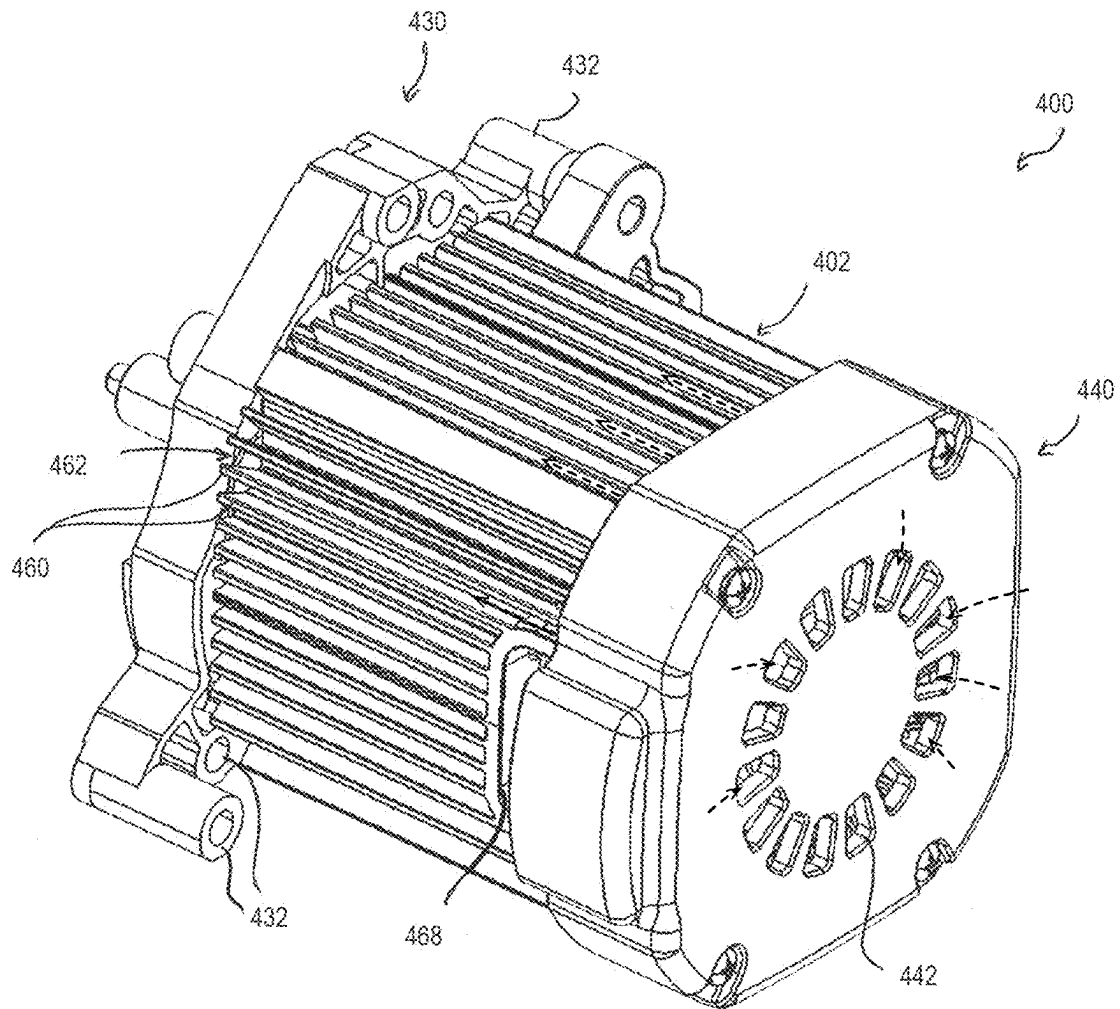


FIG. 25

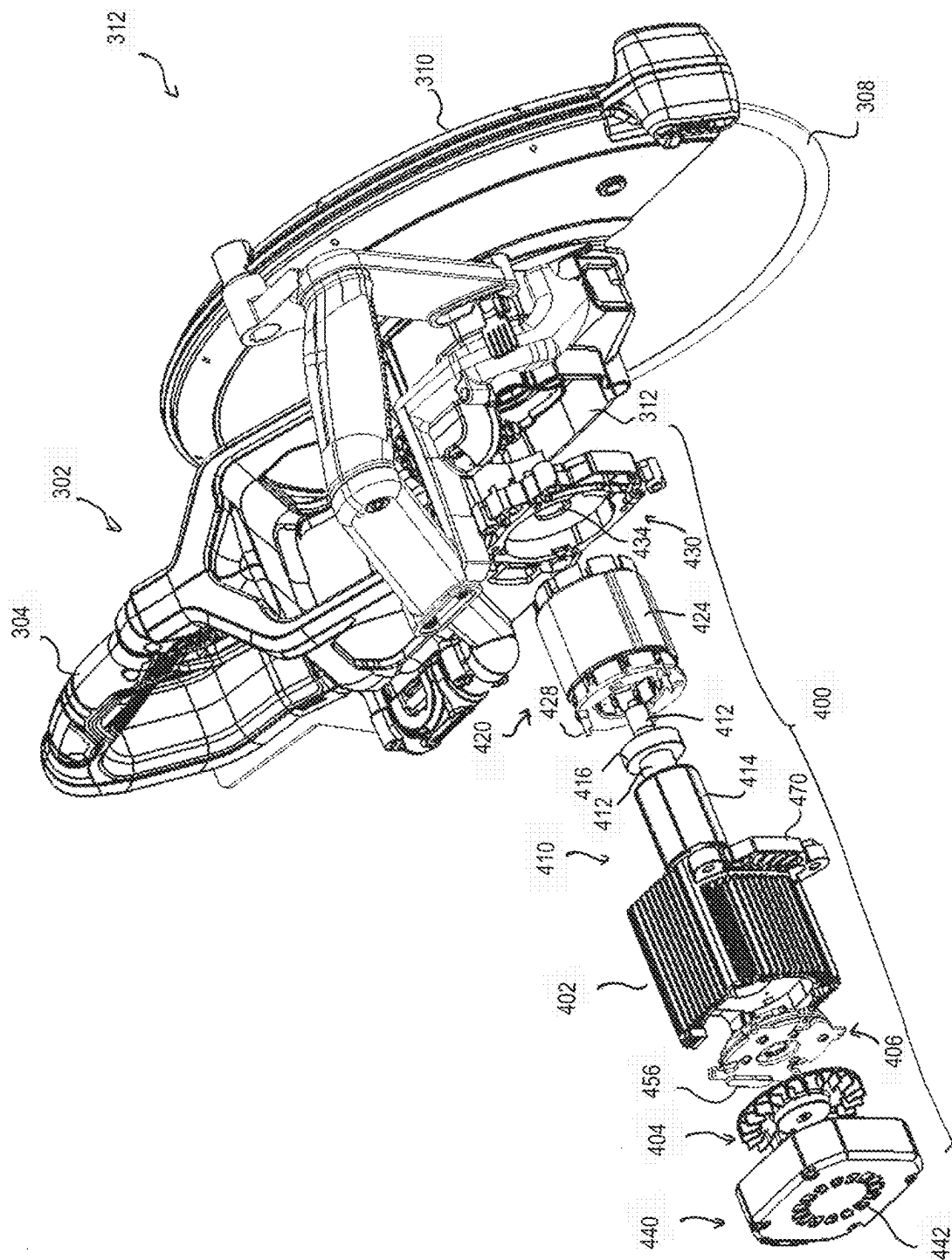


FIG. 26

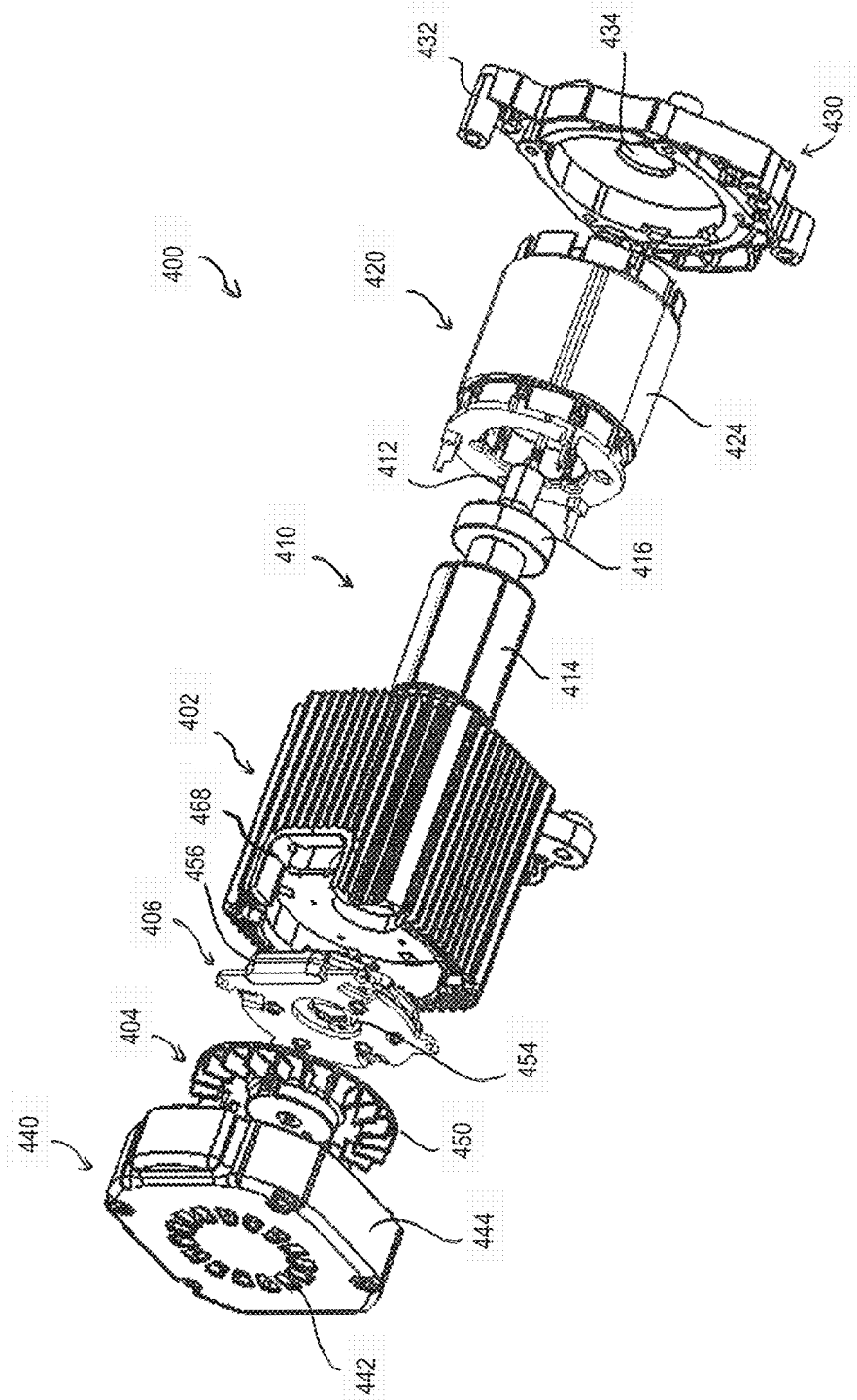


FIG. 27

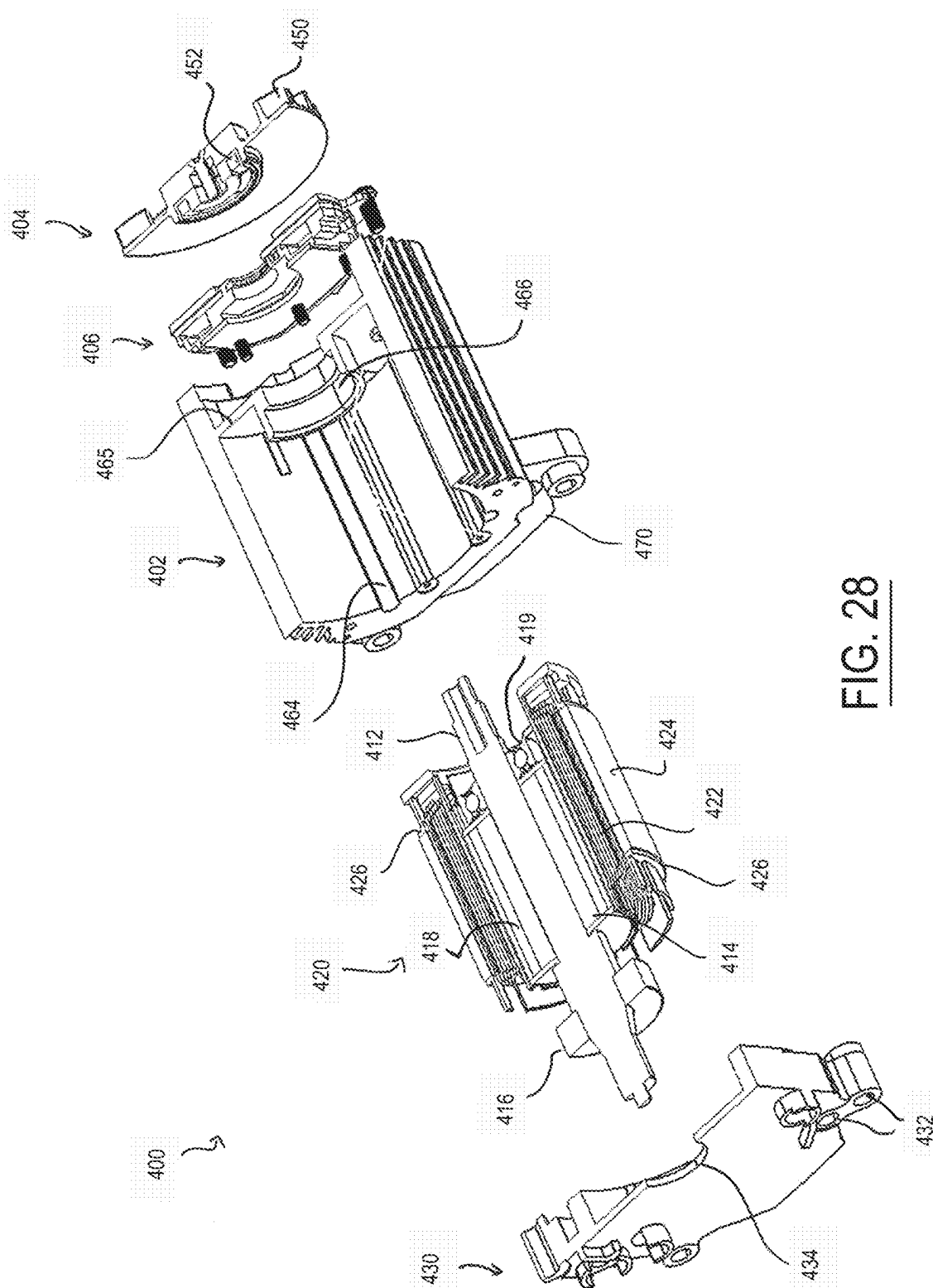


FIG. 28

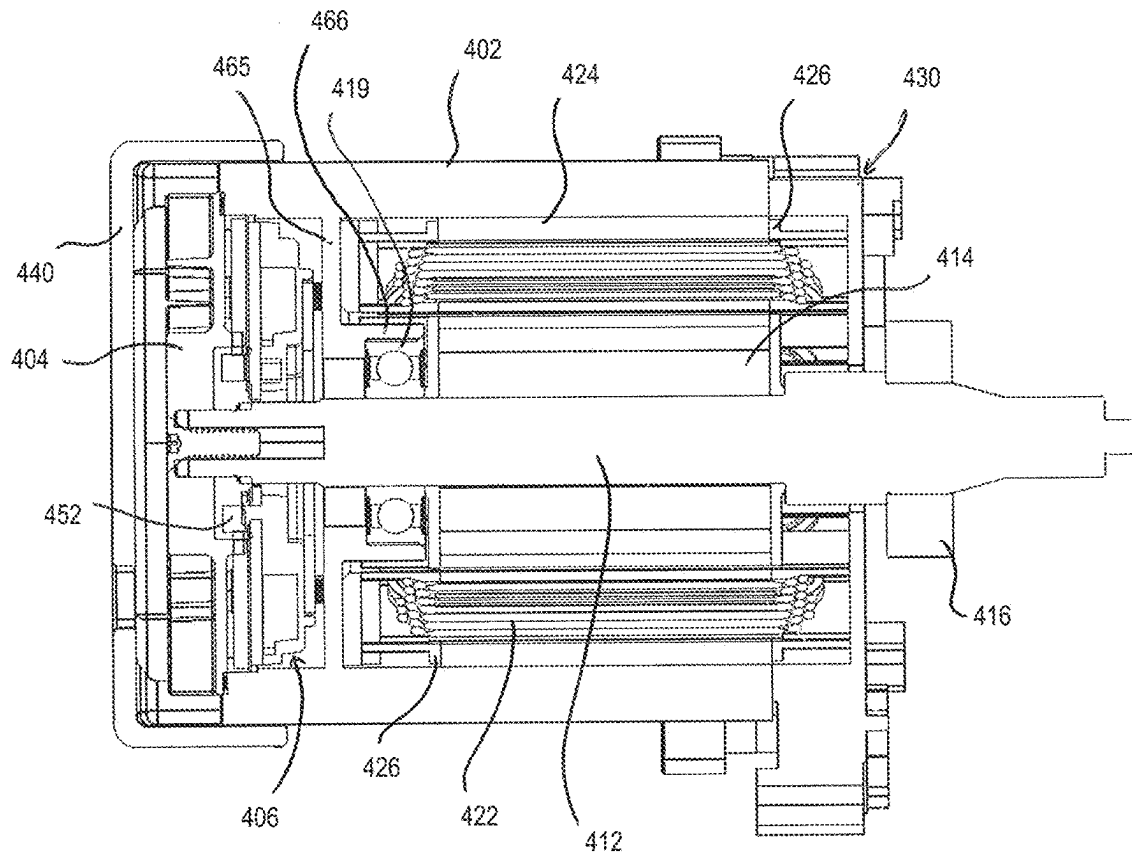


FIG. 29

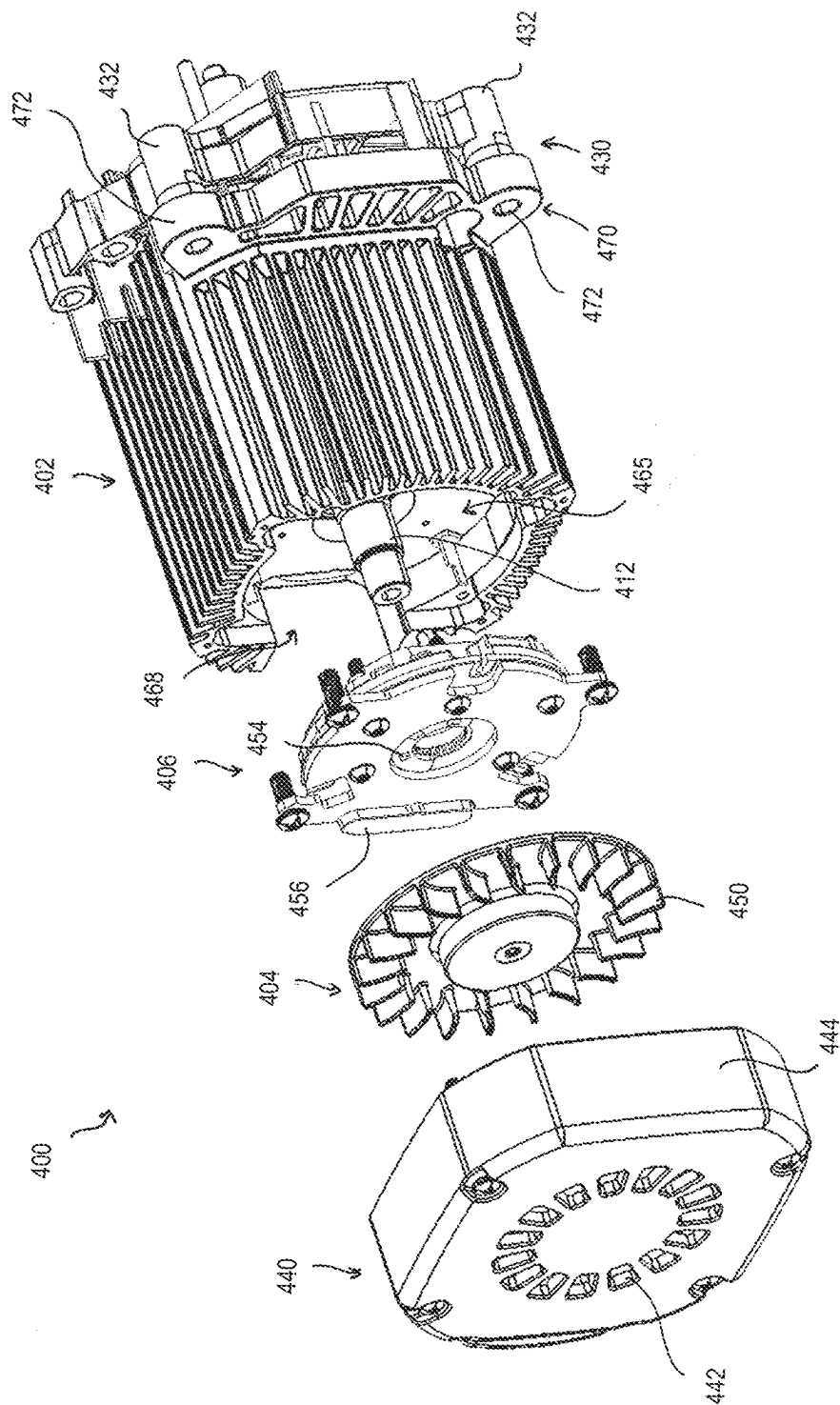


FIG. 30

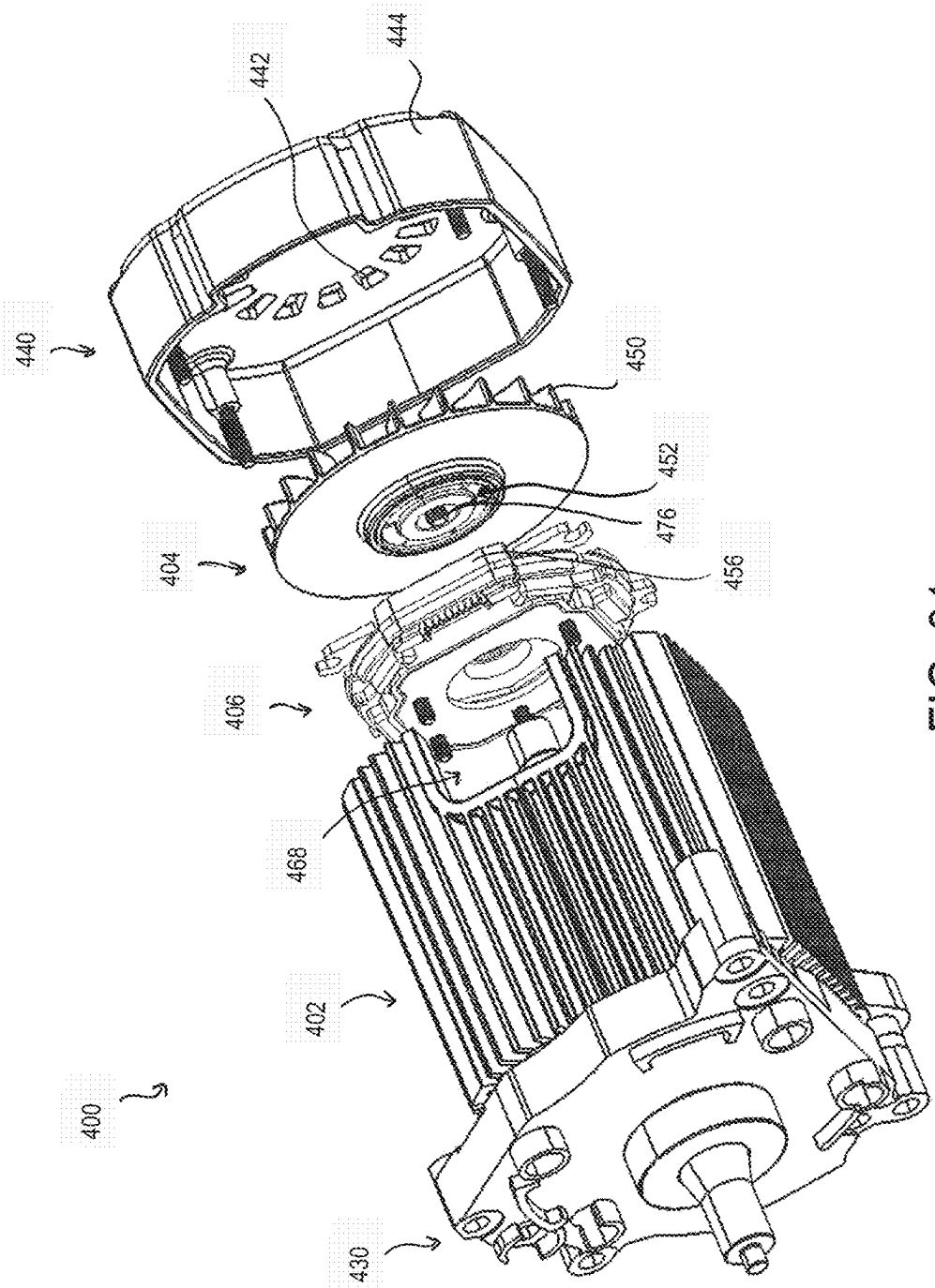


FIG. 31

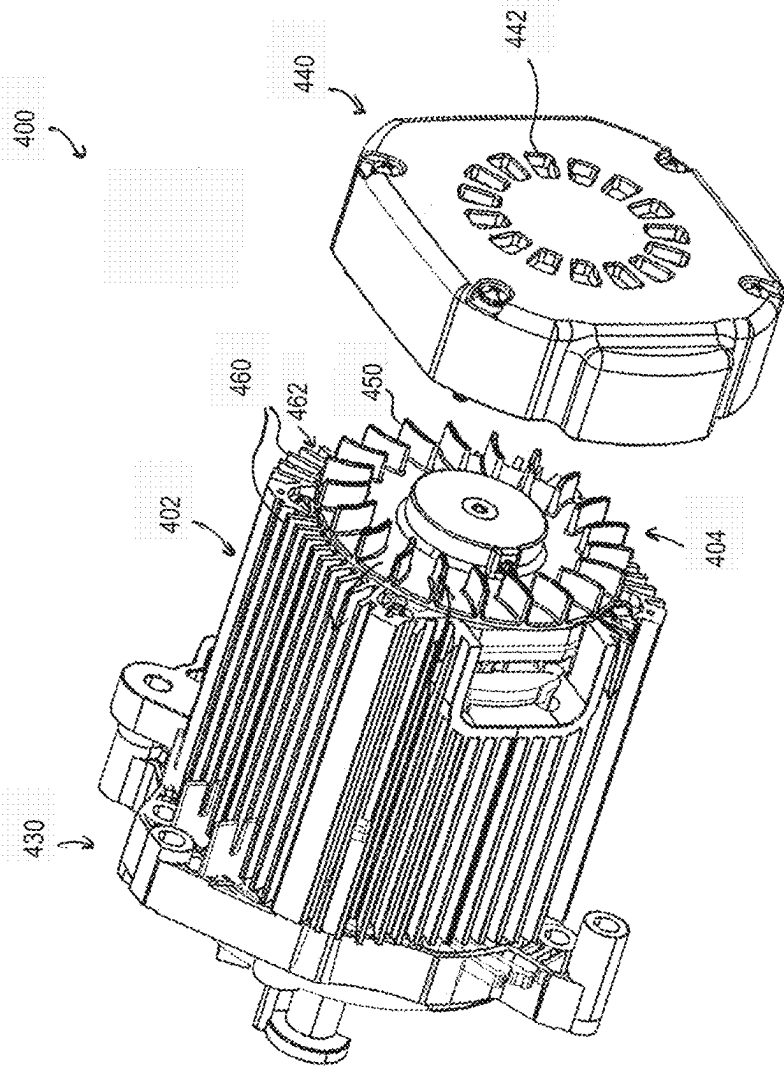


FIG. 32

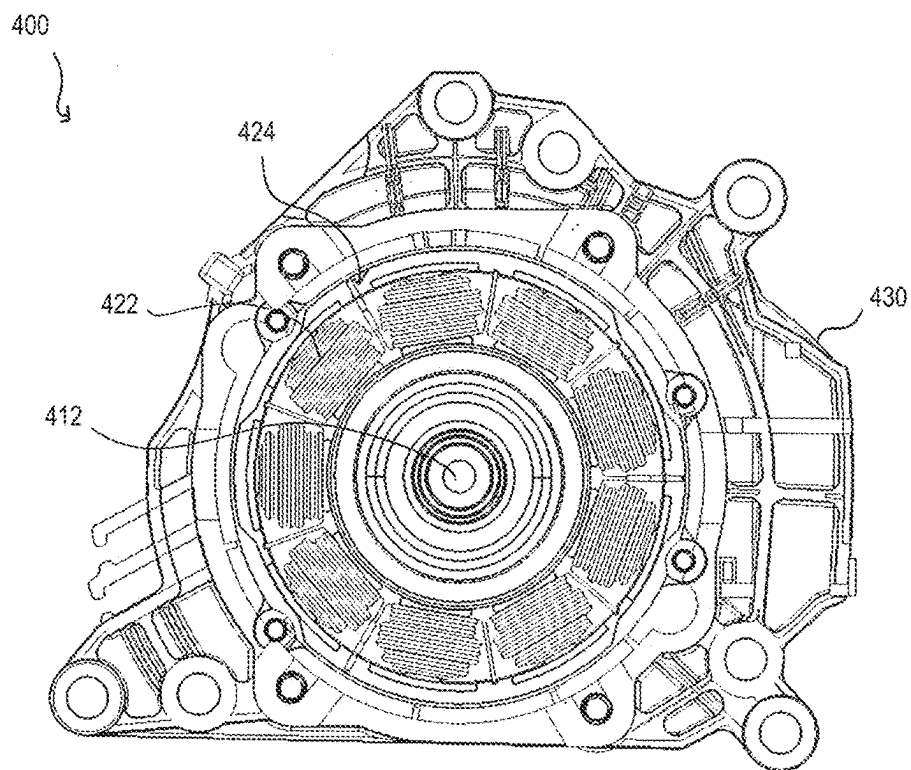


FIG. 33

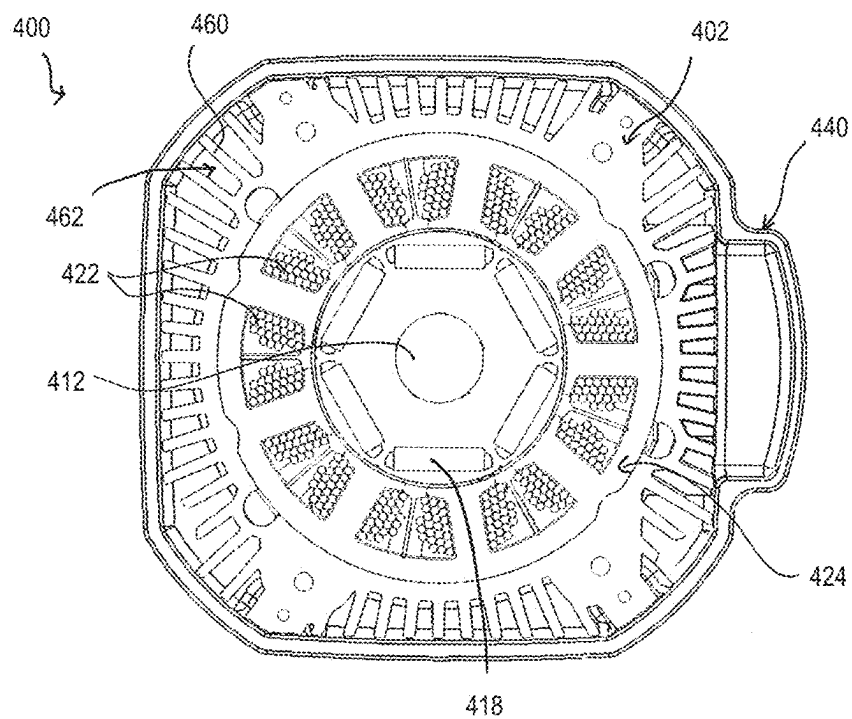


FIG. 34

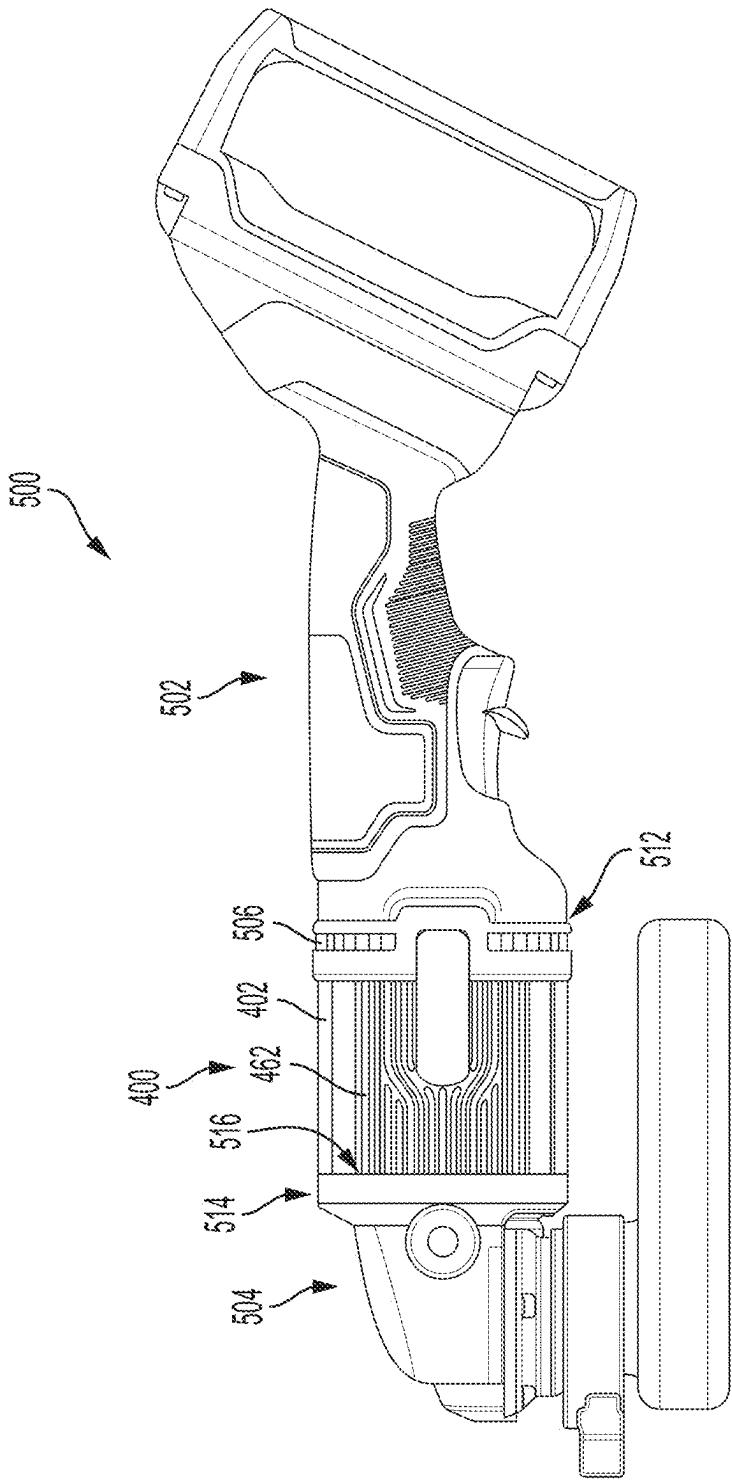


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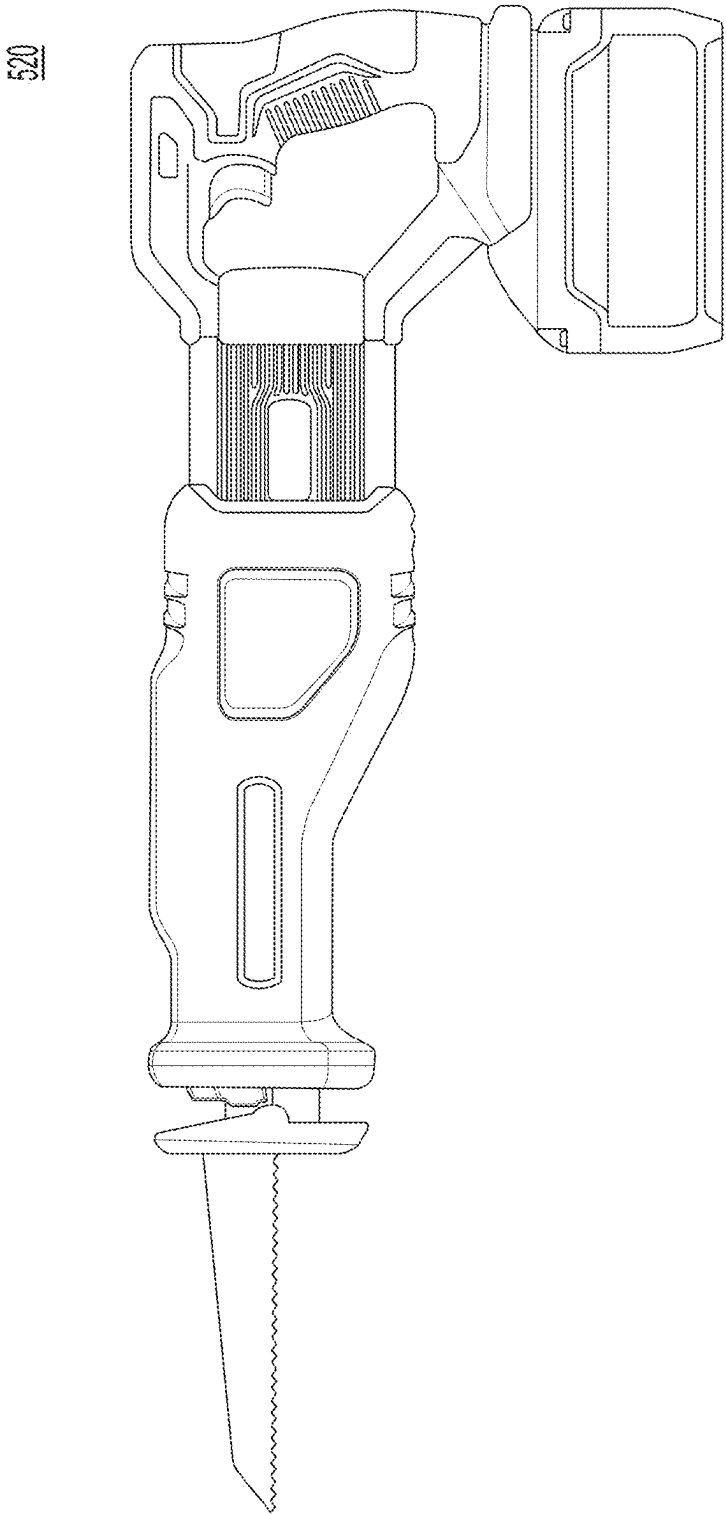


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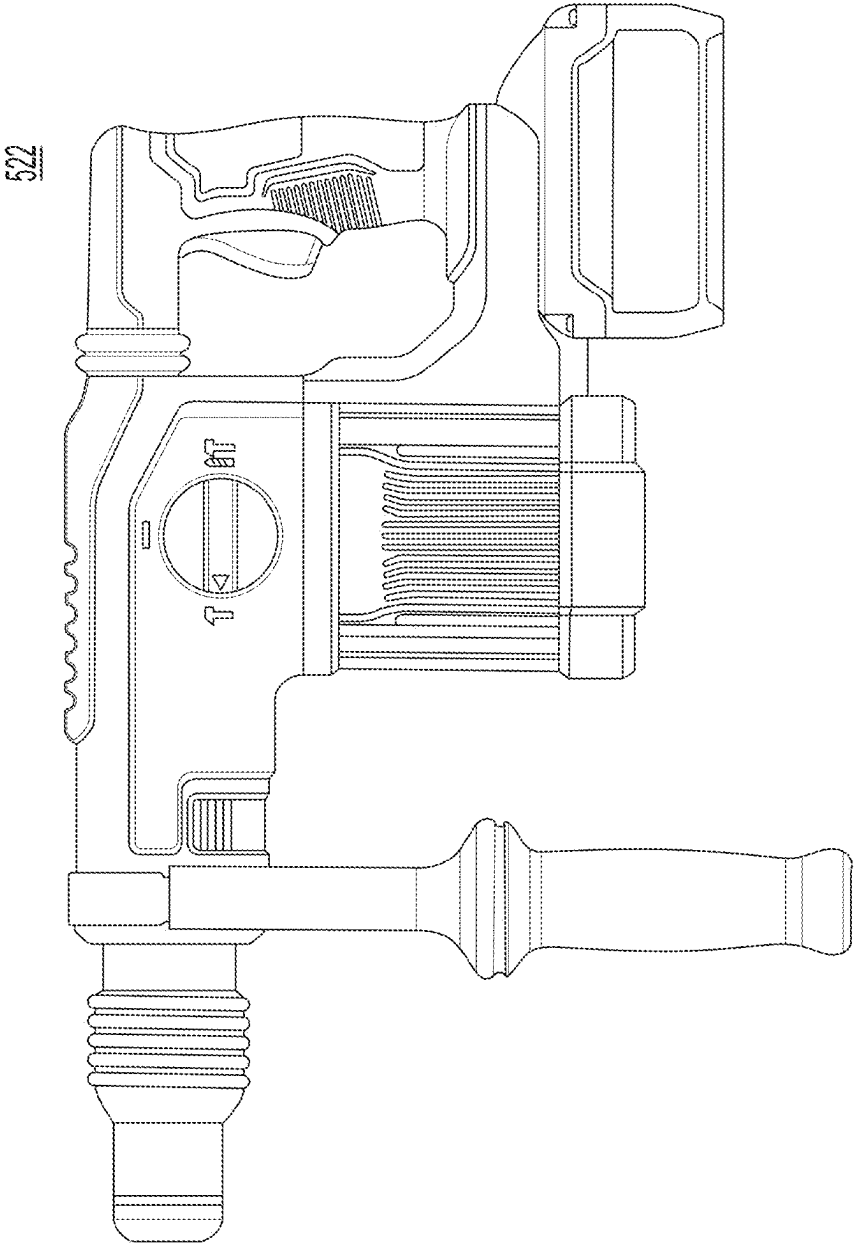


FIG. 37

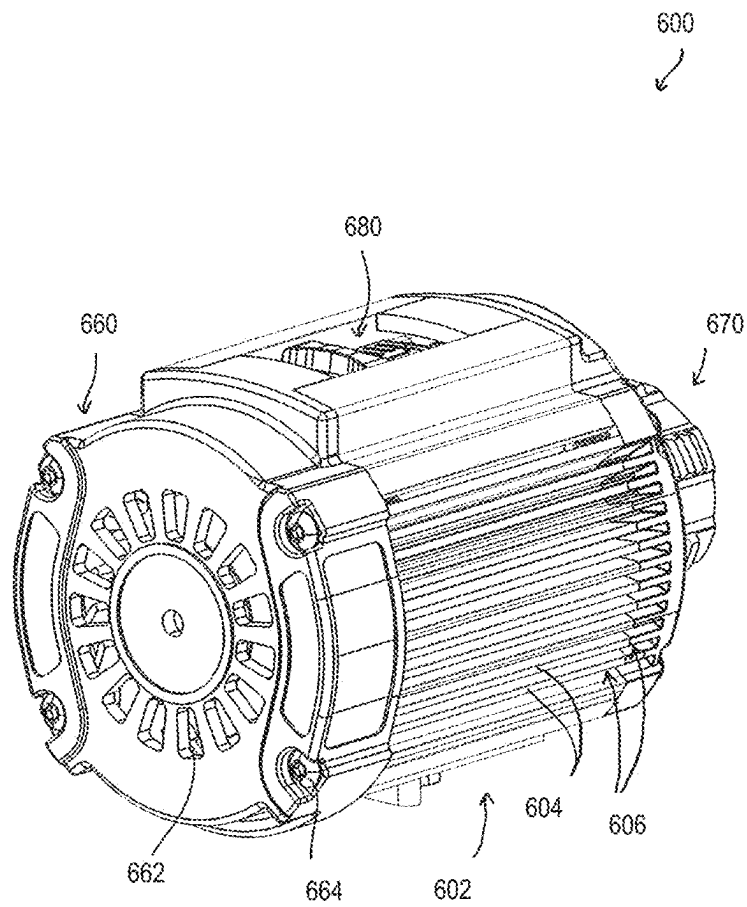
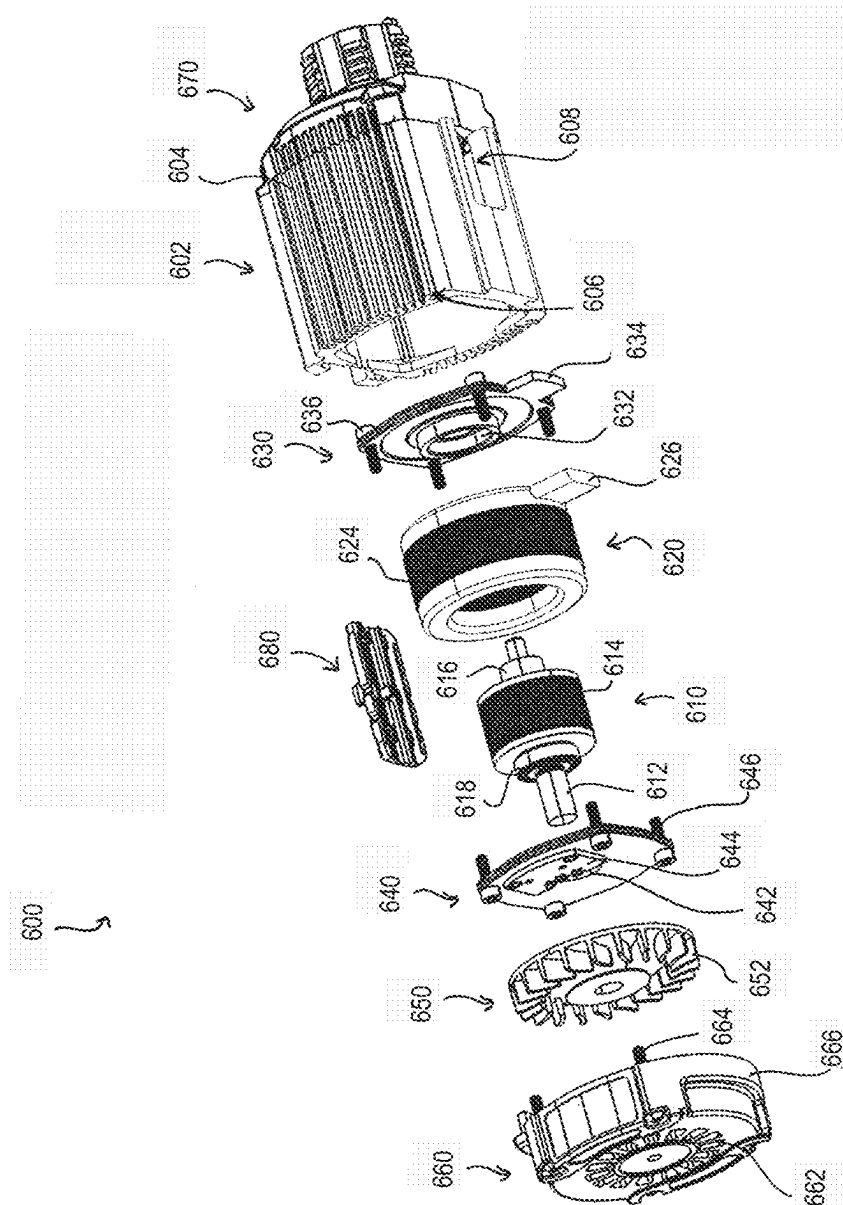


FIG. 38



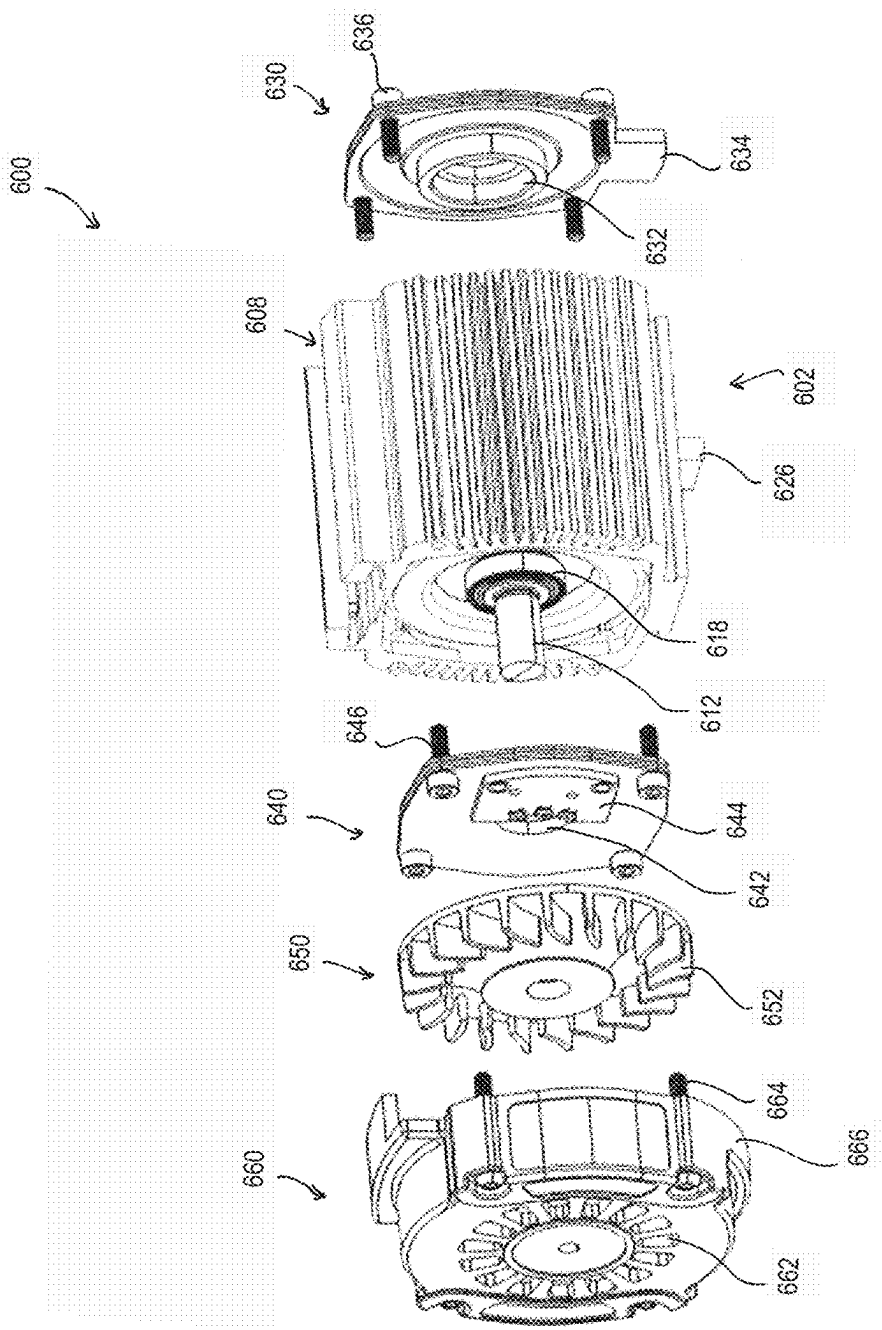


FIG. 40

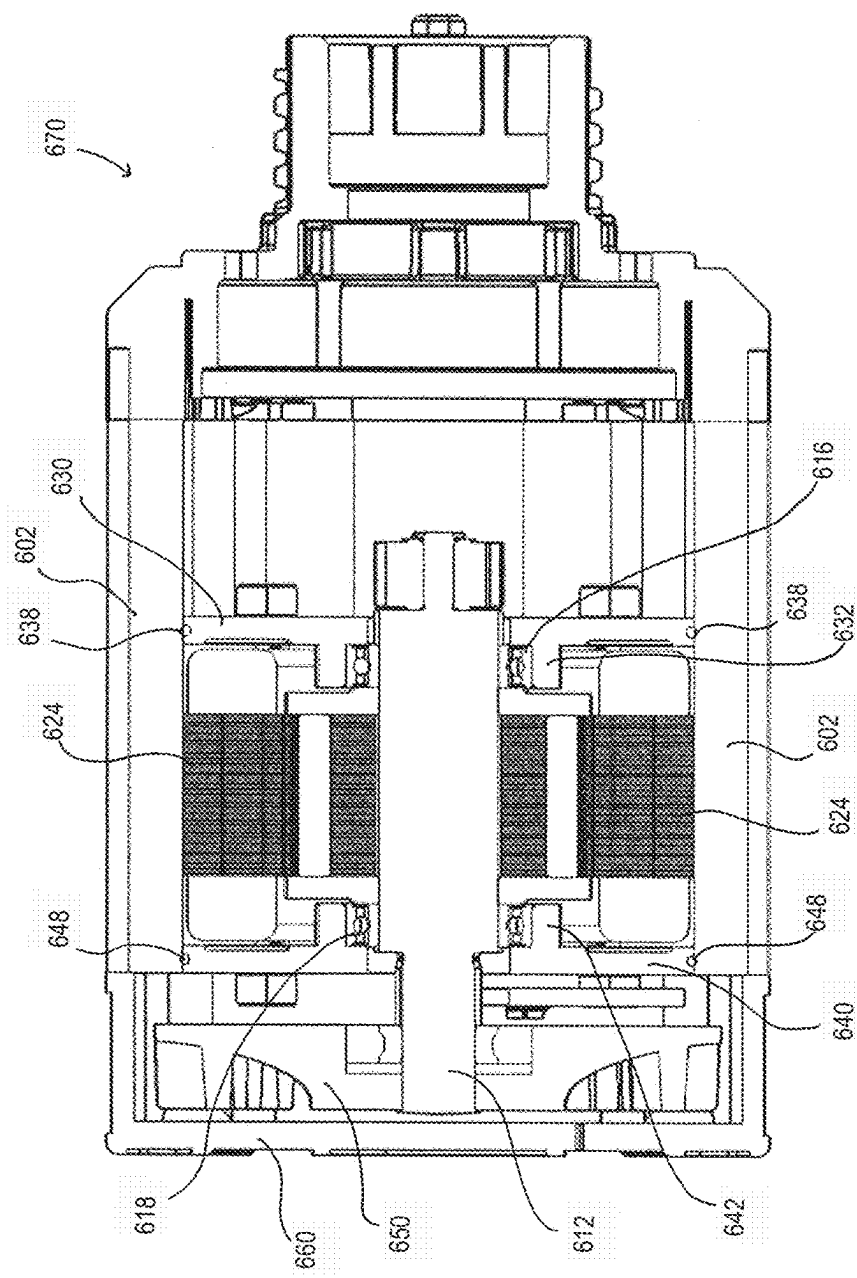


FIG. 41

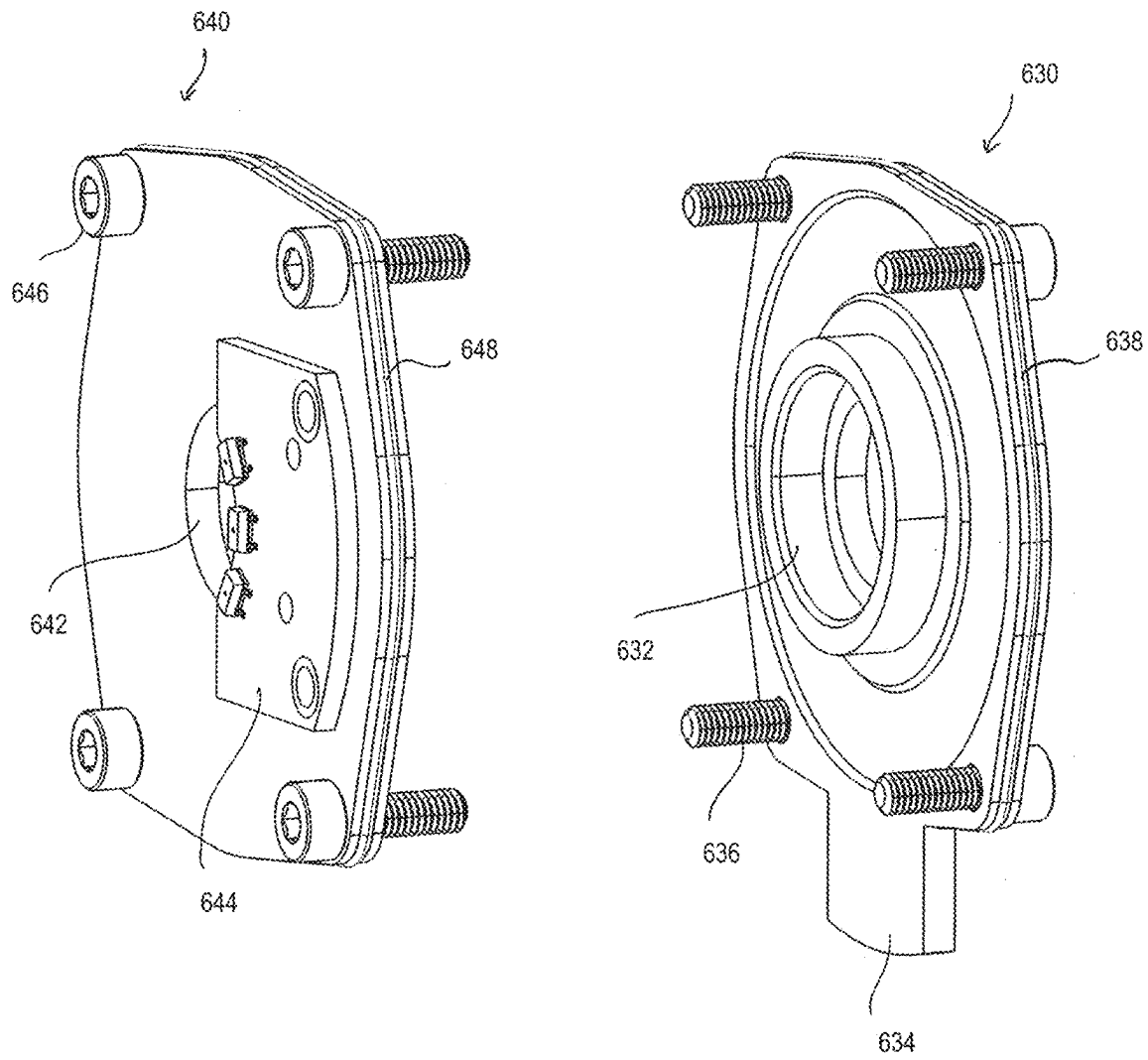


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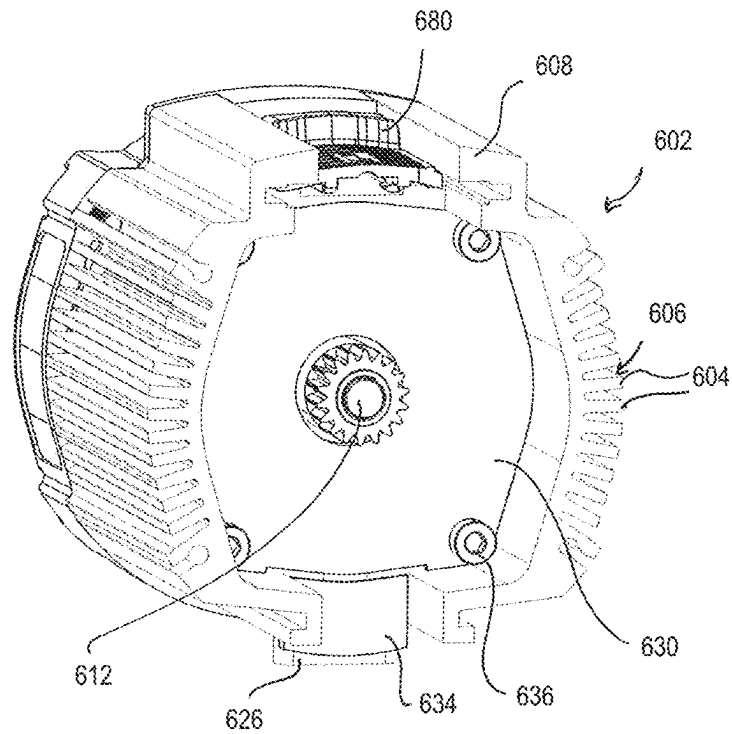


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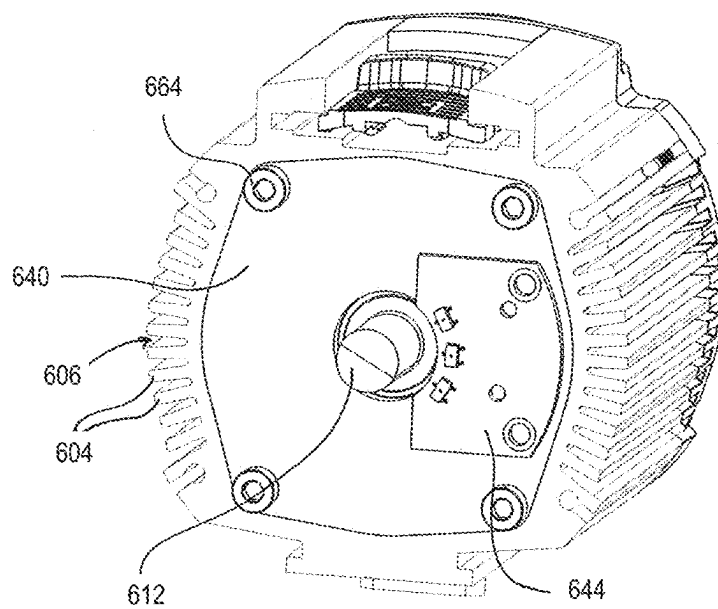


FIG. 44

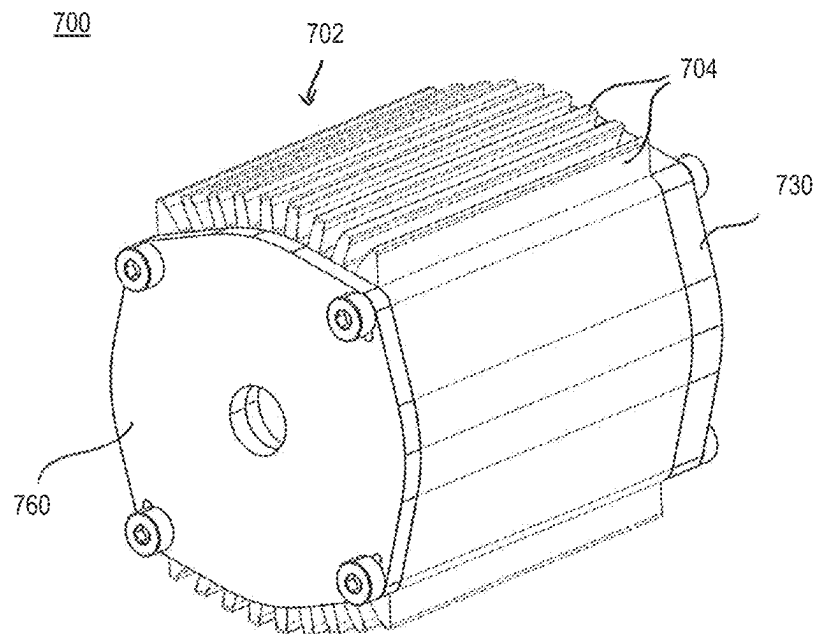


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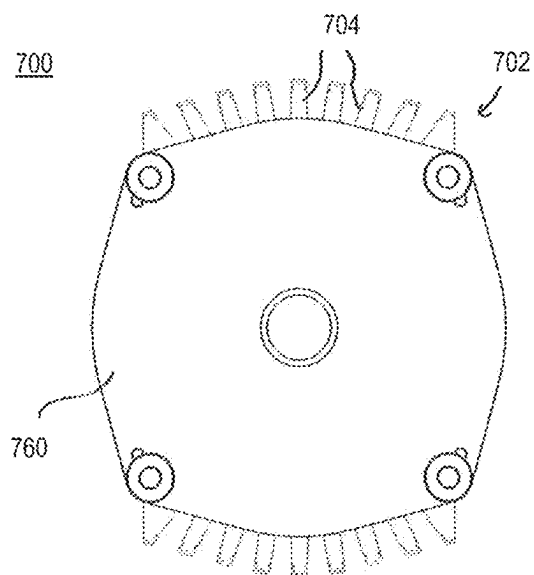


FIG. 46

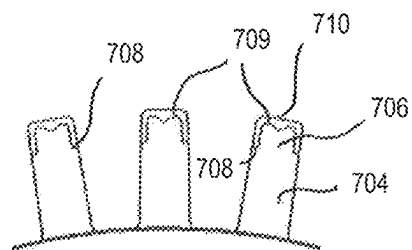


FIG. 47

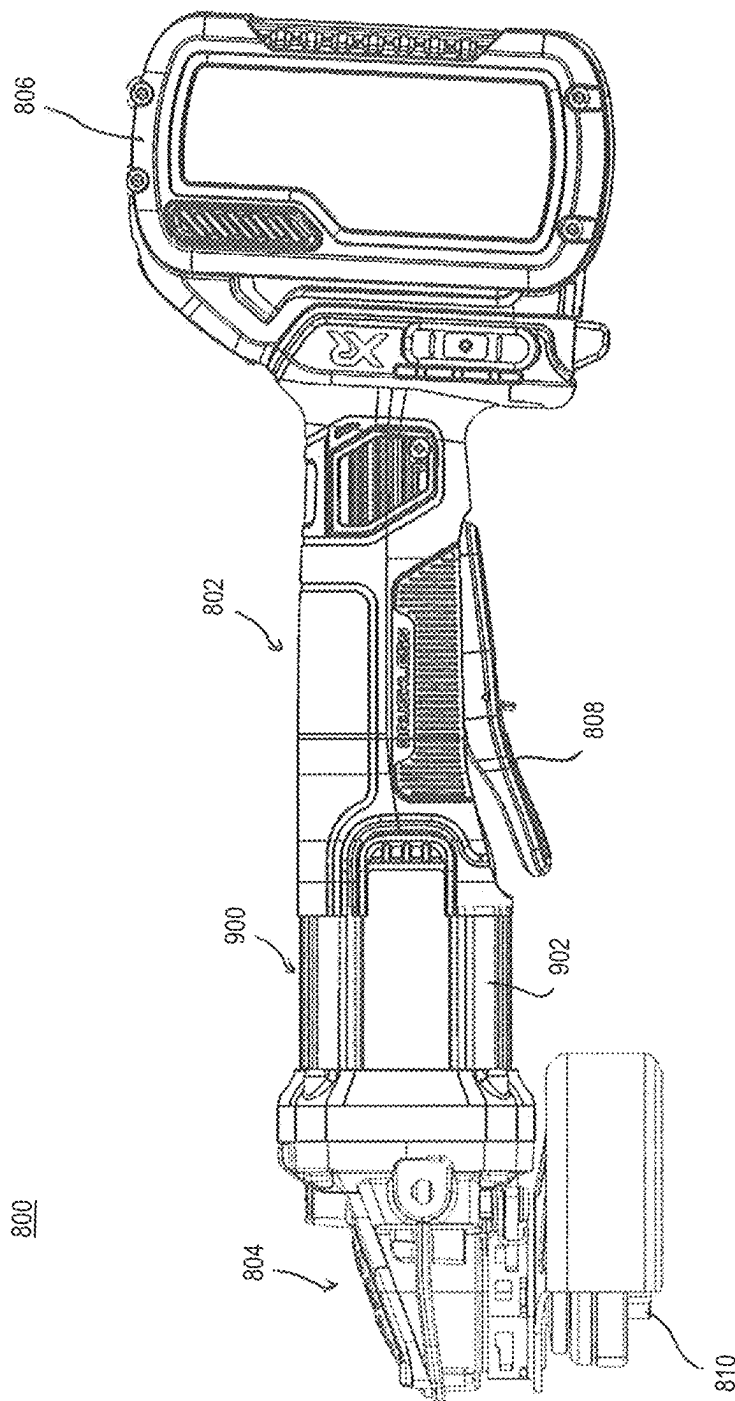


FIG. 48

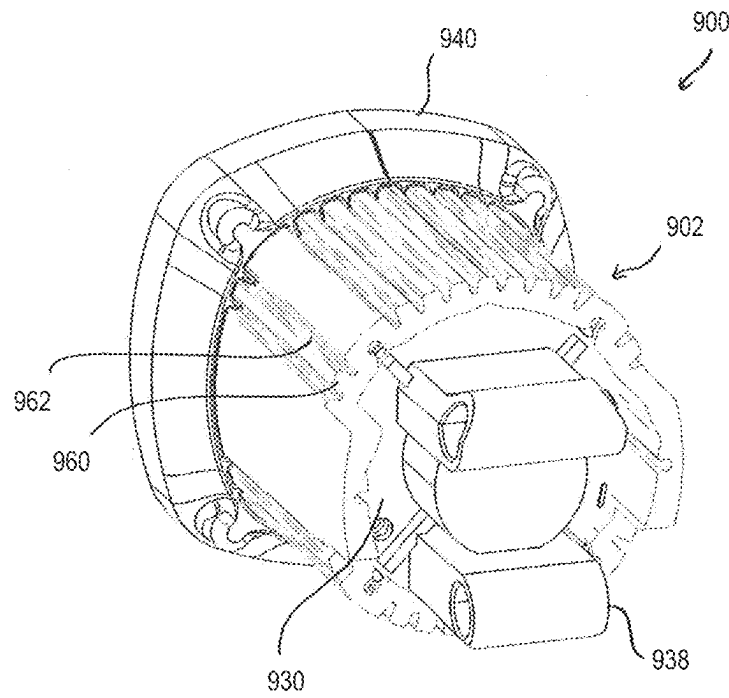


FIG. 49

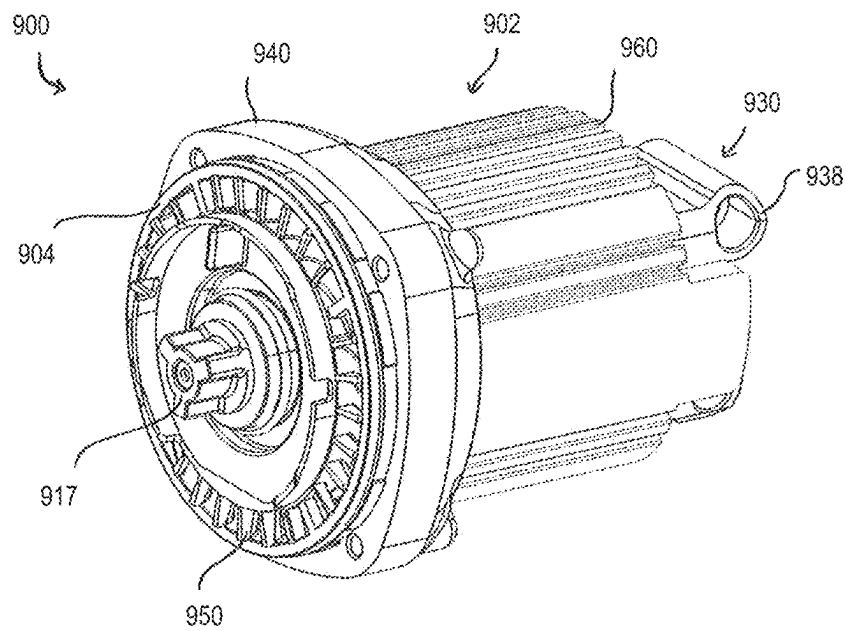


FIG. 50

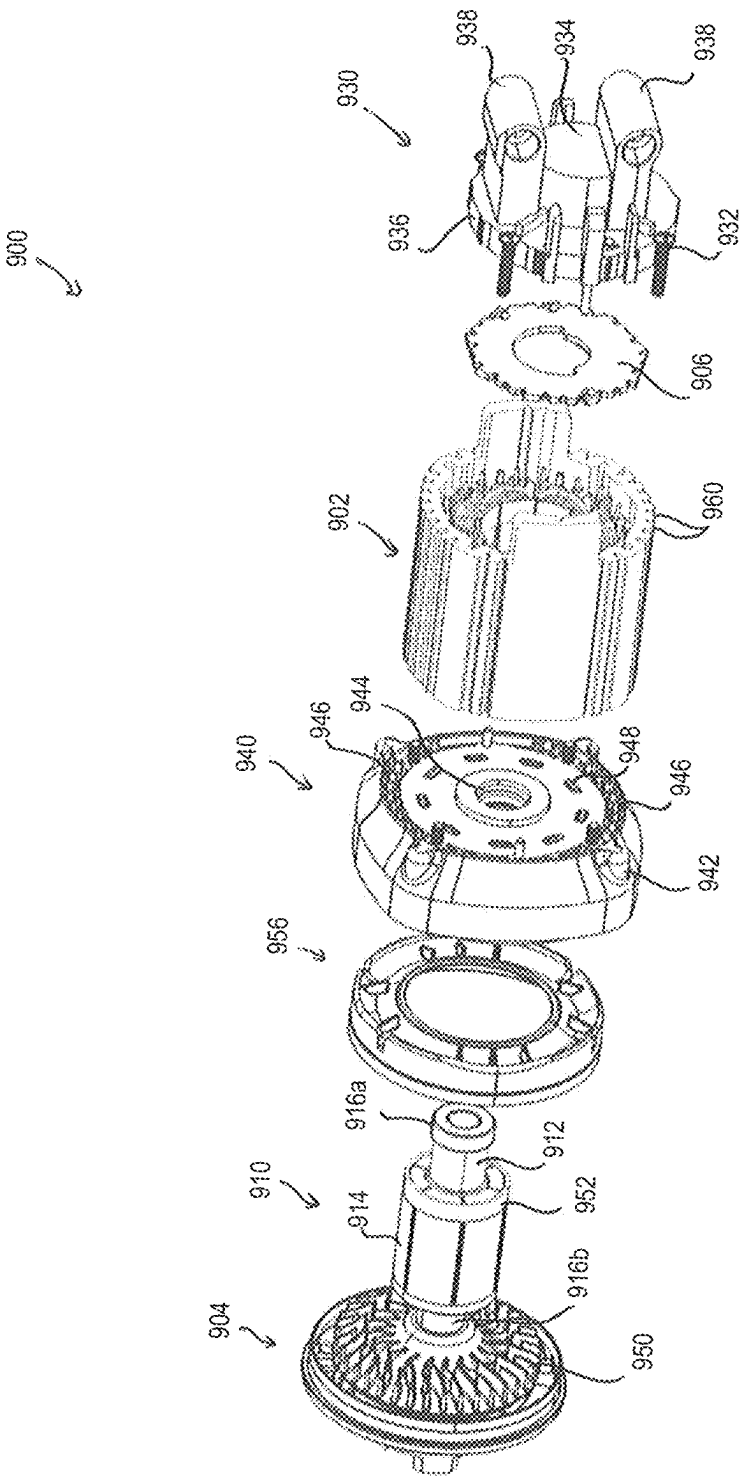


FIG. 51

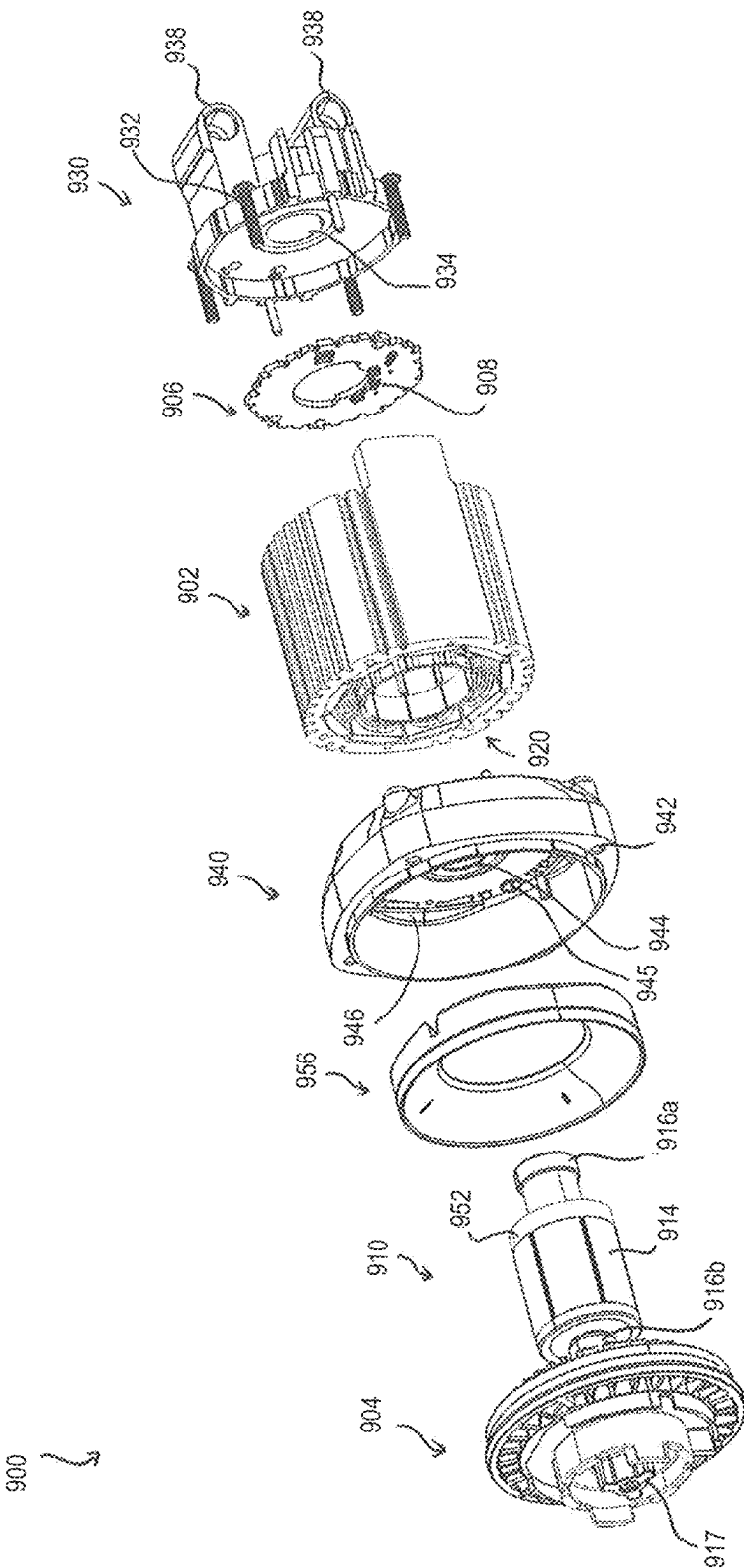


FIG. 52

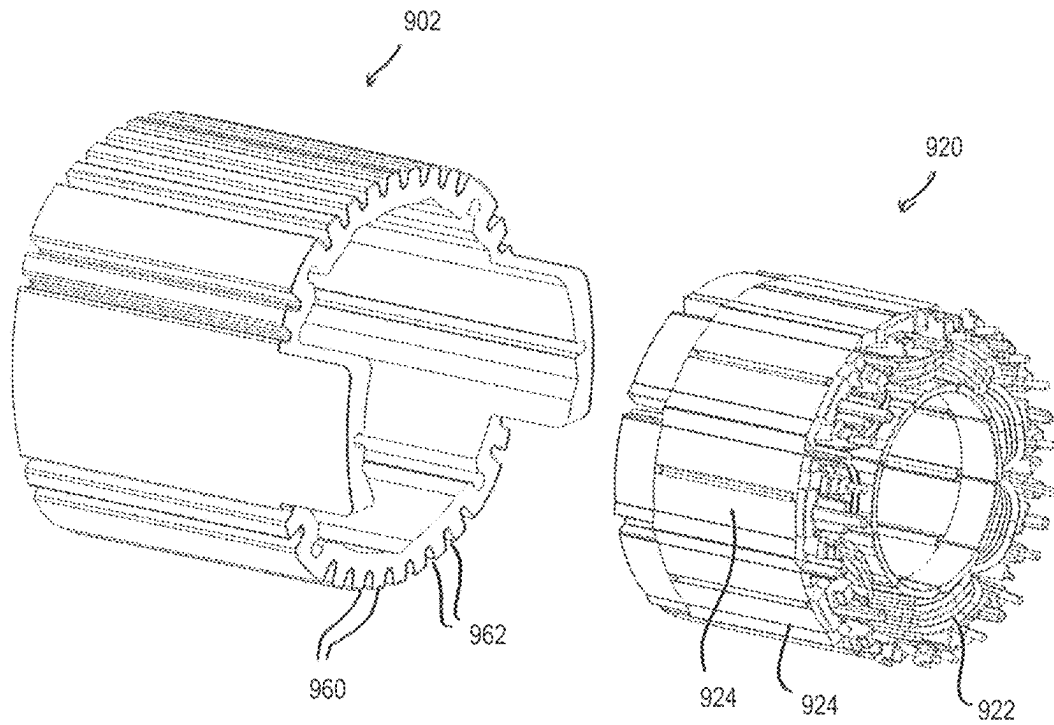


FIG. 53

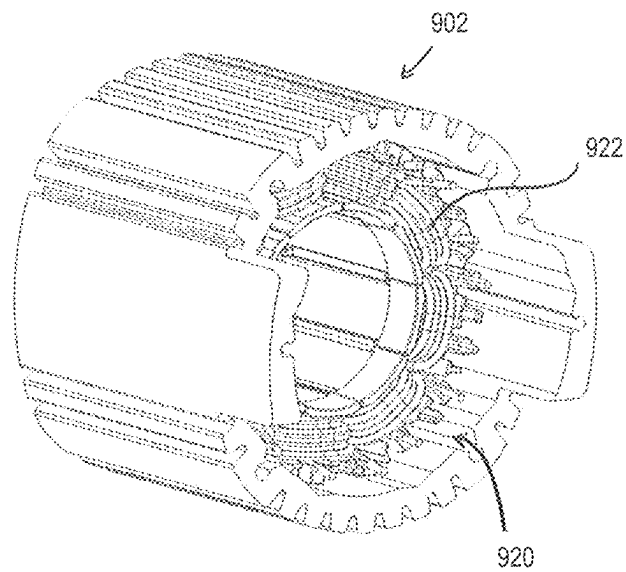


FIG. 54

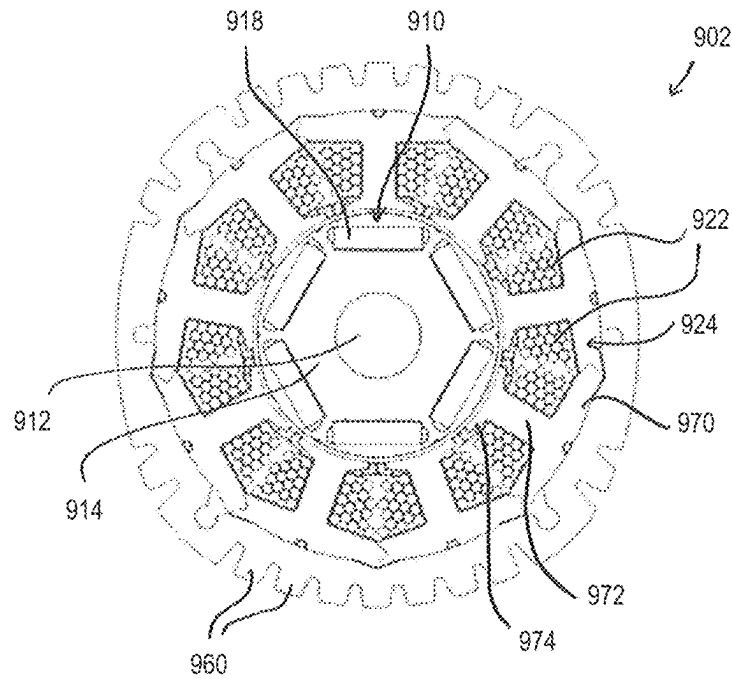


FIG. 55

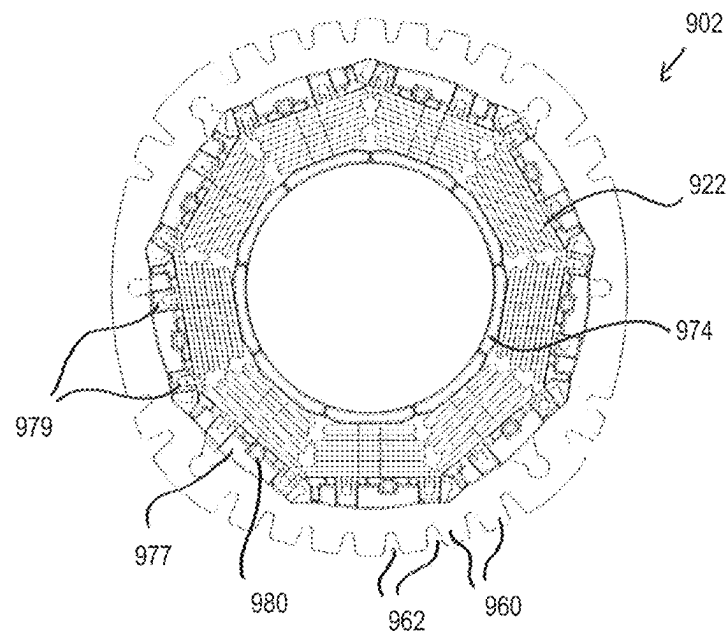


FIG. 56

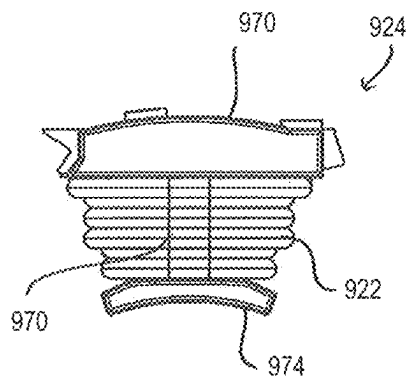


FIG. 57

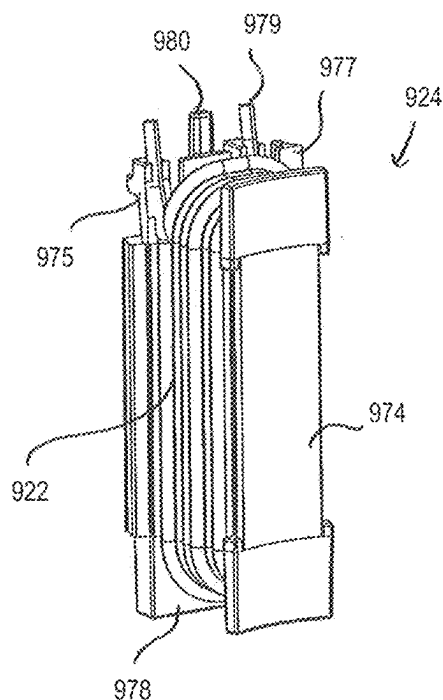


FIG. 58

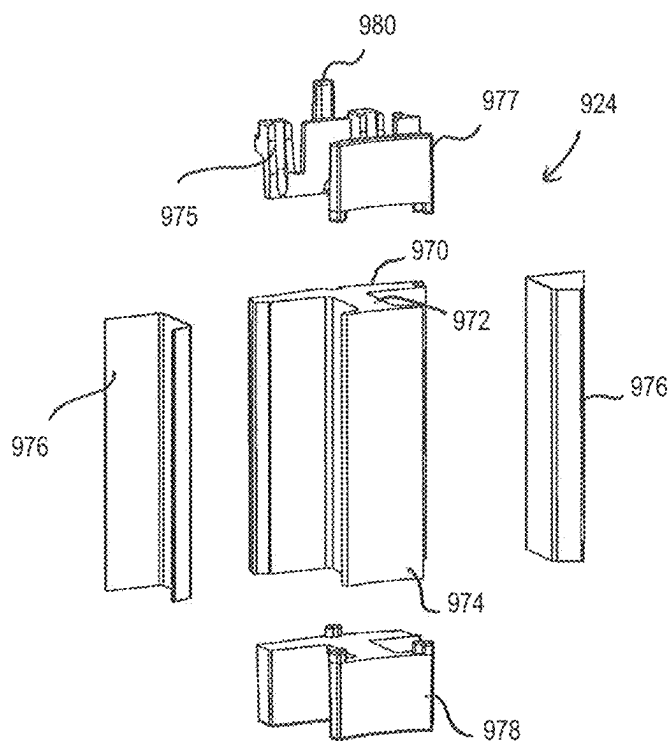


FIG. 59

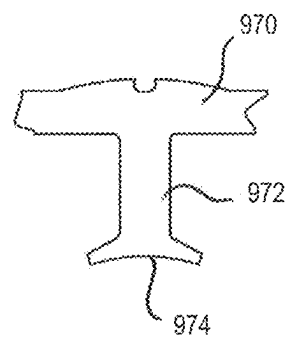


FIG. 60

- Prior Art -

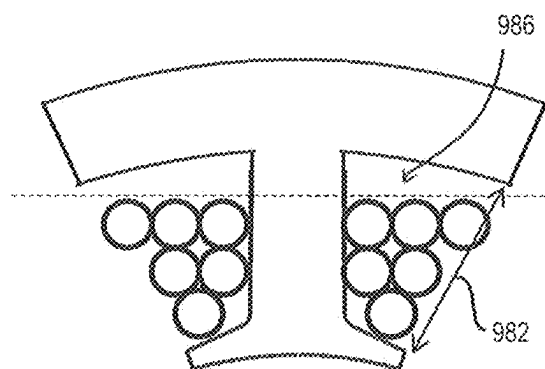


FIG. 61

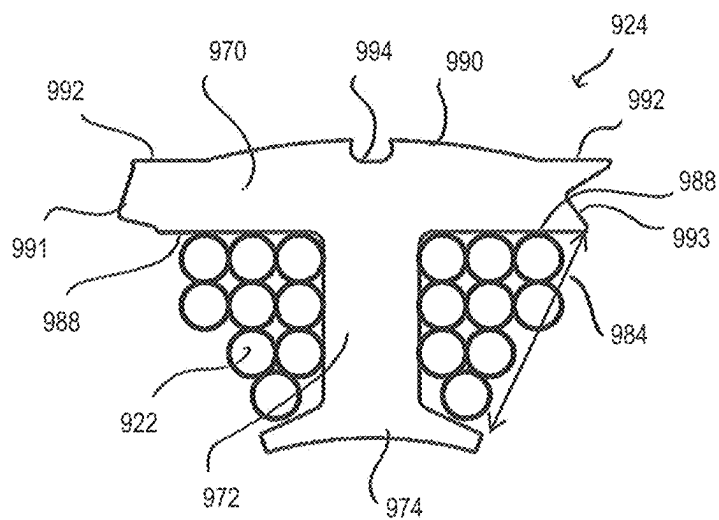


FIG. 62

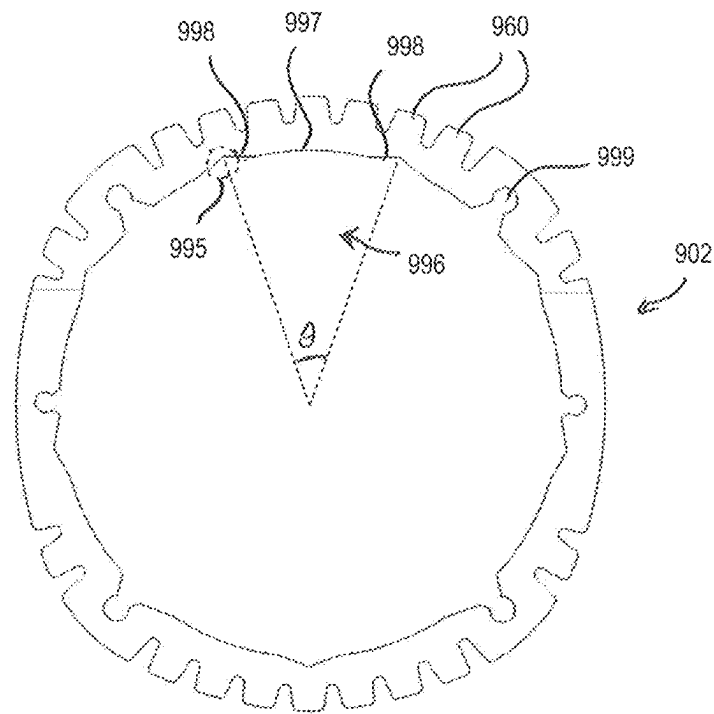


FIG. 63

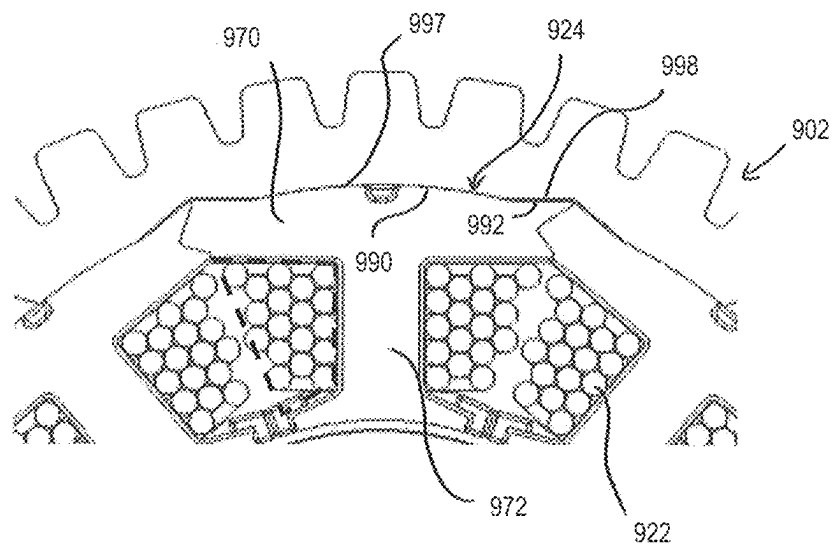


FIG. 64

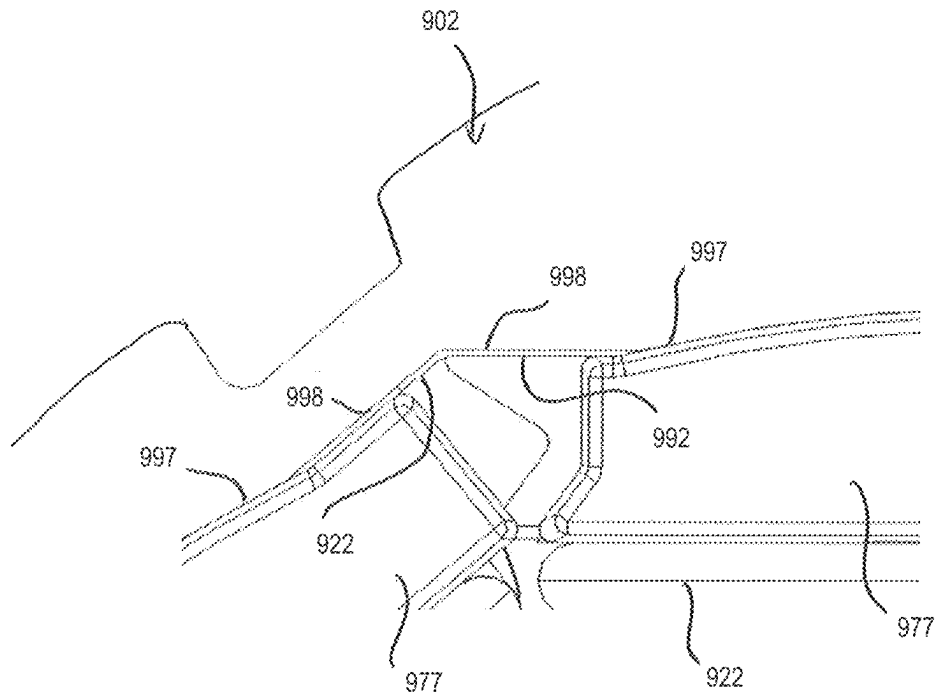


FIG. 65

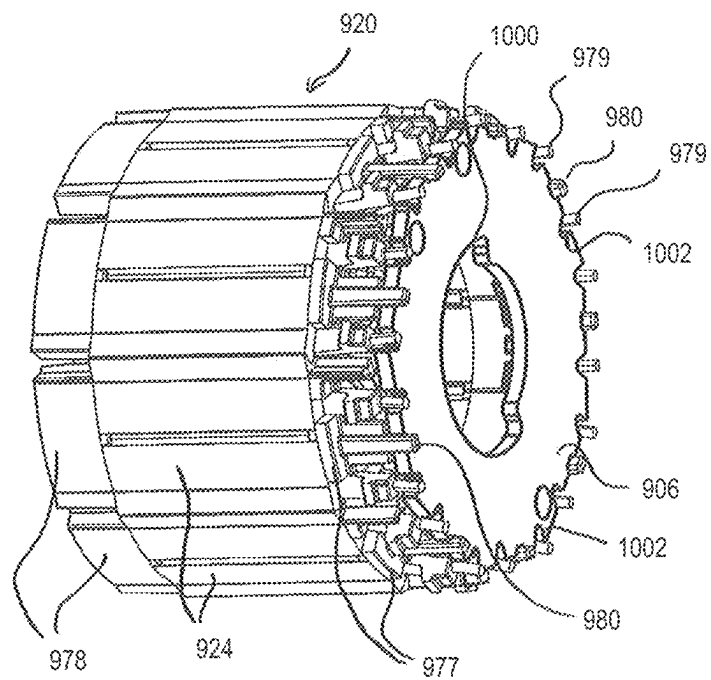


FIG. 66

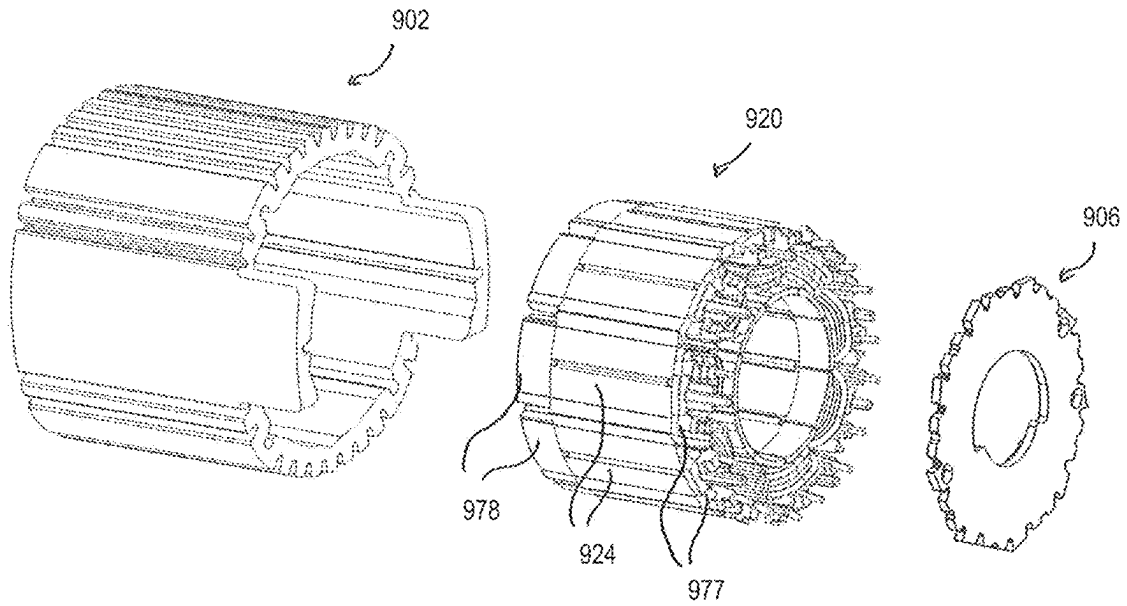


FIG. 67

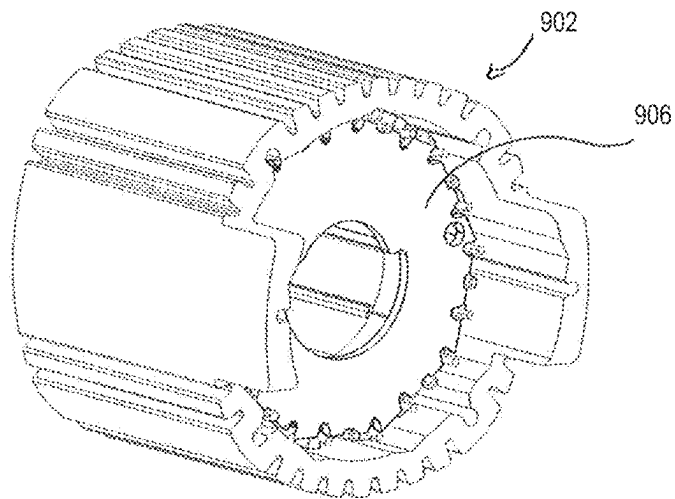


FIG. 68

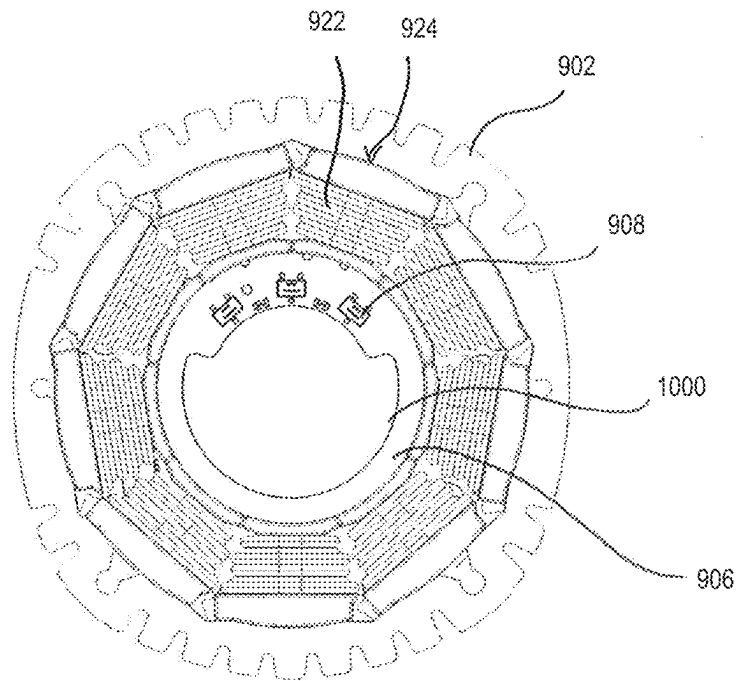


FIG. 69

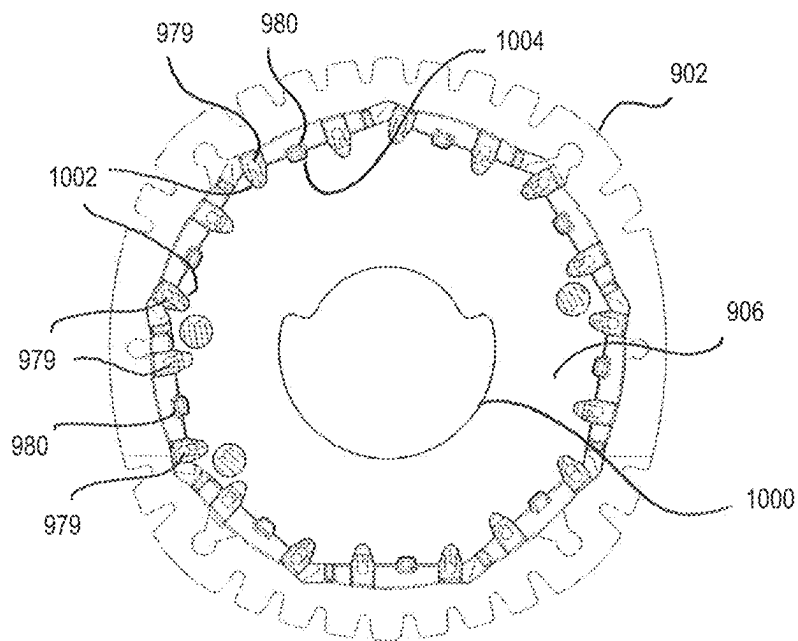


FIG. 70

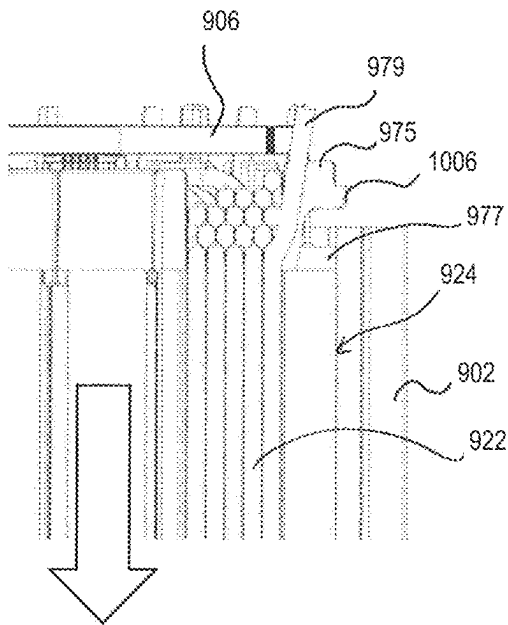


FIG. 71

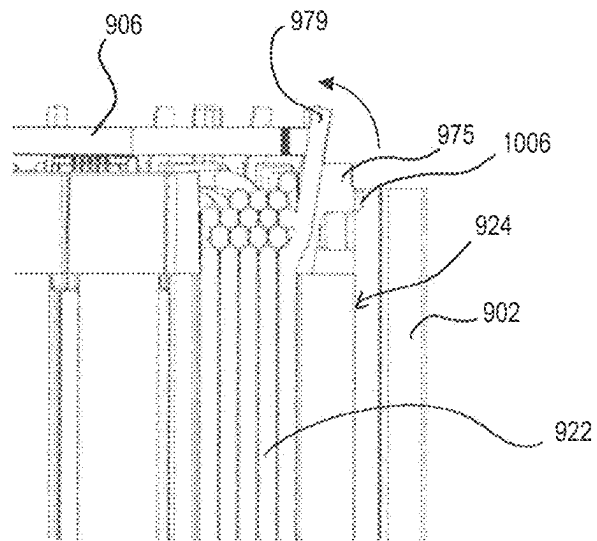


FIG. 72

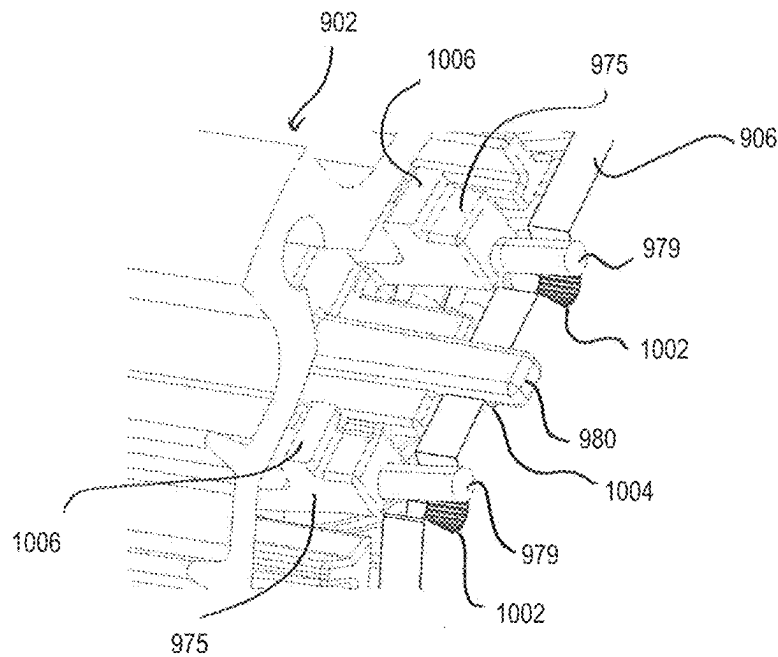


FIG. 73

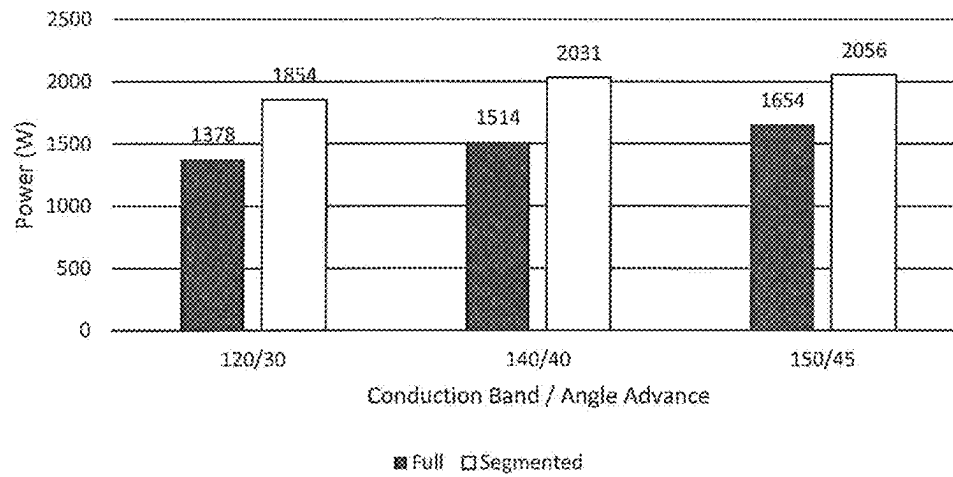


FIG. 74

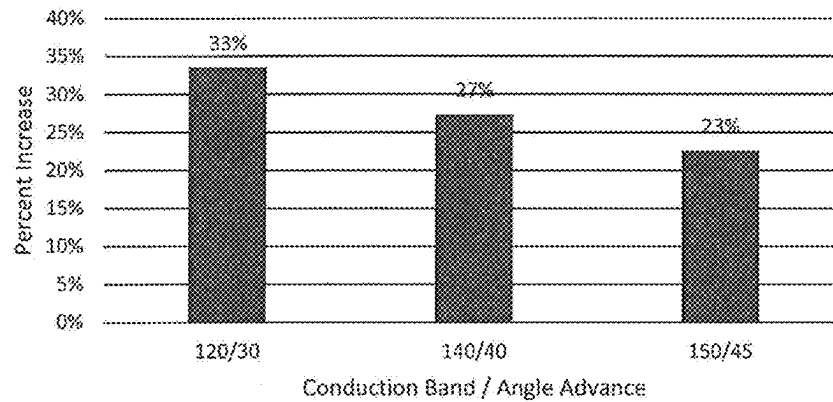
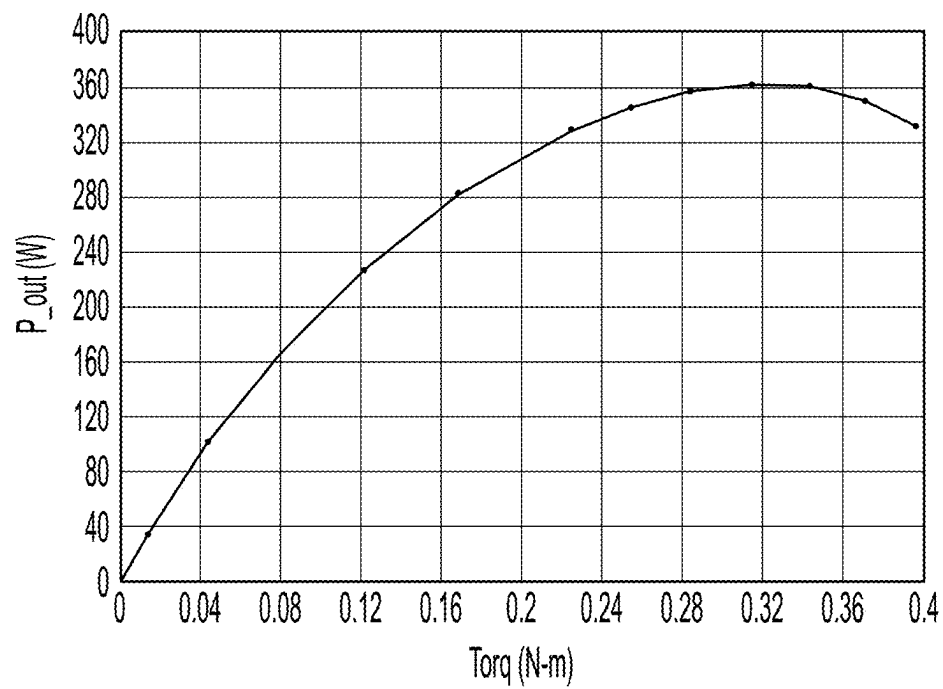


FIG. 75

FIG. 76

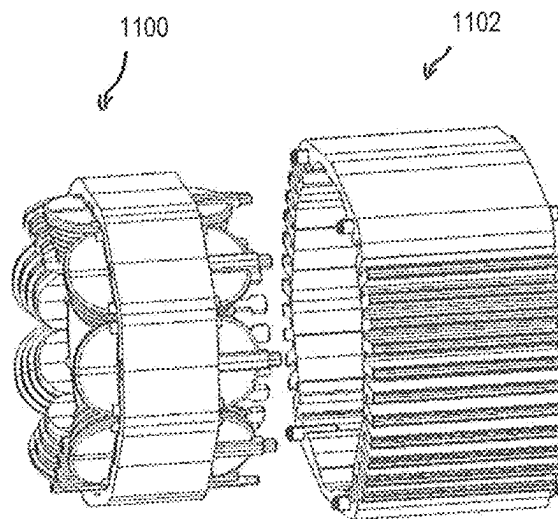


FIG. 77

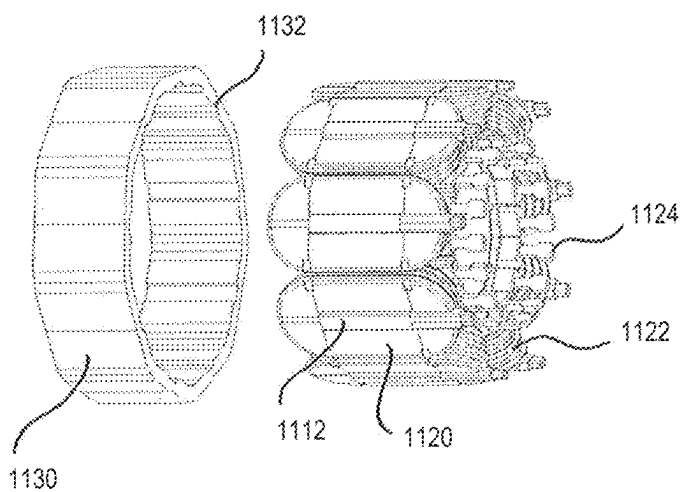


FIG. 78

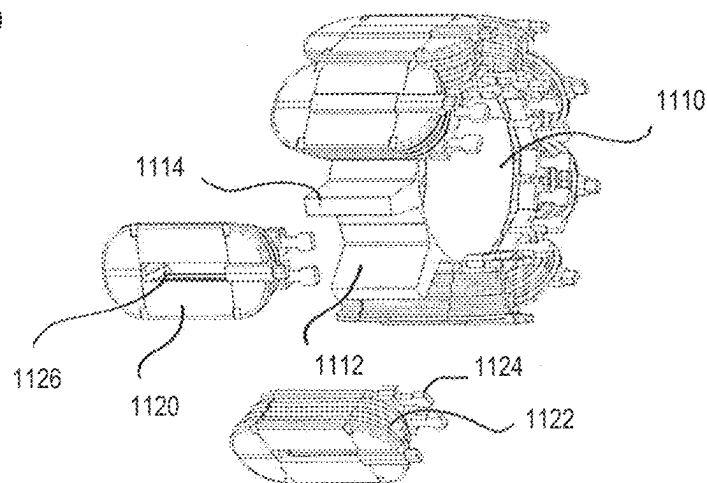


FIG. 79

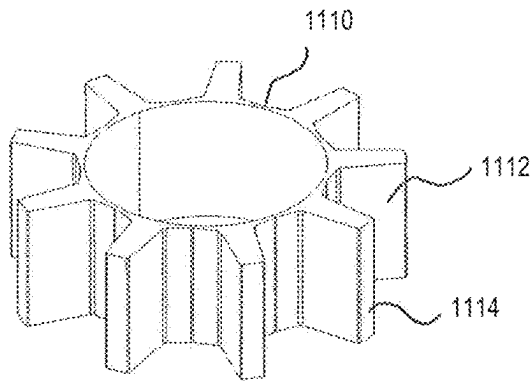


FIG. 80

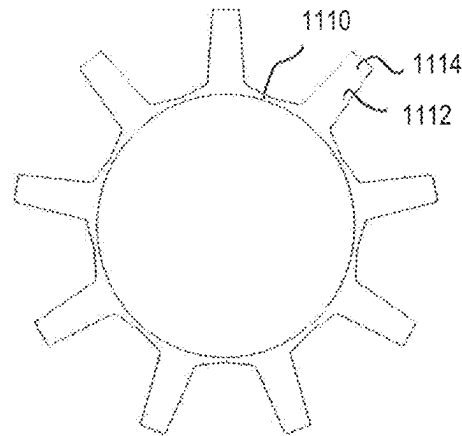


FIG. 81

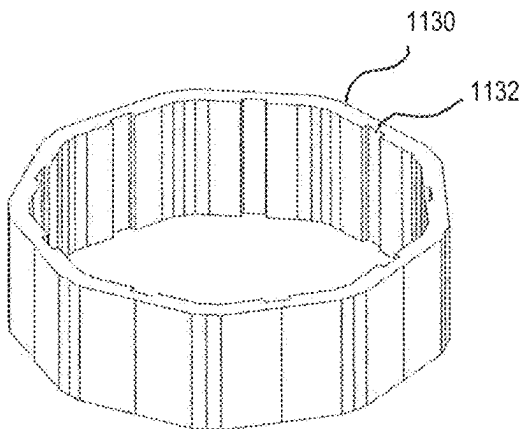


FIG. 82

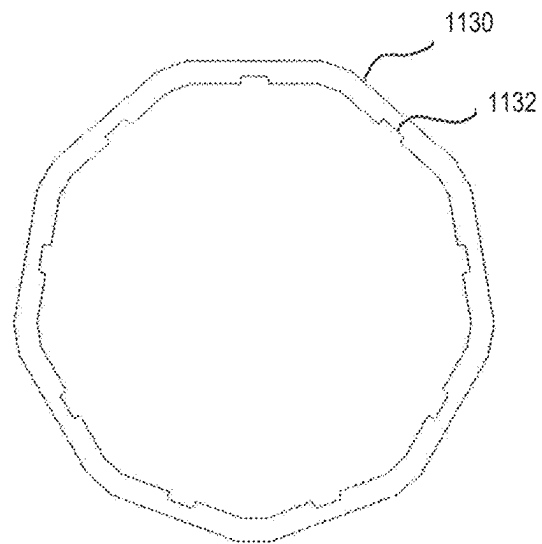


FIG. 83

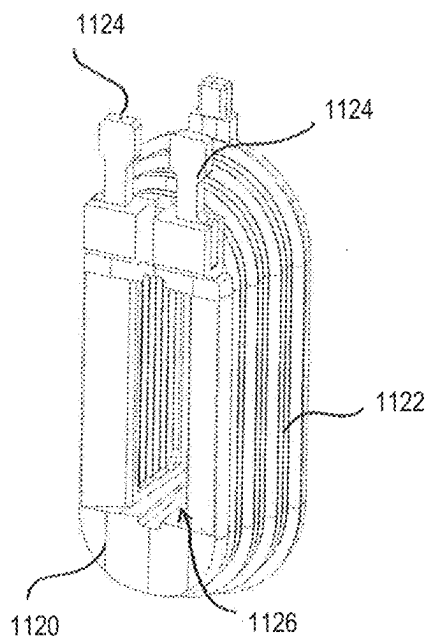


FIG. 84

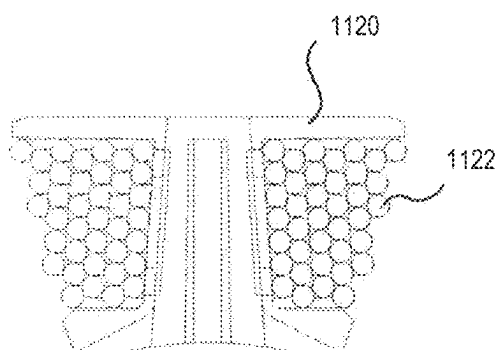


FIG. 85

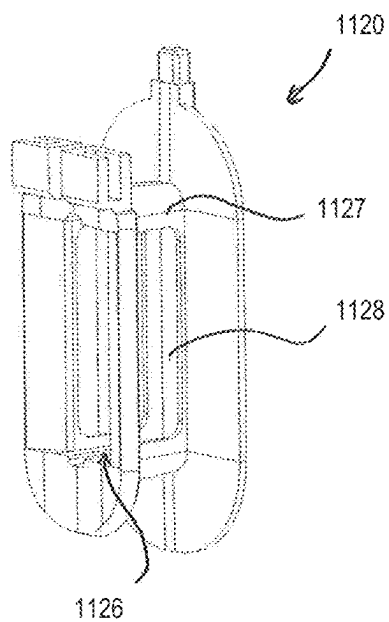


FIG. 86

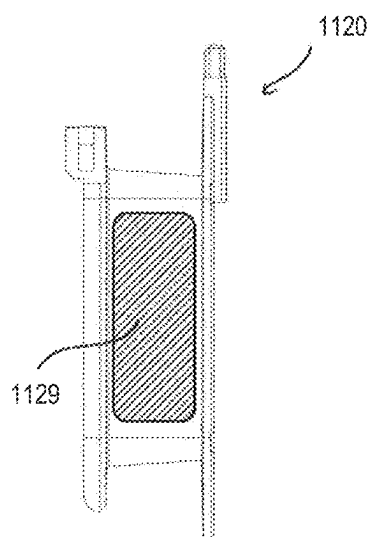


FIG. 87

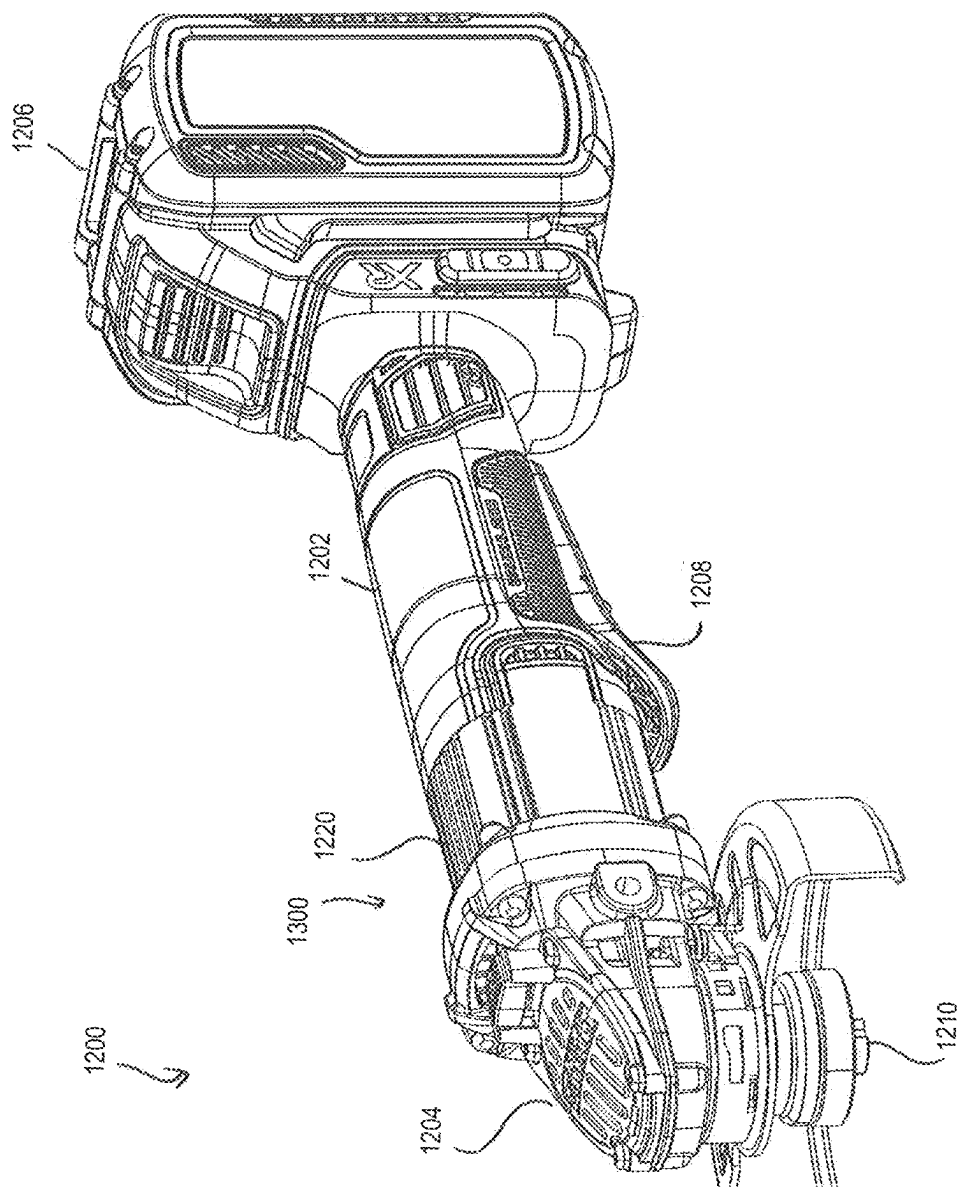


FIG. 88

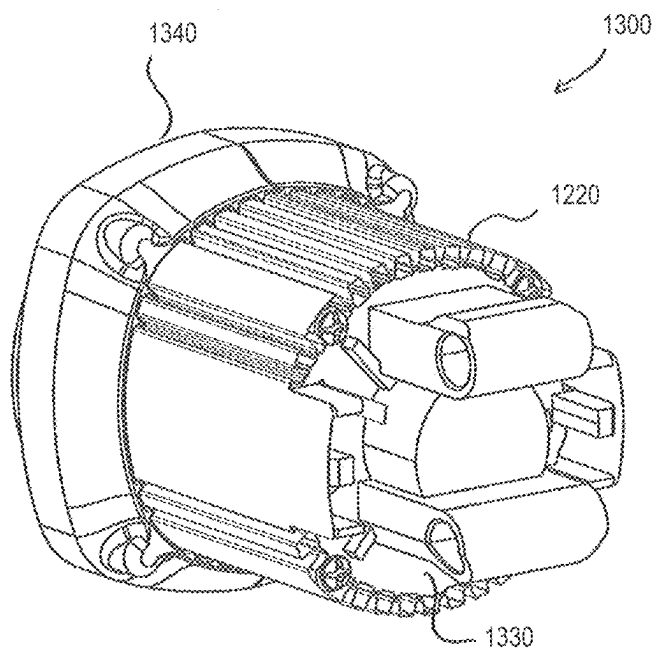


FIG. 89

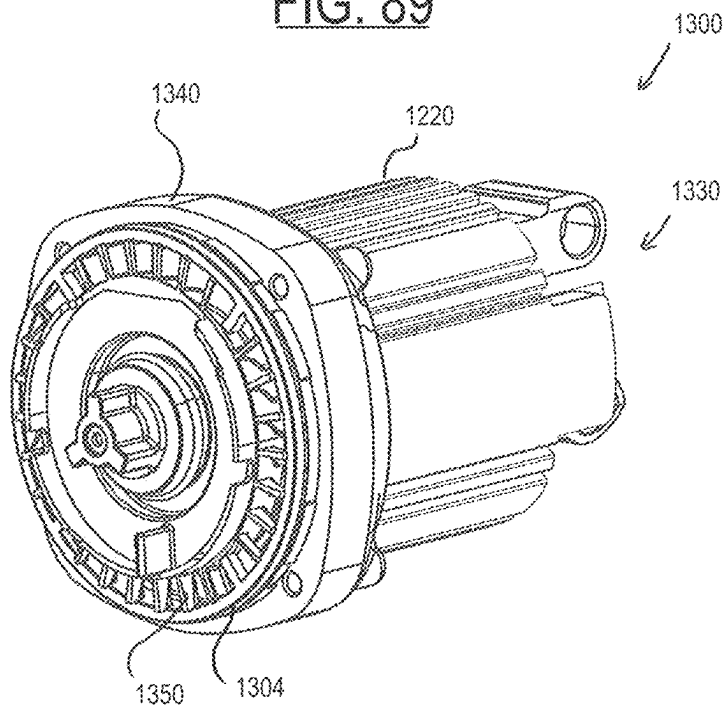


FIG. 90

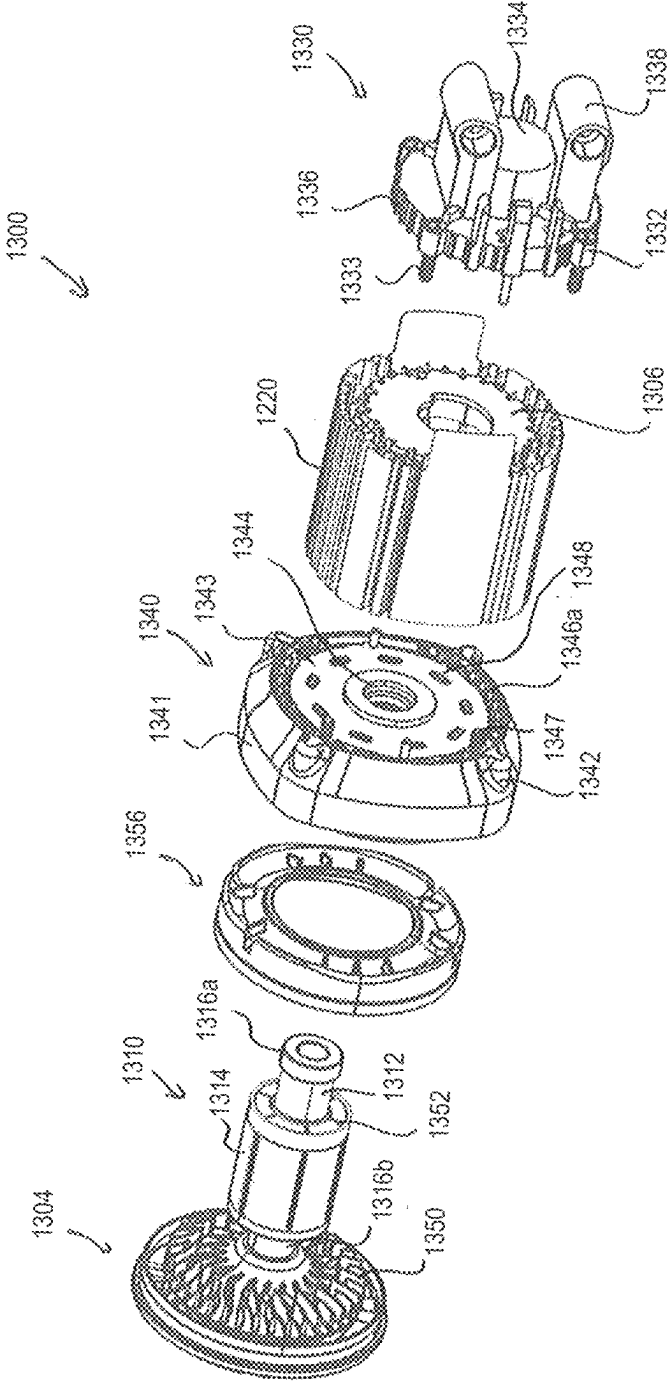


FIG. 91

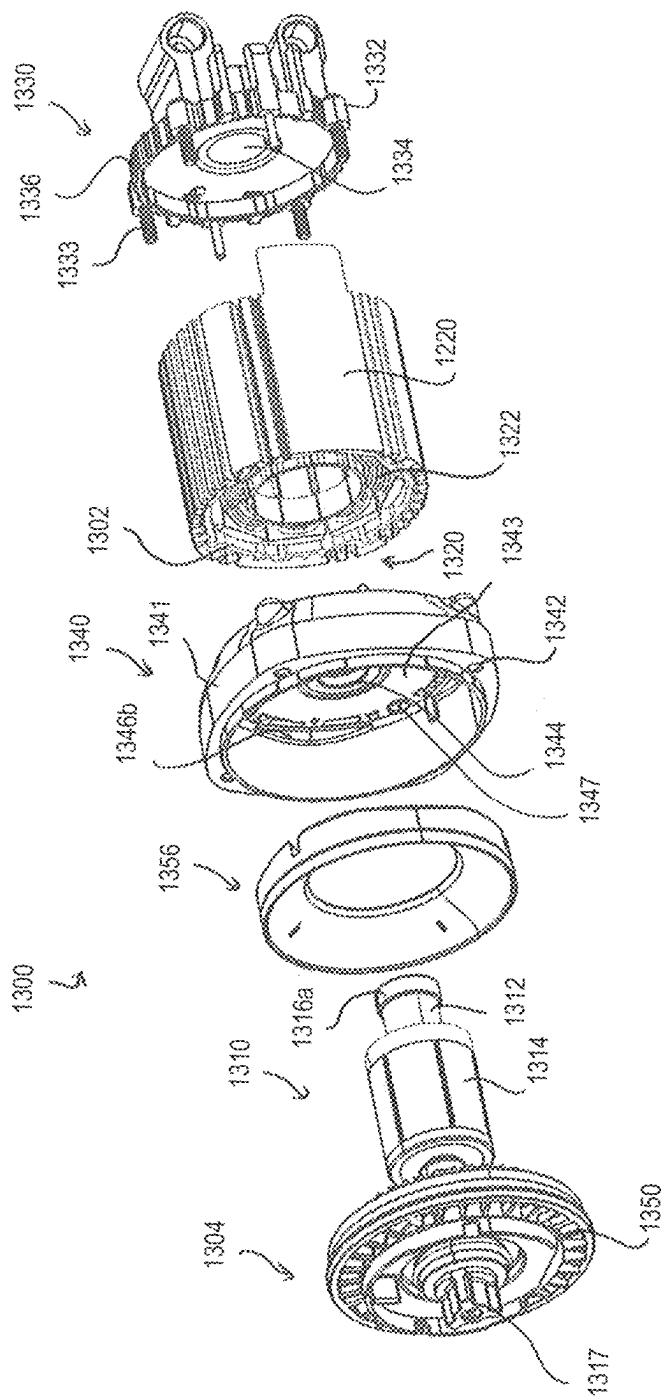


FIG. 92

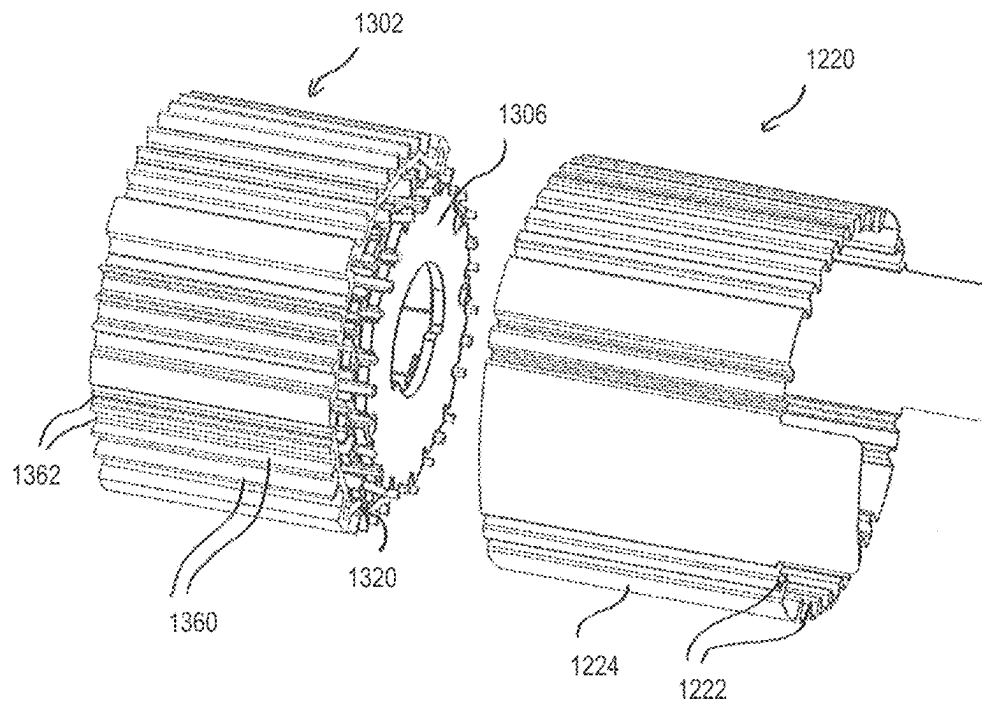


FIG. 93

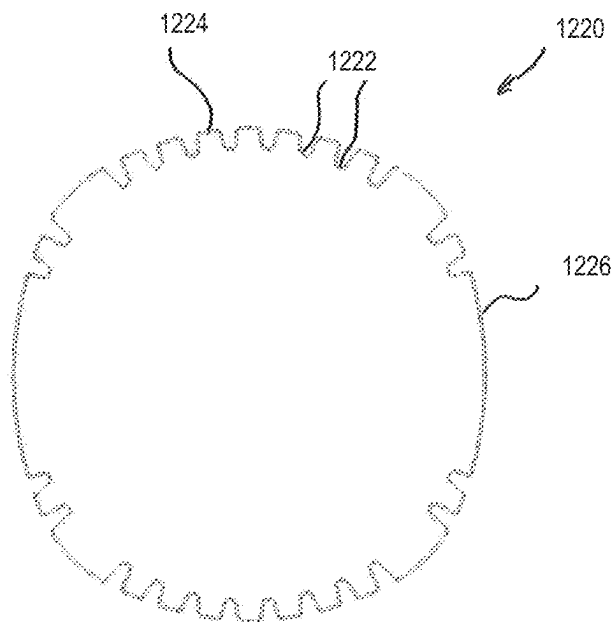


FIG. 94

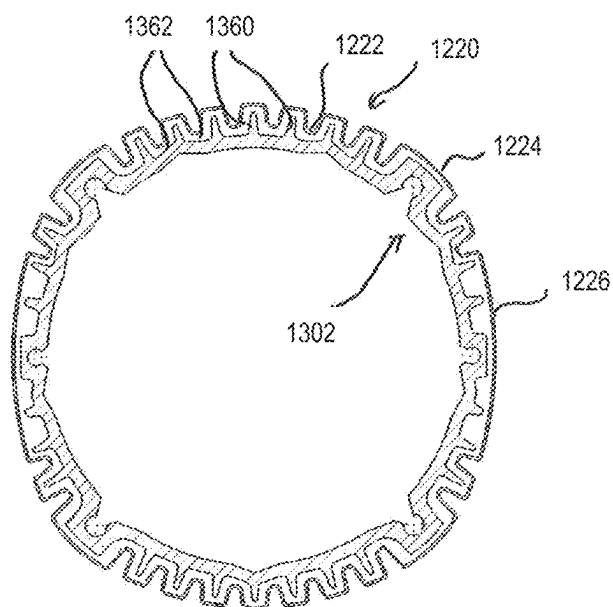


FIG. 95

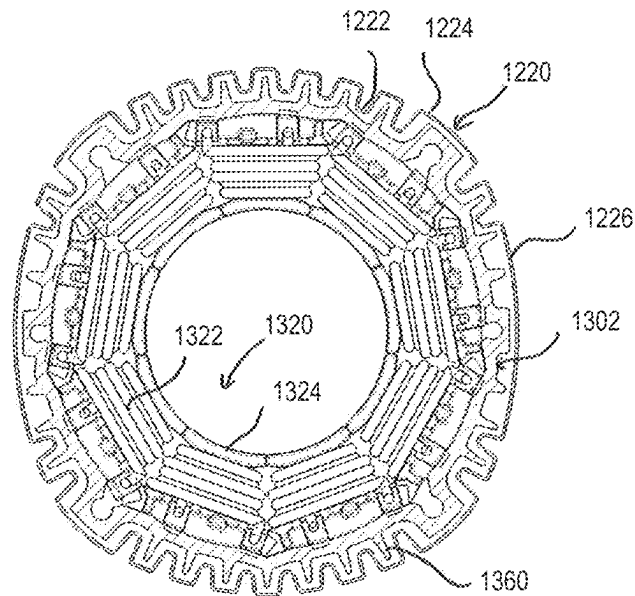


FIG. 96

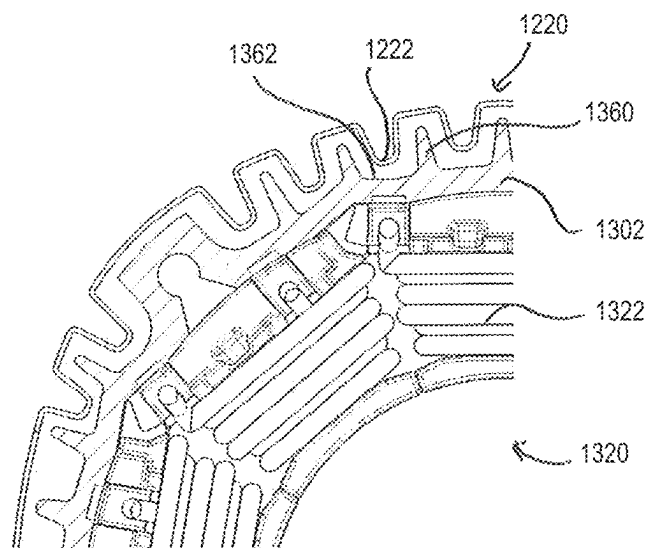


FIG. 97

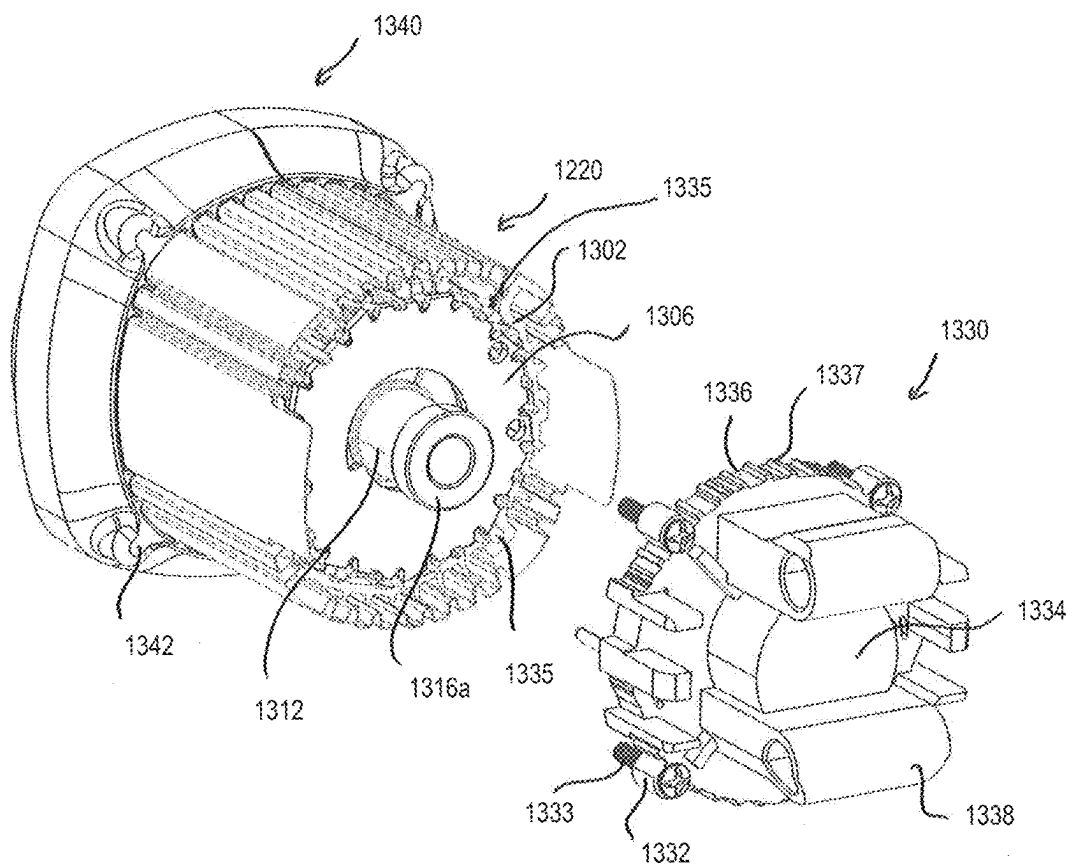


FIG. 98

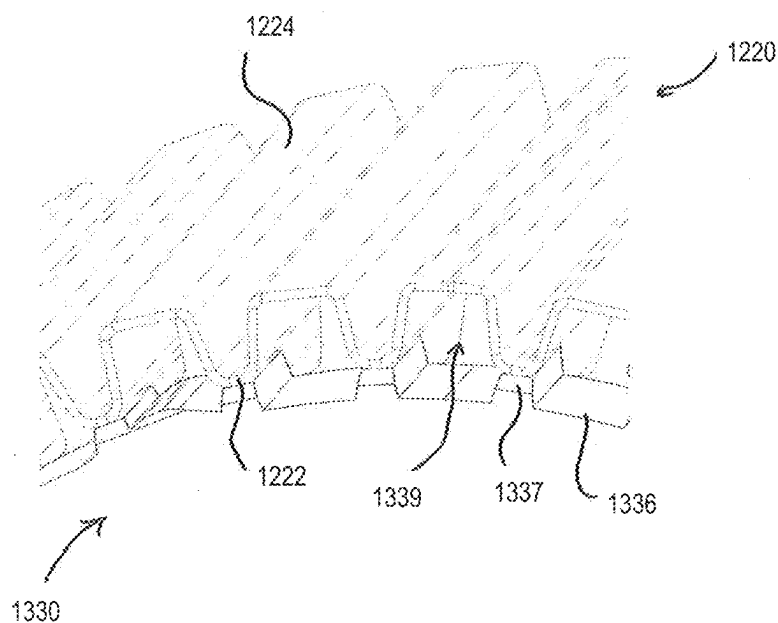


FIG. 99

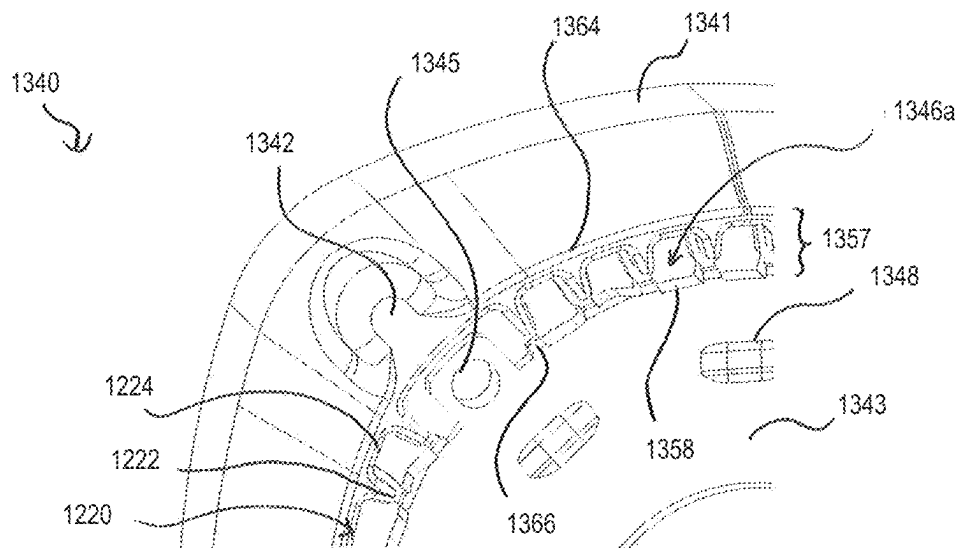


FIG. 100

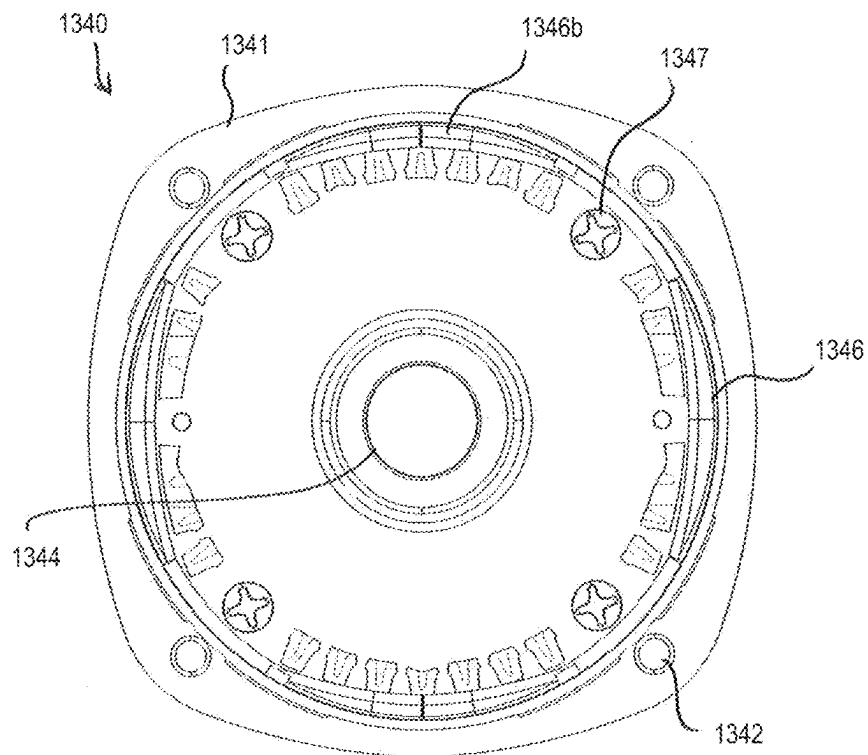


FIG. 101

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POWER TOOL**RELATED APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/411,727 filed Aug. 25, 2021, which is a continuation of U.S. patent application Ser. No. 16/281,475 filed Feb. 21, 2019, now U.S. Pat. No. 11,139,722, which claims the benefit of U.S. Provisional Application No. 62/637,810 filed Mar. 2, 2018; U.S. Provisional Application No. 62/641,008 filed Mar. 9, 2018; and U.S. Provisional Application No. 62/693,564 filed Jul. 3, 2018. The contents of all these applications are incarnated herein by references in their entireties.

FIELD OF THE DISCLOSURE

This disclosure relates to power tools. More particularly, the present invention relates to a power tool with a unique end cap design for improved air flow.

BACKGROUND

Use of brushless DC (BLDC) motors in power tools has increased significantly over the past several years. Examples of such power tool applications are U.S. Pat. No. 9,450,472, US Patent Publication No. 2016/0149463, US Patent Publication No. 2017/0106521, and US Patent Publication No. 2017/0120435, contents of all of which are incorporated herein by reference in their entireties.

As BLDC motors have been used in higher power applications such as grinders, miter saws, etc., a challenge has been to optimize power density so as to keep the power tool as compact and portable as possible while obtaining the desired power output. The goal of power tool manufacturers has been to provide the most amount of power from the motor using the least amount of space possible. The ratio of maximum power output of a motor to the volume of the motor is known as the power density. Power density can be maximized by keeping the size of the motor the same and increasing power, or by keeping the power output the same and reducing the motor volume.

One of the barriers that significantly limit the potential for optimal power density is the motor cooling techniques conventionally used in the industry. Motor coils typically generate a significant amount of heat, particularly for higher power applications. As can be seen in the references cited above, conventional BLDC motors in power tools typically include a cooling fan coupled to the rotor that generates an airflow through the stator windings. This requires a large portion of the volume of the motor stator to be dedicated to providing airflow paths between the motor windings. In most conventional tools, stator slots where stator coils are wound are required to have a maximum slot fill of 50% or less to ensure air gaps between the stator coils can sufficiently cool the coils. This results in large areas of the motor to be wasted for non-magnetic usage.

What is needed is a motor design suitable for power tool applications that provides for maximization of the power density of the motor.

SUMMARY

According to an embodiment of the invention, a motor assembly is provided including a stator and a rotor. The stator includes a stator main body defining a longitudinal axis, stator teeth projecting radially from the stator main

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body, stator windings wound around the stator teeth, and two winding terminals provided for each stator tooth extending away from the stator main body substantially parallel to the longitudinal axis. The rotor includes a rotor shaft, a rotor core mounted on the shaft, at least one rotor permanent magnet affixed to the rotor core arranged to magnetically interface with the stator windings. In an embodiment, a circuit board is oriented along a radial plane perpendicular to the longitudinal axis adjacent the stator. The circuit board includes a central through-hole through which the rotor shaft extends, at least one magnetic sensor mounted on a surface of the circuit board around the central through-hole configured to magnetically interface with the rotor, peripheral openings arranged to receive the winding terminals of the stator, and conductive routings extending from the peripheral openings to connect the stator windings within each phase of the motor in a series or a parallel configuration and the stator windings within different phases of the motor in a wye or a delta configuration.

In an embodiment, the motor includes axial retainers that axially retain the circuit board relative to an axial end of the stator.

In an embodiment, the peripheral openings include notches formed on a circumference of the circuit board. In an embodiment, the winding terminals are initially positioned at an angle of approximately 5 to 15 degrees relative to the longitudinal axis, are received in radial alignment with the notches, and are radially pressed into the notches to electrically connect to the conductive routings.

In an embodiment, at least one end insulator is located between an axial end of the stator main body and the circuit board and extending radially to insulate at least one of the stator teeth from a corresponding one of the stator windings. For each of the stator teeth, the end insulator includes two legs projecting axially and configured to support the two winding terminals in an axial direction away from the stator main body.

In an embodiment, for each of the stator teeth, the end insulator further includes an axial retainer projecting axially between the two legs configured to attach to and axially retain the circuit board relative to the axial end of the stator main body.

In an embodiment, the stator includes stator segments having segment cores mated together to form the stator main body. The stator teeth project radially from the segment cores.

In an embodiment, end insulators are mounted the stator segments to insulate the stator teeth from the stator windings. Each end insulator includes two legs projecting axially and configured to support the two winding terminals in an axial direction away from the stator main body.

In an embodiment, the motor includes a heat sink having a substantially cylindrical body disposed around the stator, where the circuit board is received inside the heat sink.

In an embodiment, the circuit board includes an outer diameter that is smaller than an outer diameter of the stator main body.

In an embodiment, the motor further includes a sense magnet mounted on the rotor shaft, where the magnetic sensor is positioned proximate the sense magnet along the longitudinal axis.

In an embodiment, a motor end cap is secured to the stator and supporting a bearing therein, where the bearing includes an inner race mounted on the rotor shaft to radially support the rotor relative to the stator.

In an embodiment, the circuit board is disposed between the motor end cap and the stator.

In an embodiment, a power tool is provided including a housing and a BLDC motor according to any of the above embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 depicts a side view of a power tool **100** constructed in accordance with the teachings of the present disclosure, according to an embodiment;

FIG. 2 depicts a perspective view of the same power tool;

FIGS. 3 and 4 respectively depict exploded perspective views of power tool, showing the internal components therein;

FIGS. 5 and 6 depict perspective partially-exploded front and rear views of the motor assembly, according to an embodiment.

FIGS. 7 and 8 depict exploded views of the motor assembly, including motor components, according to an embodiment;

FIG. 9 depicts a cross-sectional side view of the motor, according to an embodiment;

FIGS. 10 and 11 depict perspective views of the motor, prior to and after assembly of the heat sinks, respectively, according to an embodiment;

FIG. 12 depicts a perspective cut-off view of the motor, showing the heat sinks in contact with stator lamination stack, according to an embodiment;

FIG. 13 depicts a perspective view of the motor with heat sinks, showing the path of air flow through air channels, according to an embodiment;

FIGS. 14 and 15 respectively depict front and rear perspective exploded views of tool housing clam shells around the motor, according to an embodiment;

FIG. 16 depicts a perspective view of the clam shells assembled around the motor, with end cap shown at a distance, according to an embodiment;

FIG. 17 depicts a zoomed-in view of the end cap assembled at the end of the tool housing, according to an embodiment;

FIG. 18 depicts a perspective side view of the motor assembly with heat sinks, according to an embodiment;

FIG. 19 depicts a cross-sectional side view of the motor assembly without heat sinks, according to an embodiment;

FIGS. 20-23 depicts various exemplary power tools, including an impact driver, a hammer drill, a cutter, and a reciprocating saw, incorporating the partially-enveloped motor of this disclosure, according to various embodiments;

FIG. 24 depicts a perspective view of a power tool having a fully-enveloped motor, according to an embodiment;

FIG. 25 depicts a perspective view of motor assembly of power tool of FIG. 24, according to an embodiment;

FIG. 26 depicts a top perspective view of power tool, showing an exploded perspective of the motor assembly, according to an embodiment;

FIG. 27 depicts a bottom exploded view of motor assembly, according to an embodiment;

FIG. 28 depicts a top cut-off exploded view of the motor assembly from a different angle than that shown in FIG. 26, according to an embodiment;

FIG. 29 depicts a side cross-sectional view of the motor assembly, according to an embodiment;

FIGS. 30 and 31 depict partially exploded views of motor assembly following the assembly of rotor and stator within the heat sink, but prior to assembly of module and fan, according to an embodiment;

FIG. 32 depicts a perspective view of motor assembly, prior to assembly of rear end cap, according to an embodiment;

FIG. 33 depicts a perspective side view of the motor assembly from a front side of the radial wall of the heat sink, according to an embodiment;

FIG. 34 depicts a cross-sectional view of the motor assembly from a rear side of the front end cap, according to an embodiment;

FIG. 35 depicts a side view of an exemplary angle grinder including the fully-enveloped motor of this disclosure, according to an embodiment;

FIGS. 36 and 37 depict side views of an exemplary reciprocating saw and an exemplary hammer including the fully-enveloped motor of this disclosure, according to an embodiment;

FIG. 38 depicts a perspective view of a fully-sealed motor assembly, according to an embodiment;

FIG. 39 depicts a bottom exploded view of motor assembly of FIG. 38, according to an embodiment;

FIG. 40 depicts a top exploded view of the motor assembly without the transmission assembly, according to an embodiment;

FIG. 41 depicts a side cross-sectional view of the motor assembly, according to an embodiment;

FIG. 42 depicts a perspective view of front and rear bearing support structures, according to an embodiment;

FIGS. 43 and 44 depict perspective view of the front and rear bearing support structures received with the heat sink, respectively, according to an embodiment;

FIG. 45 depicts a perspective view of an exemplary motor assembly having an external heat sink, according to an embodiment;

FIG. 46 depicts an axial view of the same motor assembly, according to an embodiment;

FIG. 47 depicts a zoomed-in axial view of the heat sink, showing three of the heat sink fins with coating, according to an embodiment;

FIG. 48 depicts a side view of an exemplary power tool having a fully-enveloped segmented motor assembly, according to an embodiment;

FIGS. 49 and 50 depict front and rear perspective views of an exemplary motor assembly of FIG. 48, according to an embodiment;

FIGS. 51 and 52 depict front and rear perspective exploded views of the motor assembly, including a stator, a rotor, a heat sink, etc., according to an embodiment;

FIGS. 53 and 54 respectively depict perspective views of the stator of the motor assembly distanced from and within the heat sink, respectively, according to an embodiment;

FIG. 55 depicts an axial cross-sectional view of the heat sink, stator, and rotor, according to an embodiment;

FIG. 56 depicts an axial perspective view of the heat sink and the stator, according to an embodiment;

FIGS. 57-60 depict various views of a stator segment for the stator, according to an embodiment;

FIG. 61 depicts a conventional prior art stator segment design in which the segment core includes inner and outer arched-shaped surfaces, according to an embodiment;

FIG. 62 depicts the stator segment having an improved segment stator segment that significantly improves the slot fill, according to an embodiment;

FIG. 63 depicts an axial view of the heat sink alone, according to an embodiment;

FIG. 64 provides a partial cross-sectional axial view of the stator within the heat sink, according to an embodiment;

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FIG. 65 depicts a zoomed-in perspective axial view of two adjacent segment cores within the heat sink, according to an embodiment;

FIG. 66 depicts a perspective view of the stator with a circuit board mounted, according to an embodiment;

FIG. 67 depicts a perspective exploded view of the heat sink, stator, and circuit board, according to an embodiment;

FIG. 68 depicts a perspective view of the stator and circuit board assembled within the heat sink, according to an embodiment;

FIGS. 69 and 70 depict axial front and back assembled views of the heat sink, stator, and circuit board, according to an embodiment;

FIGS. 71 and 72 depict partial cross-sectional views of motor assembly during the assembly process, immediately prior to and after the insertion of the stator and circuit board into the heat sink, respectively, according to an embodiment;

FIG. 73 depicts a partial perspective view of the motor assembly immediately prior to the insertion of the stator and circuit board into the heat sink, according to an embodiment;

FIGS. 74 and 75 depict power output improvement of an exemplary segmented motor according to the above-described embodiments in comparison to a similarly-sized motor having a conventional stator design, according to an embodiment;

FIG. 76 depicts a power/torque diagram for another exemplary segmented motor according to the above-described embodiments;

FIG. 77 depicts a partially exploded view of a bobbin-wound stator assembly received within a fully-enveloping heat sink, according to an embodiment;

FIG. 78 depicts a partially exploded view of the bobbin-wound stator assembly alone, according to an embodiment;

FIG. 79 depicts another partially exploded view of the bobbin-wound stator assembly, according to an embodiment;

FIGS. 80-83 depict various views of the stator core and the outer ring in a bobbin-wound stator assembly, according to an embodiment;

FIGS. 84 and 85 depict perspective and cross-sectional views of a bobbin-wound stator piece, according to an embodiment;

FIGS. 86 and 87 depict perspective and side views of the bobbin-wound stator piece, according to an embodiment;

FIG. 88 depicts a side view of an exemplary power tool having a motor assembly provided within an outer heatshield, according to an embodiment;

FIGS. 89 and 90 depict front and rear perspective views of an exemplary motor assembly provided within a heatshield, according to an embodiment;

FIGS. 91 and 92 depict front and rear perspective exploded views of the motor assembly with stator and heat sink provided within the heatshield, according to an embodiment;

FIG. 93 depicts a perspective view of heatshield and heat sink housing the stator and circuit board, according to an embodiment;

FIG. 94 depicts an axial view of heatshield alone, according to an embodiment;

FIG. 95 depicts an axial view of heat sink disposed within heatshield, according to an embodiment;

FIG. 96 depicts an axial view of stator and heat sink disposed within heatshield, according to an embodiment;

FIG. 97 depicts a zoomed-in axial view of stator and heat sink disposed within heatshield, according to an embodiment;

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FIGS. 98 and 99 depict partially exploded and zoomed-in perspective views of a rear end cap received at an end of the heatshield on one end of the motor assembly, according to an embodiment; and

FIGS. 100 and 101 depict rear zoomed-in axial view and a front axial view of a front end cap of the motor assembly, according to an embodiment.

DETAILED DESCRIPTION

According to an embodiment invention, a BLDC motor design is provided that significantly improves power density for power tool applications. As described below in detail with reference to the figures, this motor incorporates an external cooling design, where one or more heat carrying elements (heat sinks) are disposed in direct contact with the outer surface of the stator. The power tool housing is shaped and designed so as to expose an outer surface of the heat sink to the outside environment. The heat generated by the coils is conducted through the stator lamination stack and dissipated through the heat sink to the outside environment.

According to an embodiment, the heat sink extends longitudinally along the power tool to make contact with transmission assembly components in addition to the motor components.

According to an embodiment, the motor may be fully sealed via a variety of techniques disclosed in this disclosure. Sealing the motor ensures that motor air flow does not enter the motor, and thus prevents environmental dust and debris from contaminating the motor components.

According to an embodiment, a fan is disposed on the motor shaft that generates cooling air flow for cooling the heat sink. In an embodiment, the fan directs air through a series of channels provided longitudinally on the outer surface of the heat sink to maximize the cooling effect of the air flow. The transmission assembly may include air inlets that align with the heat sink air channels to collect the air passing through the heat sink. Alternatively, the air through the channels may dissipate into the outside environment.

In an embodiment, the fan may be a radial fan with blades arranged opposite the motor. The fan may generate air flow that axially pushes air into the air channels of the heat sink.

In an embodiment, a rear side of the fan facing the motor may house a sense magnet in magnetic communication with positional sensors. This arrangement eliminates the need for placement of sense magnets directly on the rotor shaft or on the lamination stack, thus reducing the size of the motor.

In an embodiment, depending on the motor size and power requirements, the heat sink may be shaped to fully or partially envelope and capsule the motor. For example, for higher-power power tool applications, where motor windings carry higher voltage and are therefore likely to generate more heat, a fully-enveloping heat sink encapsulates the circumference of the stator to maximize heat transfer to the heat sink.

This improved cooling technique eliminates the need to provide large airflow paths within the stator slots for cooling the stator windings and makes that area available for additional slot fill in the form of increased magnet wires with higher number of turns, or thicker magnet wires with the same number of turns. The increased slot fill substantially increases the volumetric power density of the motor.

In an embodiment, a layer of thermally conductive dielectric material may be disposed between the stator laminations to improve thermal conductivity of the stator.

These features are described below in more detail with reference to the accompanying figures.

Partially-Enveloped Motor

A first embodiment of the invention is described herein with reference to FIGS. 1-23.

With reference to the FIG. 1, a side view of a power tool **100** constructed in accordance with the teachings of the present disclosure is shown. FIG. 2 depicts a perspective view of the same power tool **100**. FIGS. 3 and 4 respectively depict exploded perspective views of power tool **100**, showing the internal components therein. The power tool **100** in the particular example provided is an electric cordless drill, but it will be appreciated that the teachings of this disclosure is merely exemplary and the power tool of this invention could be a drill, impact driver, hammer, grinder, circular saw, reciprocating saw, or any similar portable power tool constructed in accordance with the teachings of this disclosure.

Power tool **100**, according to an embodiment, includes a tool housing **102** made up of two clam shells **102a** and **102b** that together form a motor housing **104** for housing a motor assembly **200** and a handle portion **106** for housing an integrated switch module **110**.

In an embodiment, the integrated switch module **110** integrally includes a switch circuit (not shown), such as a three-phase inverter bridge circuit having a series of high-side and low-side semiconductor switches, arranged to deliver power from a power supply such as battery pack, through terminals **116**, to the motor assembly **200**. The integrated switch module **110** further include a controller (not shown), such as a micro-controller, that controls the switching operation of the switch circuit in order to regulate the voltage being supplied to the motor assembly **200**. The integrated switch module **110** additionally includes an input unit (not shown) coupled to a trigger switch **114** to activate the controller and provide a variable-speed signal to the controller. For detailed examples of the integrated switch module **100**, reference is made to U.S. Pat. No. 9,847,194 filed Mar. 28, 2014, which is incorporated herein by reference in its entirety.

In an embodiment, a transmission assembly **108** having a gear case (not shown) is mounted to the end of the motor housing **104**. The motor assembly **200** may be coupled through the gear case to an output spindle (not shown), which is rotatably coupled to chuck **112**.

In an embodiment, a battery receptacle **118** is disposed at the end of the handle **106** opposite the motor housing **104**. The battery receptacle **118** houses the battery terminals **116** and receives a battery pack couplable to the battery terminals **116** therein. While the motor assembly **200** is powered by a battery pack in this example, it must be understood that power tool **100** may alternatively include a power cord to receive AC power from, for example, a generator or the AC grid, and may include the appropriate circuitry (e.g., a full-wave or half-wave bridge rectifier) to provide positive current to the motor **104**.

In an embodiment, the motor assembly **200** includes a brushless DC (BLDC) motor **202**, a pair of heat sinks **120a** and **120b** disposed on two sides of the motor **202** and held by the clam shells **102a** and **102b** along the outer peripheral surface of the motor **202** to partially envelope an outer periphery of the motor assembly **200**, and a fan **130** disposed at an axial end of the motor **202** opposite the transmission assembly **108**. In an embodiment, the clam shells **102a** and **102b** each includes distal air intakes **140** disposed at the end of the motor housing **104**, radially around a center axial of the motor housing **104**, facing and adjacent the fan **130**. The clam shells **102a** and **102b** also include two longitudinal side openings **142a** and **142b** formed around the heat sinks **120a**

and **120b**, through which the heat sinks **120a** and **120b** are exposed to the outside environment. In an embodiment, an end cap **150** is secured to the end of the motor housing **104** in close proximity to the air intakes **140**.

Motor assembly **200** is now described in detail with reference to FIGS. 5-9, according to an embodiment. FIGS. 5 and 6 depict perspective partially-exploded front and rear views of the motor assembly **200**, including the motor **202**, heat sinks **120a** and **120b**, and fan **130**, according to an embodiment. FIGS. 7 and 8 depict exploded views of the motor assembly **200**, including motor **202** components, according to an embodiment. FIG. 9 depicts a cross-sectional side view of the motor **202**, according to an embodiment.

In an embodiment, the motor **202** includes a rotor **210**, a stator **220**, and front and rear bearing support structures **230** and **240**.

Rotor **210**, in an embodiment, includes a rotor shaft **212** on which a rotor lamination stack **202** housing several permanent magnets **218** is mounted. Fan **130** is mounted on the rotor shaft **212** on one side of the rotor lamination stack **214**, and a gear **216** for engagement with the transmission assembly **108** is mounted on the rotor shaft **212** on the other side of the rotor lamination stack **214**. Rotation of the rotor lamination stack **214** causes the rotation of the rotor shaft **212**, the fan **130**, and the gear **216** within the motor assembly **200**. Front and rear rotor bearings **218a**, **218b** are also mounted on the rotor shaft **212**, with front bearing **218a** disposed between the lamination stack **214** and gear **216**, and rear bearing **218b** disposed between the lamination stack **214** and the fan **130**.

Stator **220**, in an embodiment, is disposed around the rotor lamination stack **214** and includes stator windings **222** wound around teeth of a stator lamination stack **224** and two stator end insulators **226**. The stator windings **222** are connected around the stator in, for example, a wye or a delta configuration, and are electrically connected to and energized by integrated switch module **110**. Windings **222** are thereby electrically commutated to generate a magnetic force on the rotor **210**, thereby rotating the rotor **210** within the stator **220** in a desired speed and direction of rotation.

Front bearing support structure **230**, according to an embodiment, includes a substantially disc-shaped planar radial plate **232** defining a bearing pocket **234** therein. Rear bearing support structure **240**, according to an embodiment, similarly includes a substantially disc-shaped planar radial plate **242** defining a bearing pocket **234** therein. The front and rear bearings **218a** and **218b** are press-fitted inside the bearing pockets **234** and **244** in a way that the outer races of the bearing **218a** and **218b** are non-rotatably supported by the bearing support structures **230** and **240**, respectively.

In an embodiment, a positional sensor board **246**, including a series of positional sensors **248** (e.g., hall sensors), is mounted on the rear bearing support structure **240** facing the rotor **210**. The positional sensor board **246** is positioned in close proximity to a sense magnet ring **250** housed by an end cap **252** of the rotor adjacent the rotor lamination stack **214**. Positional sensors **248** sense the rotational position of the rotor **210** via the sense magnet ring **250**, and the positional sensor board **246** provides the positional information of the to the integrated switch module **110** via connector **254**.

In an embodiment, front bearing support structure **230** includes one or more peripheral arcuate walls **236** that extend longitudinally from radial plate **232** around two opposite sides of the stator **220**. Arcuate walls **236** are sized to form-fittingly receive the stator **220** therein, with the outer surface of the stator lamination stack **224** in contact with,

and securely held in place between, the arcuate walls **236**. In an embodiment, rear bearing support structure **240** similarly includes one or more peripheral arcuate walls **256** that extends longitudinally from radial plate **242** along the outer surface of the stator **220** and mates with a corresponding arcuate wall **236** of the front bearing support structure **230**. In the illustrated example, the lower one of the arcuate walls **256** includes a gap to accommodate the positional sensor board **246** and the connector **254**. In an embodiment, a series of fasteners **260** axially secure the front and rear bearing support structures **230** and **240** together around the stator **220**.

In an embodiment, front bearing support structure **230** further includes two side openings **237** peripherally formed between the arcuate walls **236**. Similarly, in an embodiment, rear bearing support structure **240** includes two side openings **257** peripherally formed between the arcuate walls **256**. These peripheral openings **237** and **257** together allow the heat sinks **120a** and **120b** to come into contact with two sides of the outer surface of the stator lamination stack **224** for improved heat transfer from the motor **200**, as described herein.

Fan **130**, in an embodiment, is a high pressure radial fan, which as explained above, is mounted on the rotor shaft **212** to rotate with the rotor **210**. Fan **130** includes a series of blades **270** extending longitudinally around a peripheral portion of a planar fan plate **272**. The fan plate **272** is press-fitted on the rotor shaft **212**. The fan **130** includes a first surface **274** facing the radial plate **242** of the rear bearing support structure **240**, and a second surface **276** from which the blades **270** project longitudinally. The fan blades **270** are oriented such that they suck air from a middle portion of the fan **130** and push air radially out around the fan **130**.

FIGS. **10** and **11** depict perspective views of the motor **200**, prior to and after assembly of the heat sinks **120a** and **120b**, respectively. As shown in these figures, and with continued reference to FIGS. **5-8**, arcuate walls **236** and **256** respectively include guide channels **238** and **258** that align along the axial edges of the peripheral openings **237/257**. The heat sinks **120a** and **120b** include rail guides **261** that are slidably received within the guide channels **238** and **258** in a such a way that an inner surface **162** is in direct contact with the outer surface of the stator **224**.

In an embodiment, outer surfaces **264** of heat sinks **120a** and **120b** opposite the stator **220** include a series of longitudinal fins **266** forming a series of longitudinal air channels **268** therebetween. Air channels **268** are arranged to guide air flow generated by the fan **130** along the outer surfaces **264** of the heat sinks **120a** and **120b**, as discussed below in further detail.

FIG. **12** depicts a perspective cut-off view of the motor **200**, showing the heat sinks **120a** and **120b** in contact with the stator lamination stack **224**, according to an embodiment.

FIG. **13** depicts a perspective view of the motor **200** with heat sinks **120a** and **120b** fully inserted into the guide channels **238** and **258**, showing the path of air flow generated by the fan **130** through the air channels **268**.

FIGS. **14** and **15** respectively depict front and rear perspective exploded views of tool housing clam shells **102a** and **102b** around the motor **202**, according to an embodiment.

FIG. **16** depicts a perspective view of the clam shells **102a** and **102b** assembled around the motor **202**, with end cap **150** shown at a distance.

FIG. **17** depicts a zoomed-in view of the end cap **150** assembled at the end of the tool housing **102**.

As shown in these figures, clam shells **102a** and **102b** each includes distal air intakes **140** disposed at the end of the motor housing **104**, radially around a center axial of the motor housing **104**, facing and adjacent the fan **130**. The air intakes **140** are oriented to let air towards a middle portion of the fan **130**. In other words, the air intakes **140** are positioned at a closer radially orientation with respect to the center axis of the motor housing **104** than the fan blades **270**. This allows the fan blades **270** to radially repel the incoming air to the outer periphery of the fan **130**, where air is then pushed into and guided through air channels **268** of the heat sinks **120a** and **120b**.

In an embodiment, clam shells **102a** and **102b** also include two longitudinal side openings **142a** and **142b** formed around the heat sinks **120a** and **120b**, through which the heat sinks **120a** and **120b** are exposed to the outside environment. In an embodiment, side openings **142a** and **142b** are shaped to receive the outermost fins **266** there-through such that the ends of the fins **266** radially align with, or substantially close to, the outer surfaces **280** of the clam shells **102a** and **102b**. In an embodiment, air inlets **280** are formed between the tool housing **102** and the air channels **268** that guide the air into the air channels **268**. In an embodiment, a radial wall **282** of each of the clam shells **102a** and **102b** that surrounds the fan **130** aligns with the ends of the fins **266** of heat sinks **120a** and **120b** to form the air inlets **280** at the ends of the air channels **268**.

In an embodiment, end cap **150** is attached to the back end of the tool housing **102** in such a way that it maintains a small radial air gap (e.g., approximately 1 mm) between the end cap **150** and the end of the tool housing **102**, as shown in FIG. **17**. This allows air to radially enter through the air gap from outside the tool into the air intakes **140**. In an embodiment, two side caps **284** project axially from the end cap **150** to partially cover the ends of the heat sinks **120a** and **120b**, thus ensuring that the air pushed through the air inlets **280** is properly channeled through air channels **268** of the heat sinks **120a** and **120b**.

It is noted herein that while the cooling mechanism described above includes a radial fan **130** pushing air through the air channels **268** of the heat sinks **120a** and **120b**, the cooling mechanism of this disclosure may employ other types of fan (e.g., a low pressure in-line fan) or other air flow arrangements (e.g., one where the fan pulls air through the air channels of the heat sinks).

Furthermore, while the fan **130** described above is mounted on the rotor shaft **212**, it is envisioned that use of a separately-powered fan, alone or in combination with fan **130**, for cooling the heat sinks **120a** and **120b** is within the scope of this disclosure. In an embodiment, the separately-powered fan may be operated even when the tool is not in use or when the rotor operates at very low speed to continue thermal dissipation and avoid high temperatures that may result from a stalled rotor stall or a low speed operation.

Furthermore, while air is used as the fluid exchange medium in the above-described embodiment, the principle elements of the invention extend to systems cooled with other gases or liquids. For example, in a liquid cooled power tool system, a cooling system include pump, a coolant reservoir, and other elements well known in the art may be incorporated to cool the heat sinks **120a** and **120b**.

It is also noted that while the heat sinks **120a** and **120b** are provided as discrete components in the above-described embodiment, heat sinks **120a** and **120b** may be alternatively provided as parts of other power tool components, for

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example, the transmission assembly **108** or the integrated switch module **110**. In an exemplary embodiment, the heat sinks **120a** and **120b** may be integrated into the transmission assembly **108** and later assembled into the motor assembly **200** together with the transmission assembly **108**.

In an embodiment, the heat sinks **120a** and **120b** may be integrally formed with the stator **220**. For example, the heat sinks, including the fins **226** and air channels **268**, may be integrally stamped in the outer geometry of the stator lamination stack **224**.

Furthermore, depending on the motor size and power output requirement, it is envisioned that the motor can be sufficiently cooled without a heat sink and by external cooling of the outer surface of the stator using the methods disclosed herein. Conduction of the heat generated by the stator windings through the stator lamination stack and cooling of the outer surface of the stator outside the motor envelope is thus within the scope of this disclosure.

In an embodiment, in order to enhance the cooling of the motor **200**, particularly the rotor **210** and its components, the fan **130**, including the fan plate **272** and/or the fan blades **270** may be made of thermally conductive material such as metal.

The external end surfaces of the heat sink fins **266** are accessible for contact by the user. Given that the heatsinks may get hot and may exceed permissible limits for user interaction, in an embodiment, additional measures and construction may be employed. Examples of this may include application of a surface coating (such as a polymer) on the end surfaces of the fins **266**. In an embodiment, the surface coating increases the thermal resistance of the end surfaces of the fins **262**, limits direct contact by the user to the end surfaces of the fins **262**, and/or contains airflow directly within the heat sink air channels **268**.

It is also noted that while the heat sinks **120a** and **120b** described above are held in place by motor housing **104** and the motor assembly **200**, the heat sinks may be fully supported by only the enclosure and structural components of the motor assembly **200** alone, or by retention features provided in the motor housing **104** alone.

Additionally, in an embodiment, the boundary thermal resistance between the stator **200** and the heatsinks **120a** and **120b** may be improved by known construction methods, for example, using a thermal bonding compound, via interference and/or compression fits, or using fasteners that securely draw the heatsink against the stator.

In an embodiment, the motor assembly **200** may be fully sealed using a variety of techniques. In an exemplary embodiment, a gasket, adhesive, or other sealant material may be disposed within the mating surfaces of arcuate walls **236** and **256**, and the mating surfaces of guide channels **238** and **258**. In an embodiment, a similar sealant may also be disposed between the edges of the heat sinks **120a**, **120b**, and the guide channels **238**, **258**, and/or the arcuate walls **236**, **256**. In an embodiment, the opening of the rear bearing support structure **240**, through which connector **254** is disposed, is further sealed via a resin or adhesive material, or a plug sized to be fitted within the opening.

Referring now to FIG. **18** a perspective side view of the motor assembly **200** and heat sinks **120a** and **120b** is depicted. FIG. **19** depicts a cross-sectional side view of the motor assembly **200**. As shown in these figures, due to the external cooling of the motor assembly **200**, the number of turns for each winding **222** around stator teeth **292** and through the slots **294** may be significantly increased to a slot fill over 50% or more, preferably to a slot fill of 60% or more, more preferably to a slot fill of 70% or more, more

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preferably to a slot fill of 80% or more, and even more preferably to a slot fill of 90% or more. This increased slot fill has the effect of increasing the volumetric power density of the motor to levels unachievable with conventional internally-cooled BLDC motors.

According to an embodiment, to improve the thermal bond between the stator winding wires and the stator lamination stack, a variety of techniques may be utilized. These include, but are not limited to, impregnation of the coils with varnish or epoxy, winding the coils in corporation with thermal adhesives and/or filters, use of bondable wire, or overmolding the stator windings.

According to an embodiment, in order to maximize the slot fill, a variety of winding techniques or stator designs may be utilized. For example, instead of conventional needle winders that require insertion of a winding needle into the stator slots to guide the magnet wires, a precision guide winding machine may be used. A precision guide winding machine guides the wires through the stator slots without the need to insert a winding needle into the slot. This allows more magnet wire to be packed around the stator teeth with the slots.

In an embodiment, the gap **296** between adjacent stator teeth **292** may be significantly reduced for better retention of the stator windings within the slots **294**.

In an embodiment, the thickness of the stator core **290** may be reduced in comparison to conventional BLDC motors to enhance heat transfer between the heat sinks **120a** and **120b** and the stator windings **222**.

According to an embodiment, as shown in FIGS. **18** and **19**, with use of high slot fill as described above, the motor power density can be further improved by increasing the number of stator slots and correspondingly the number of rotor poles. In the illustrated example, nine (9) stator poles are arranged, three of which are commonly coupled to a single phase of the motor **200**, and the rotor is provided with six (6) permanent magnet **218**.

FIGS. **20-23** depicts various exemplary power tools, including an impact driver **295**, a hammer drill **297**, a cutter **298**, and a reciprocating saw **299**, incorporating the partially-enveloped motor **200** of this disclosure. In an embodiment, each of these tools, as well as tool **100** of FIGS. **1-4**, is powered by a battery pack having a maximum voltage of approximately 10V to 30V, preferably approximately 20V.

It should be understood that the partially-enveloped motor design described here is not limited to power tools having aforementioned power tools with the aforementioned voltage ratings, and can be used in various power tool having different power requirements. The size and thickness of the heat sinks **120a** and **120b**, including its fins and air channels, may be varied depending on the motor power requirements and heat dissipation.

Fully-Enveloped Motor

A second embodiment of the invention is described herein with reference to FIGS. **24-37**.

FIG. **24** depicts a perspective view of a power tool **300**, according to an embodiment of the invention. The power tool **300** in the particular example provided is an electric cordless circular saw. Power tool **300**, according to an embodiment, includes a tool housing **302** including a handle portion **304** and a battery receptacle **306** for receiving a battery pack therein. A motor assembly **400**, described below in detail, is provided to drive a saw blade **308**. A saw guard **310** partially enclosing the blade **308** is mounted around a collar (not shown) to protect the blade **308**. For more details of this particular power tool, reference is made

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to US Patent Publication No. 2014/0366383, which is incorporated herein by reference in its entirety.

In an embodiment, power tool **300** has a higher power output requirement than power tool **100** of the previous embodiment, and thus requires a generally-larger motor assembly that generates more heat as compared to the previous embodiment. It will be appreciated that the power tool **300** of FIG. **24** is merely exemplary and the power tool of this embodiment could be any other type of power tool, particularly medium to high rated voltage power tools receiving 40V to 120V max voltage battery packs, preferably a single 60V max voltage battery pack or two 60V max voltage battery packs arranged in series. Alternatively, the power tool of this embodiment may be one powered by an alternating current (AC) power supply, or a combination of AC and DC power supplies, and include a rectifier circuit to cover AC voltage to DC voltage suitable to energizing the motor **400**. These power tools include, but are not limited to, high power drills and impact drivers, hammer drills, reciprocating saws, table saws, grinders, miter saw, mixers, chain saws, etc., or any similar portable power tool constructed in accordance with the teachings of this disclosure.

FIG. **25** depicts a perspective view of motor assembly **400** of power tool **300**, according to an embodiment. FIG. **26** depicts a top perspective view of power tool **300**, showing an exploded perspective of the motor assembly **400**. FIG. **27** depicts a bottom exploded view of motor assembly **400**. FIG. **28** depicts a top cut-off exploded view of the motor assembly **400** from a different angle than that shown in FIG. **26**, according to an embodiment. FIG. **29** depicts a side cross-sectional view of the motor assembly **400**, according to an embodiment. Motor assembly **400** is herein described with reference to these figures.

According to an embodiment, similarly to the previously embodiment, motor assembly **400** includes a rotor **410** having a rotor shaft **412** and a rotor lamination stack **414** housing a series of permanent magnets **418**, and a stator **420** having a series of stator windings **422** wound on slots of a lamination stack **424** and two end insulators **426**. In an embodiment, a rotary fan **404** is mounted on one end of the rotor shaft **412** to rotate with the rotor shaft **412**. It should be noted that motor assembly **400** includes many of the same or similar features as motor assembly **200** previously disclosed, and many of the details and alternative and/or additional embodiments disclosed above with reference to motor assembly **200** are applicable in this embodiment.

In an embodiment, a front end cap **430** is provided on one end of the motor assembly **400** for mounting the motor to a mounting bracket **312** of the power tool **300**. Front end cap **430** is sized to be fittingly received within mounting bracket **312**, and includes peripheral receptacles **432** for fastening the motor assembly **400** to the mounting bracket **312** via a series of fasteners (not shown). Front end cap **430** also includes a center through-hole **434** through which the rotor shaft **412** extends out.

In an embodiment, a driver **416** is mounted near an end of the rotor shaft **412** extending out of the front end cap **430** through-hole **434**. Driver **416**, in an embodiment, is press-fitted into a corresponding opening (not shown) of an output spindle (not shown) of the power tool **300** to rotatably drive the saw blade **308**. The driver **416** thus radially and axially supports the rotor shaft **412** relative to the front end cap **430**, and thus relative to the stator **420**. Accordingly, a front side bearing is not provided in this particular example. It should be understood, however, that in an alternative embodiment, through-hole **434** of the front end cap **430** may be as a bearing support pocket and the rotor shaft **412** may be

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correspondingly provided with a front bearing received within the bearing support pocket.

In an embodiment, a rear end cap **440** is provided on an end of the motor assembly **400** opposite the front end cap **430**. Rear end cap **440** is provided with air intakes **442** oriented to let air in towards a middle portion of the fan **404**. Fan **404** includes blades **450** facing the rear end cap **440** that generate a radial air flow, as discussed below in detail. Fan **404** also houses a sense magnet ring **452** facing the stator **420** and rotor **410**. The sense magnet ring **425** includes one or more magnets corresponding to rotor **410** permanent magnets **418** and rotates with the rotor shaft **412**.

In an embodiment, an electronic switch and control module **406** is provided between fan **404** and stator **420**. In an embodiment, module **406** may include a disc-shaped printed circuit board on which a controller and power switches for controlling the supply of power to the motor assembly **400** are disposed. Module **406** energizes the stator windings **422** via a series of motor terminals **428** received from the stator **420**. Module **406** may also include a series of positional sensors **454** mounted on the printed circuit board facing the fan **404**. Positional sensor **454** sense the magnetic position of sense magnet ring **425**, and thus the rotor **410**, and provide that information to the controller. In an embodiment, module **406** further includes a power terminal **456** that receives battery current from the tool battery receptacle **306**. For detailed examples of electronic switch and control module **406**, reference is made to US Patent Publication No. 2017/0106522 filed on Oct. 13, 2016, and U.S. patent application Ser. No. 15/708,484 filed on Sep. 19, 2017, contents of both of which are incorporated herein by reference in their entireties.

In an embodiment, motor assembly **400** includes an outer heat sink **402** having a generally cylindrical body surrounding the stator **420**. Heat sink **402** includes a cylindrical opening that is sized to fittingly receive the stator **420** therein, the inner surface of the heat sink **402** being in thermal and physical contact with the outer surface of the stator lamination stack **424**. Since heat sink **402** covers substantially the entire periphery of the stator lamination stack **424**, it provides for optimal heat transfer from the stator **420**.

The outer surface of heat sink **402**, in an embodiment, includes a series of longitudinal fins **460** forming a series of longitudinal air channels **462** therebetween. Air channels **462** are arranged to guide air flow generated by fan **404** along the outer surface of heat sink **402** parallel to the longitudinal axis of the motor **400**. Specifically, as fan **404** rotates, it generates air flow from the air intakes **442** in a radially outwardly direction. Peripheral walls **444** of the rear end cap **440** baffle and guide the air into air channels **462** of the heat sink **402**. In an embodiment, when fully assembled, rear end cap **440** covers the end of the heat sink **402** in contact with the outer edges of fin **460** to guide the air longitudinally into the air channels **462**. Fins **460** increase the outer surface area of the heat sink **402**, while passage of cooling air through the air channels **462** provides for improved heat dissipation from the heat sink **402**.

In an embodiment, air traveling through air channels **462** dissipates into the outside environment at or near the front end cap **430**. In an embodiment, part of the air may be guided through the front end cap **430** and mounting bracket **312** to enter into the spindle assembly (not shown) of the power tool **300** for cooling of the power tool **300** components.

In an embodiment, heat sink **402** may be provided integrally with the stator **420** as a single piece. For example, the

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heat sink 402, including the fins 460 and air channels 462, may be integrally stamped in the outer geometry of the stator lamination stack 424.

In an embodiment, the inner surface of heat sink 402 is provided with piloting and positioning features 464 for proper of the stator 420 within the heat sink 402. In an embodiment, stator 420 is provided with corresponding axial ribs or grooves that align with the piloting and positioning features 464 of the heat sink 402.

In an embodiment, fully envelopment of the stator 420 and rotor 410 within the heat sink 402 in the manner described herein provides an arrangement in which the heat sink 402 is the principal enclosure and support structure for the motor 400 while maximizing interfacial contact with the stator 420 and the heat sink 402 for optimal thermal conduction. In an embodiment, the boundary thermal resistance between the stator 420 and the heat sink 402 may be improved by, for example, application of a thermal bonding compound and interference, and/or placement of compression fits, etc., between the outer surface of the stator 420 and the inner surface of the heat sink 402.

In an embodiment, the inner surface of heat sink 402 is provided with a radial wall 465. In an embodiment, radial wall 465 is provided to axially fix the relative positions of various motor components. In this example, radial wall 465 spatially separates the rotor 410 and stator 420 from the module 406 and the fan 404, receiving the stator 420 form-fittingly from one end heat sink 402 and receiving the module 406 form-fittingly from the other end. Additionally, in an embodiment, radial wall 465 may include a bearing pocket 466 for housing a bearing 419 of the rotor 410. In that manner, heat sink 402, in cooperation with driver 416, provides radial and axial support for the rotor 410 with respect to the stator 420.

In an embodiment, provided peripherally on one end of the heat sink 402 facing the rear end cap 440, an opening 468 is provided. When module 406 is received within the heat sink 402, power terminals 456 of module 406 are accessible through the opening 468.

In an embodiment, one end of the heat sink 402 facing the front end cap 430 is provided with one or more mounting structures 470. Mounting structure 470 includes through-holes 472 through which fasteners (not shown) secure the heat sink 402 to receptacles 432 of the front end cap 430 and/or the mounting bracket 312.

Referring to FIGS. 30 and 31, partially exploded views of motor assembly 400 is depicted following the assembly of rotor 410 and stator 420 within the heat sink 402, but prior to assembly of module 406 and fan 404, according to an embodiment. FIG. 32 depicts a perspective view of motor assembly 400, prior to assembly of rear end cap 440, according to an embodiment.

As shown in these figures, module 406 may be form-fittingly received within the inner surface of the heat sink 402. Alternatively and/or additionally, module 406 may include a series of fasteners 474 that securely fasten the module 406 to radial wall 465 of the heat sink 402.

In an embodiment, fan 404 includes a center receptacle 476 into which the end of the rotor shaft 412 is secured. Once mounted on the end of the rotor shaft 412, fan 404 substantially aligns with the end of the heat sink 402 with fan blades 450 being disposed outside the envelope of the heat sink 402. This arrangement allows the fan blades 450 to generate airflow that is directed to air channels 462 on the outer surface of the heat sink 402, and not into the inner envelope of the heat sink 402.

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In an embodiment, the rotor shaft 412, the radial wall 465, and the heat sink 402 are sized to ensure that fan 404 is positioned in close proximity and parallel to the module 406 to facilitate proper magnetic sensing of the sense magnet ring 452 of the fan 404 by positional sensors 454 of module 406.

In an embodiment, power terminals 456 of module 406 (which in an embodiment is provided as a part of a connector which may include communication signals for other functions in addition to power transfer) are slidably received within opening 468 and covered by peripheral wall 444 of the rear end cap 440. In an embodiment, the area around the terminals may then be covered by resin or other sealant material.

Referring now to FIG. 33, a perspective side view of the motor assembly 400 from a front side of the radial wall 465 of the heat sink 402 is depicted. FIG. 34 depicts a cross-sectional view of the motor assembly 400 from a rear side of the front end cap 430. As shown in these figures, due to the external cooling of the motor assembly 400, the number of turns for each winding 422 around stator teeth may be significantly increased to a slot fill over 50% or more, preferably to a slot fill of 60% or more, more preferably to a slot fill of 70% or more, more preferably to a slot fill of 80% or more, and even more preferably to a slot fill of 90% or more. This increased slot fill has the effect of increasing the volumetric power density of the motor to levels unachievable with conventional internally-cooled BLDC motors.

According to an embodiment, to improve the thermal bond between the stator winding wires and the stator lamination stack, a variety of techniques may be utilized. These include, but are not limited to, impregnation of the coils with varnish or epoxy, winding the coils in corporation with thermal adhesives and/or filters, use of bondable wire, or overmolding the stator windings.

According to an embodiment, in order to maximize the slot fill, a variety of winding techniques or stator designs may be utilized. For example, instead of conventional needle winders that require insertion of a winding needle into the stator slots to guide the magnet wires, a precision guide winding machine may be used. A precision guide winding machine guides the wires through the stator slots without the need to insert a winding needle into the slot. This allows more magnet wire to be packed around the stator teeth with the slots.

In an embodiment, the gap between adjacent stator teeth may be significantly reduced for better retention of the stator windings within the slots.

In an embodiment, the thickness of the stator core may be reduced in comparison to conventional BLDC motors to enhance heat transfer between the heat sink 402 and the stator windings 422.

According to an embodiment, with use of high slot fill as described above, the motor power density can be further improved by increasing the number of stator slots and correspondingly the number of rotor poles. In the illustrated example, nine (9) stator poles are arranged, three of which are commonly coupled to a single phase of the motor 400, and the rotor is provided with six (6) permanent magnet 418.

FIGS. 35-37 depict various exemplary power tools, including a grinder 500, a reciprocating saw 520, and a hammer 522, incorporating the fully enveloped motor 400 of this disclosure. In an embodiment, each of these tools, as well as tool 300 of FIG. 24, is powered by a battery pack having a maximum voltage of approximately 40V to 80V, preferably approximately 60V.

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It should be understood that the fully-enveloped motor design described here is not limited to power tools having higher power output requirements, and can be used in various power tool having different power requirements. The size and thickness of the heat sink 402, including its fins and air channels, may be varied depending on the motor power requirements and heat dissipation.

In an embodiment, as shown in FIG. 35, motor assembly 400 may be provided along a main axis of the power tool 500 where heat sink 402 is gripped by the user. In this exemplary power tool, a handle portion 502 of the tool is disposed on one side of the motor assembly 400, and gear case 504 is disposed on the other side of the motor assembly 400 opposite the handle portion 502. A front end 512 of the handle portion 502, the heat sink 402, and a rear end 514 of the gear case 504 are provided with approximately the same girth for a substantially uniform tool body. Instead of a rear end cap 440 with axial air intakes 442, in this embodiment, a series of radial air intakes 506 are disposed between the front end 512 of handle portion 502 and the heat sink 402. One of more baffles (not shown) guide the incoming air from the air intakes 506 towards a center portion of the fan, where the air is radially redirected and pushed into the air channels of the heat sink 400 along the tool body. In an embodiment, the front end 514 of the transmission housing may be provided with additional air vents 516 arranged in fluid communication with air channels 462 of the heat sink 402, through which the air traveling through the air channels 462 enters the gear case 504 for cooling the gear case 504 components.

In an embodiment, a surface coating may be applied to the outer surfaces of the fins 460 to increase the thermal resistance of the outer surface of the heat sink 402. This prevents the gripping area of the heat sink 402 from getting too much and burning a user's hand. The coating may also assist in containing airflow within the heat channels 462.

In the embodiments described above, external cooling via the partially-enveloping heat sinks (first embodiment) or fully enveloping heat sink (second embodiment) eliminates the need for air from entering the enclosure of the motor. As a result, in the embodiments of the invention, the motor enclosure may be designed so as to provide total enclosure, i.e., closure or sealing of openings to prevent fluid communication between the inside components of the motor and the outside environment. This is of particular interest with power tool motors as contamination from the environment, e.g., rain, mud, oil, concrete aggregate, metal ingestion, etc., are all common causes of motor failure.

This may be accomplished by providing sealing materials applied or disposed between mating surfaces of various motor components. For example, in the partially-enveloped design of the first embodiment, the mating surfaces of the front and rear bearing support structures and the heat sinks may be provided with sealing material. Similarly, in the fully-enveloped design of the second embodiment, the mating surfaces of the front end cap and the heat sink, and the mating surfaces of the electronic switch and control module and the inner surface of the heat sink, may be provided with sealing material. Such material may include, but are not limited to, adhesives, grease, gaskets, O-rings, compressive fits, etc., provided for increased levels of sealing.

In addition, in an embodiment, various openings such as those provided for disposition of connectors and terminals may be sealed with application of material such as grommets, adhesives, potting, and other gap filling and/or coating measures with the ultimate goal of total motor enclosure.

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Fully Enveloped and Sealed Motor

Referring now to FIGS. 38-44, a third embodiment of the invention is describe herein. In this embodiment, front and rear support structures for the rotor bearings are provided within a sealing material (e.g., O-rings) within a fully-enveloping heat sink. The front and rear bearing support structures, in addition to providing positional support for the rotor relative to the stator, fully seal the axial ends of the rotor and stator assemblies from fluid communication with the outside environment.

FIG. 38 depicts a perspective view of a motor assembly 600, according to an embodiment. FIG. 39 depicts a bottom exploded view of motor assembly 600, according to an embodiment. FIG. 40 depicts a top exploded view of the motor assembly 600 without the transmission assembly, according to an embodiment. FIG. 41 depicts a side cross-sectional view of the motor assembly 600, according to an embodiment. Motor assembly 400 is herein described with reference to these figures.

Similarly to above-described embodiments, motor assembly 600 a rotor 610 having a rotor shaft 612 and a rotor lamination stack 614 housing a series of permanent magnets, and a stator 620 having a series of stator windings wound on slots of a lamination stack 624. In an embodiment, a rotary fan 650 is mounted on one end of the rotor shaft 612 to rotate with the rotor shaft 612.

It should be noted that motor assembly 600 includes many of the same or similar features as motor assemblies 200 and 400 previously disclosed, and many of the details and alternative and/or additional embodiments disclosed above with reference to motor assemblies 200 and 400 are applicable in this embodiment.

In an embodiment, similarly to above-described embodiments, motor assembly 600 includes a heat sink 602, similarly to embodiments described above, is substantially-cylindrical with outer fin 604 defining longitudinal air channels 604 in between formed on the outer surface of the heat sink 602. Heat sink 602 includes a peripheral opening 608 for disposition of motor connector and/or terminal, as described below.

In an embodiment, heat sink 602 additionally includes a switch receptacle 608 that slidably receives a mode/speed selector switch 680 therein. Switch 680 may be used in some power tool applications such as, for example drills.

In an embodiment, a transmission housing 670 is disposed on a front end of the heat sink 602, and a rear end cap 660 is disposed on a rear end of heat sink 602.

In an embodiment, a rear end cap 660 is provided with air intakes 662 oriented to let air in towards a middle portion of the fan 650. Fan 650 includes blades 652 facing the rear end cap 660 that generate a radial air flow towards away from a center of the fan 650. Peripheral walls 666 of the rear end cap 660 redirect the air generated by the fan 650 axially into the air channels 606 of the heat sink, similarly to embodiments disclosed above. Fan 650 also houses a sense magnet ring (not shown) facing the stator 620 and rotor 610, similarly to embodiments disclosed above.

In an embodiment, front and rear bearings 616 and 618 are mounted on the rotor shaft 612 on two sides of the rotor lamination stack 620. Front and rear bearing support structures 630 and 640 are provided adjacent the ends of the stator 620 and rotor 610 to spatially receive and support the front and rear bearings 616 and 618, respectively. Front bearing support structure 630 includes a bearing pocket 632 that form-fittingly receives an outer race of the front bearing 616. Similarly, rear bearing support structure 640 includes a bearing pocket 642 that form-fittingly receives an outer race of the rear bearing 618. Front and rear bearing support

structures **630** and **640** are respectively provided with fasteners **636** and **646** that securely fasten them to the ends of the stator **620**.

In an embodiment, rear bearing support structure **640** includes a positional sensor board **644**, which as described in the embodiments disclosed above, faces the sense magnet ring of the fan **650**.

The stator **626** includes a connector **620** projecting radially from an end of the stator **620**. Connector **626** may accommodate motor terminals or other communication signals to and/or from motor switches and/or the controller (not shown). In an embodiment, front bearing support structure **630** includes a connector tab **634** corresponding to the shape of the connector **626**.

During assembly of motor assembly **600**, stator **610** and rotor **620** are together received from a front end of the heat sink **602** into an inner envelope of the heat sink **602**, with connector **626** received within the opening **608** of the heat sink **602**. Front bearing support structure **630** is also received from the front end of the heat sink **602**, with connector tab **634** received through the opening **608** adjacent or in contact with the connector **626**. Rear bearing support **640** is received through a rear end of the heat sink **604**.

FIG. **42** depicts a perspective view of front and rear bearing support structures **630** and **640**, according to an embodiment. FIGS. **43** and **44** depict perspective view of the front and rear bearing support structures **630** and **640** received with the heat sink **602**, respectively, according to an embodiment. In an embodiment, as shown in this figure, and with continued reference to FIGS. **38-41**, front and rear bearing support structures **630** and **640** are sized to be form-fittingly received within the inner envelope of the heat sink **602** on the two sides of the rotor **610** and stator **620**. Thus, front and rear bearing support structures **630** and **640**, together with the heat sink **602**, substantially seal the rotor **610** and stator **620**. Additionally, in an embodiment, front and rear bearing support structures **630** and **640** are each provided with a seals **638**, **648** disposed around the outer periphery thereof. Seals **638**, **648** may be adhesives, grease, gaskets, O-rings, compressive fits, or other sealants, that substantially seal any gaps between the support structures **630**, **640** and the heat sink **602**, thus providing an air-tight enclosure around the rotor **610** and stator **620** components.

In an embodiment, a seal may also be applied around the connector **626** and/or connector tab **634**. In an embodiment, the remaining open area of opening **608** may be covered with a plug or other sealant material.

In an embodiment, after assembly of the front and rear bearing support structures **630** and **640**, fan **650** is mounted on an end of the rotor shaft **612**, and rear end cap **660** is secured to the rear end of the heat sink **601**. In an embodiment, peripheral walls **666** of the rear end cap **660** partially covers the end of the heat sink **602** for more efficient baffling of the air flow into the air channels **606** of the heat sink **602**. Rear end cap **660** may be secured to the heat sink **602** via a series of fasteners **664** received into the ends of some of the air channels **606**.

In an embodiment, transmission assembly **670** may be mounted to the front end of the heat sink **602**. Transmission assembly **670** may include air inlets (not shown) that collect and intake air traveling through air channels **606** for cooling of transmission components. In addition, in an embodiment, at least some of the transmission components may be positioned within the inner envelope of the heat sink **602** forward from the front bearing support structure **630** to utilize the heat sink **602** for cooling of these components.

Stator Having Improved Heat Conduction

Conventionally, the steel used in construction of stator lamination has a thermal conductivity of 20 to 25 Watts per meter-Kelvin (W/m-k). This thermal conductivity defines the ability of the motor stator to conduct heat from the interior of the motor to the surface where excess heat may be transferred to the surrounding environment through radiation, conduction or convection. A temperature gradient will form between the inner core of the motor and the outside area of the case of the motor. The magnitude of the temperature gradient is dependent on the thermal conductivity of the stator laminations. It is desirable to reduce the temperature gradient as much as possible in order to increase the efficiency and reliability of the motor. Other materials such as aluminum, copper and carbon have higher thermal conductivities than steel, however they lack the necessary magnetic properties of the steel used in the stator laminations.

Stator laminations are sheets of steel on the order of 100-1000 microns. A protective coating of resin and/or cellulose ester (i.e., lacker material) or other high dielectric material is typically disposed between adjacent layers of the steel lamination to reduce electrical Eddy currents in the steel laminations that contribute motor inefficiency.

According to an embodiment, in order to increase the thermal conductivity of the stator while substantially maintaining its magnetic properties and efficiency, steel laminations of the stator lamination stack are interleaved with materials having high thermal conductivity. Specifically, in an embodiment, the thermal conductivity of the stator is improved by replacing the conventional dielectric materials between adjacent stator laminations with dielectric materials that have high thermal conductivity properties. Various forms of carbon, including graphite, graphene, carbon Nano tubes, diamonds, and carbon fiber have thermal conductivities as high as 2000 W/m-k. With thermal conductivities nearly 100 times greater than the stator steel, only a relatively thin layer of the carbon-based material between adjacent laminations is sufficient to significantly increase thermal conductivity of the stator.

In an embodiment, since carbon alone does not possess desirable dielectric properties, the layer of carbon-based material is applied to a thin film having dielectric properties.

In an embodiment, carbon material may be dispersed into a resin system and applied through a coating process. Alternatively, in an embodiment, carbon material may be deposited directed as a dry powder. In yet another embodiment, premade sheets of carbon material, such as pyrolytic Graphite, may be provided between adjacent laminations.

It was found by the inventors of this application that with application of thin layers of pyrolytic Graphite between adjacent laminations, thermal conductivity of the stator lamination stack can be significantly improved depending on the thickness of the pyrolytic Graphite sheet (PGS) being applied. In an embodiment, as shown in Table A below, the thickness of a single PGS may be between 2.8% to 28% the thickness of a single steel lamination. Thus, an increase of merely 2.8% to 28% in the lamination stack length results in improved thermal conductivity of between 1.7 times to 4.7 times.

TABLE A

Stator Steel Lamination Thickness (m)	360.0E-6	360.0E-6	360.0E-6	360.0E-6	360.0E-6	360.0E-6	360.0E-6
Stator Steel Lamination Thermal conductivity (W/m-k)	23.0E+0	23.0E+0	23.0E+0	23.0E+0	23.0E+0	23.0E+0	23.0E+0
PSG Sheet Thickness (m)	100.0E-6	70.0E-6	50.0E-6	40.0E-6	25.0E-6	17.0E-6	10.0E-6
PGS Graphite Sheets (W/m-k)	700.0E+0	1.0E+3	1.3E+3	1.4E+3	1.6E+3	1.9E+3	2.0E+3
Thermal conductivity Improvement % increase in thickness	4.7	4.7	4.4	3.8	2.9	2.4	1.7
	27.8%	19.4%	13.9%	11.1%	6.9%	4.7%	2.8%

It will be appreciated that while PGS is used herein by way of example, other carbon-based material and depositing method described above are likely to yield similar results in improved thermal conductivity.

In an embodiment, where materials such as the PGS are used between laminations, the sheets can be oversized so as to provide increased contact area between motor windings on the inside the stator, and heat sinks on the outside of the motor.

The slots for the motor windings are often lined with a material such as paper to help protect the windings during assembly. In an embodiment, constructing the motors using PGS in place of, or in addition to, the paper liners also improves the thermal performance of the motor, as it provides improved thermal conductivity between the motor windings and the stator structure.

Selectively Coated Heat Sink

Another aspect of the invention is described herein with reference to FIGS. 45-47, according to an embodiment.

A motor assembly with an external heat sink, as described in various embodiments of this disclosure, forms the external face of a power tool that is accessible by users. While cooling methods previously described sufficiently cool the heat sink for many applications, in some high-power application, or in applications where power tool is continually used for extended periods of time, the heat sink temperature may reach levels unsafe or uncomfortable for users to touch and operate.

According to an embodiment, in order to reduce thermal transfer to a user's hand, extremities of the external heat sink are selectively coated and/or covered by a layer of thermally-resistive material. In an embodiment, the extremities of the heat sink are coated selectively, for example at the tips of the heat sink fins, such that most of the surface area of the heat sink is left uncoated convectively transfer heat to the outside environment.

In an embodiment, examples of thermally-resistive material may include, but are not limited to, plastic, ceramic, rubber, paint, powder coating, etc. Alternatively, in an embodiment, thermally-resistive material may be tape, molding, extruded plastic, etc., manufactured separately and attached to the extremities of the heat sink. In yet another embodiment, the coating may be in the form of a process, e.g., anodization, applies to the surface of the heat sink, changing the physical characteristics of the surface properties of the heat sink.

FIG. 45 depicts a perspective view of an exemplary motor assembly 700 having an external heat sink 702 having fins 704, and front and rear bearing support structures 730 and

760. FIG. 46 depicts an axial view of the same motor assembly 700. This motor assembly 700 is depicted here by way of an example, and the principles provided in this section can apply to any of other embodiments of the invention. Motor assembly 700 is not described in great detail herein, and it should be understood that the teachings of the previous embodiments are applicable to motor assembly 700.

In an embodiment, FIG. 47 depicts a zoomed-in axial view of the heat sink 700, showing three of the heat sink fins 704. In an embodiment, the end portion 706 of each fin 704 is provided with recessed side surfaces 708 and a double-humped end surface 709. The coating 710 is applied at the end portion 706 around the recessed side surfaces 708 and the doubled-humped end surface 709. This fin 704 geometry increases the bonding of the coating 710 at the end portion 706, and provides a uniform and consistent perimeter profile between the coating 710 and the rest of the fin 704. In an embodiment, the fin 704 geometry may alternatively and/or additionally include undercuts, scores, texturing/knurling, and/or reliefs to enhance the bonding of the coating 710 to the end portion 706.

In an embodiment, the recessed side surfaces 708 occupy approximately 20-30% of the height, and 10-40% of the width of the fins.

In an embodiment, the coating may be applied by selective masking of the heat sink 702 via, for example, tape or wax, before application of the coating material. Alternatively, the coating may be applied to the entire surface area around the fins and then selectively removed by secondary scraping. In yet another embodiment, coating may be selectively deposited by a known process or material such as ultra-violet curing and/or laser.

Fully-Enveloped Segmented Motor

Another aspect of the invention is described herein with reference to FIGS. 48-76. In an embodiment, as described in detail here, stator assembly of the motor may be provided with a segmented construction. An external heat sink is provided for thermal management of the motor as well as mechanical retention and constraint of the segments.

FIG. 48 depicts a side view of an exemplary power tool 800 having a fully-enveloped segmented motor assembly 900. While power tool 800 in a grinder by way of example, it will be appreciated that the teachings of this disclosure with respect to this embodiment are merely exemplary, and the power tool of this invention could be any type of corded or cordless power tool, such as a drill, impact driver, hammer, hammer drill, circular saw, reciprocating saw, multi-tool, or any similar power tool constructed in accor-

dance with the teachings of this disclosure. In an embodiment, power tool **800** may include a handle **802** disposed on one side of the motor assembly **900**, and a gear case **804** disposed on another side of the motor assembly **900** opposite the handle **802**. In an embodiment, a battery pack **806** is removably mounted to the end of the handle **802**. A trigger mechanism **808** is also mounted on the handle **802**. In an embodiment, the handle **802** houses a controller and other electronics (not shown) for driving the motor assembly **900**. In an embodiment, the motor assembly **900** rotatably drives a motor spindle (not shown) within the gear case **804**, which in turns causes rotation of output spindle **810**.

FIGS. **49** and **50** depict front and rear perspective views of an exemplary motor assembly **900**, according to an embodiment. FIGS. **51** and **52** depict front and rear perspective exploded views of the motor assembly **900**, according to an embodiment.

According to an embodiment, similarly to the previously-described embodiments, motor assembly **900** includes a rotor **910** having a rotor shaft **912** and a rotor lamination stack **914** housing a series of permanent magnets **918**, front and rear bearings **916a** and **916b**, and a sense magnet ring **952**. In an embodiment, the motor assembly **900** also includes a stator **920** having segmented windings within an outer heat sink **902** for external cooling and mechanical constraints of the stator **920**, as described later in detail. In an embodiment, a rotary fan **904** having blades **950** is mounted on one end of the rotor shaft **912** to rotate with the rotor shaft **912**.

It should be noted that motor assembly **900** includes many of the same or similar features as motor assemblies **200** and/or **400** previously disclosed, and many of the details and alternative and/or additional embodiments disclosed above with reference to motor assemblies **200** and/or **400** are applicable in this embodiment.

In an embodiment, the outer heat sink **902** may be constructed according to the embodiments described above, particularly a fully-enveloping heat sink include a series of outer longitudinal fins **960** forming longitudinal air channels **962** therebetween on its outer surface. In an embodiment, outer heat sink **902** includes a generally cylindrical body surrounding the stator **920**. Heat sink **902** includes a cylindrical opening that is sized to fittingly receive the stator **920**, with the inner surface of the heat sink **902** being in thermal and physical contact with the outer surface of the stator **920**. Since heat sink **902** covers substantially the entire periphery of the stator **920**, it provides for optimal heat transfer from the stator **920** to the outside environment.

In an embodiment, the outer geometry of the heat sink **902** is sized to provide a substantially uniform profile on the power tool **800** between the handle **802** and the gear case **804**.

In an embodiment, air channels **962** are arranged to guide air flow generated by fan **904** along the outer surface of heat sink **902** parallel to the longitudinal axis of the motor **900**. Fins **960** increase the outer surface area of the heat sink **902**, while passage of cooling air through the air channels **962** provides for improved heat dissipation from the heat sink **902**. In an embodiment, air channels **962** form air inlets with the handle **802** to receive incoming air from inside the handle **802**. In an embodiment, air traveling through air channels **962** dissipates into the outside environment at or near the front end cap **940**. In an embodiment, part of the air may be guided through the front end cap **440** into the gear case **804** for cooling the gear case components.

In an embodiment, heat sink **902** may be constructed of non-ferrous material in order to minimize the impact of eddy current losses and heating from the magnetic operation of

the motor assembly **900**. In an embodiment, the heat sink **902** may be constructed of high-thermal conductivity material, such as aluminum, to facilitate the transfer of heat from the stator **920** to the outside environment. In an embodiment, the heat sink **902** may be of an extruded, machined, or die cast construction.

In an embodiment, a rear end cap **930** is provided on one end of the motor assembly **400** for mounting the motor **900** to the power tool handle **802**. An outer periphery **936** of the rear end cap **930** is sized to be fittingly received within an end of the heat sink **902**, i.e., via press-fitting. Alternatively, in an embodiment, rear end cap **930** includes peripheral receptacles **932** for fastening the rear end cap **930** to the end of the heat sink **902** via a series of fasteners. The fasteners may be fastened into corresponding threaded receptacles in the heat sink **902**, or within the air channels **962**. In an embodiment, front end cap **930** further includes a center bearing pocket **934** that receives the rear bearing **916a** of the rotor shaft **912**.

In an embodiment, a rear portion of the rear end cap **930** is provided with a pair of passthrough bosses **938**. The handle **802** of power tool **800**, which is ordinarily made up on two clam shells, form around the passthrough bosses **938** and includes features that secure the passthrough bosses **938** with respect to the handle **802**.

In an embodiment, a front end cap **940** is provided on an end of the motor assembly **900** opposite the rear end cap **930**. Front end cap **940** is provided with outer receptacles **942** for fastening the front end cap **940** to the gear case **804** of the power tool **800** via fasteners. Front end cap **940** is also provided with inner receptacles **945** for fastening the front end cap **940** to the heat sink **902** via fasteners. The latter fasteners may be fastened into corresponding threaded receptacles in the heat sink **902**, or within the air channels **962**. Front end cap **940** further includes a through-hole **944** provided as a bearing support for front bearing **916b** of the rotor shaft **912**. Alternatively, through-hole **944** may be a pass-through opening through which the rotor shaft **912** freely extends and is piloted in the gear case **804** via a driver **917**.

In an embodiment, front end cap **940** may also be provided with a series of peripheral openings forming air inlets **946**. Air inlets **946** are aligned with the air channels **962** of the heat sink **902** to receive air traversing through the air channels **962**. A baffle **956** direct air from the air inlets **946** towards a middle portion of the fan **904**. Fan **904** includes blades **955** that generate air flow through the air channels **962**, through the air inlets **946**, in the direction of the fan **904**. In an embodiment, the air flow generated by the fan **904** may be expelled out of air exhaust ports (not shown) in the gear case **804**.

In an embodiment, front end cap **940** may be provided with a series of recesses **948** around the through-hole **944**. Recesses **948** are positioned where the ends of the stator windings **922** facing the front end cap **940** to account with tolerances associated with the stator windings **922**.

In an embodiment, the heat sink **902** and rear and front end caps **930** and **940** together substantially encapsulate and seal the stator **920** and rotor **910** from the outside environment, preventing or at the very least minimizing air flow, and in particular contaminated air flow, from entering the magnetic areas of the motor assembly **900**. In an embodiment, additional sealants, such as for example, gaskets, adhesive, etc. may be applied as needed to the rear and front end caps **930** and **940** to form a water-sealed containment around the magnetic areas of the motor assembly **900**.

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In an embodiment, circuit board **906** is received within the heat sink **902**. In an embodiment, circuit **906** may include disc-shaped and include positional sensors for sensing the rotary position of the sense magnet ring **952** are disposed. It should be noted, however, that power switches for powering the phase of the motor **900**, and/or controller for controlling the switching operation of the power switches, may also be disposed on the circuit board **906**. In an embodiment, circuit board **906** may also include metal tracks for connecting the stator windings, as described below in detail. Circuit board **906** electronically communicates with other electronic components disposed in the power tool handle **802** via pins or wires passing through the rear end cap **930**.

Referring now to FIGS. **53-56**, various views of the stator **920** received within the heat sink **902** are provided. FIGS. **53** and **54** respectively depict perspective views of the stator **920** distanced from and within the heat sink **902**, respectively, according to an embodiment. FIG. **55** depicts an axial cross-sectional view of the heat sink **902**, stator **920**, and rotor **910**, according to an embodiment. FIG. **56** depicts an axial perspective view of the heat sink **902** and the stator **920**, according to an embodiment.

In an embodiment, stator **920** includes a series of segmented windings **922** wound over discrete stator segments **924**. The windings **922** are wound on each stator segment **924** independently via a winding machine. The stator segments **924** are shaped to mate in a circular fashion together forming the stator **920** after the windings **922** have been wound. In an embodiment, the heat sink **902** provides radial constraint to securely hold the stator segments **924** together.

FIGS. **57-60** depict various views of the stator segments **924** are depicted. As shown herein, and with continued reference to FIGS. **53-56**, each stator segment **924** includes a segment core **970** arranged to form a segment of the outer periphery of the stator **920**, and a segment pole **972** extending radially inwardly from the segment core **970** on which the windings **922** are disposed. In an embodiment, arched-shaped pole ends **974** extend laterally from the ends of the segment poles **972** around the rotor **910**. In an embodiment, the segment core **970**, segment pole **972**, and pole ends **974** may be made of metal laminations. In an embodiment, two insulating members **976** are disposed on the sides of the segment poles **972** to insulate the windings **922** from the stator segments **924**. Two end insulators **977** and **978** are also disposed at the ends of the of the stator segments **924** for the same purpose. The windings are wound around the end insulators **977**, **978** and the insulating member **976**. In an embodiment, end insulator **977** further includes two legs **975** that receive two winding terminals **979**, and an axial retainer **980** for attachment of the circuit board **906**, as described later in detail.

Referring to FIGS. **61** and **62**, an embodiment of the invention for increasing the slot fill within each stator segment **924** is described.

FIG. **61** depicts a conventional prior art stator segment design in which the segment core includes inner and outer arched-shaped surfaces. The ends of the segment core are flat and at a normal angle to the inner and outer surfaces of the segment core. When the stator segments are assembled together, the inner and outer surfaces of the stator segments together form circular inner and outer surfaces for the stator core. While this arrangement provides smooth surfaces for easy and high-precision assembly, it limits the number of turns of the windings that can be fitted within each slot. Specifically, the curvature of the inner surface of the segment core limits the opening **982** through which the wind-

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ings are received, preventing the windings from being fitted into space **986** using commonly-used winding machines.

FIG. **62** depicts the stator segment **924** having an improved segment stator segment that significantly improves the slot fill, according to an embodiment of the invention. In an embodiment, segment core **970** of the stator segment **924** is provided with flat inner surfaces **988** disposed substantially perpendicularly to the segment pole **972**. This construction provides a wider opening **984** compared to the prior art design to allow a higher number of windings to be disposed around the segment pole **972**. This construction optimizes the accessible area for wire fill, and therefore increases power density of the motor.

Furthermore, in an embodiment, to ensure that the thickness of the segment core **970** is not too small around the edges, the outer surface of the segment core **970** is provided with two flat edge portions **992** disposed on two sides of an arch-shaped portion **990**. In an embodiment, arch-shaped portion **990** is aligned with a radial axis of the segment pole **972**, and a notch **994** is provided as a part of the arch-shaped portion **990** for proper piloting and alignment of the stator segments **924** within the heat sink **902**. In an embodiment, flat edge portions **992** are substantially parallel to the flat inner surface **988**. In an embodiment, each flat edge portion **992** extends approximately 10-20% of the width of the segment core **970**. In an embodiment, first and second ends **991** and **993** of the segment core **970** are provided with mating geometric surfaces for improved constraining and retention of the stator segments **924**.

Referring to FIGS. **63-65**, an inner profile of the heat sink **902** optimized for receiving the stator **920** is described herein.

FIG. **63** depicts an axial view of the heat sink **902** alone, according to an embodiment. FIG. **64** provides a partial cross-sectional axial view of the stator **920** within the heat sink **902**. FIG. **65** depicts a zoomed-in perspective axial view of two adjacent segment cores **970** within the heat sink **902**.

As shown in these figures, and with continued reference to FIGS. **53-56**, in an embodiment, heat sink **902** is provided with internal geometry designed for proper radial constraining of the above-described stator segments **924**. Specifically, in an embodiment, instead of a fully circular inner surface, the inner surface of the heat sink **902** includes a series of sectors **996**, one of which is denoted in FIG. **63** within angular position **6**. The sectors **996** correspond to the stator segments **924** is number and surface profile. In an embodiment, each sector **996** includes an arched portion **997** and two flat side portions **998**. Arched portions **997** have the same curvature as the arched-shaped portions **990** of the stator segments **924** and are arranged to form-fittingly receive the arched-shaped portions **990** of the stator segments **924** therein. In an embodiment, each arched portion **997** has a radius of curvature that shares the same center as the center of the heat sink **902** outer surface.

In an embodiment, flat side portions **998** extend laterally from two sides of the arched portion **997**. Abutting flat side portions **998** of adjacent sectors **996** form recessed portions **995** therebetween in the inner surface of the heat sink **902**. As shown in FIG. **64**, recessed portions **995** receive flat edge portions **992** of adjacent stator segments **924** therein.

In an embodiment, as shown in FIG. **65**, flat side portions **998** are angled to provide a small angular gap (e.g., approximately 1 degrees, up to 0.5 mm in width) between the flat edge portions **992** of the stator segments **924** and flat side portions **998** of the corresponding sectors **996**. This arrangement accounts for tolerances associated with the heat sink

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902 and stator segments 924 and ensures that the arched-shaped portions 990 of the stator segments 924 come into contact with the arched portions 997 of the heat sink 902 for proper diametric insertion of the stator segments 924 within the heat sink 902.

By form-fitting insertion of the stator segments 924 within the sectors 996 of the heat sink 902, via for example press-fitting, the heat sink 902 provides radial as well as axial constraint to hold the stator segments 924 together around the rotor 910 with a high level of precision. Furthermore, securing the rear and front end caps 930 and 940 to the ends of the heat sink 902 as previously described provides axial and radial retention between the rotor 910 and the stator 920 with a small radial gap therebetween.

In an embodiment, a series of notches 999 may be provided at different positions on the inner surface of the heat sink 902. Welding material may be welded inside the notches 999 for enhanced bonding of the stator 920 to the heat sink 902 and increased relief on the arched portions 997 of the heat sink 902. Notches 999 ensure that the welding material does not interfere with the stator 920 insertion operation.

Referring now to FIGS. 66 to 73, assembly of circuit board 906 to the stator 920 is described herein, according to an embodiment of the invention.

FIG. 66 depicts a perspective view of the stator 920 with circuit board 906 mounted, in an embodiment. FIG. 67 depicts a perspective exploded view of the heat sink 902, stator 920, and circuit board 906, according to an embodiment. FIG. 68 depicts a perspective view of the stator 920 and circuit board 906 assembled within the heat sink 902, according to an embodiment. FIGS. 69 and 70 depict axial front and back assembled views of the heat sink 902, stator 920, and circuit board 906, according to an embodiment.

As previously described briefly, circuit board 906 includes a series of positional sensors 908 disposed around a center through-hole 1000. In addition, circuit board 906 includes conductive routings (not shown) that connect the stator windings 922 of the respective stator segments 924 together. In an embodiment, the conductive routings may be configured to couple corresponding stator windings 922 within each phase of the motor 900 in a series or parallel configuration. Furthermore, the conductive routings may couple stator windings 922 of different phases of the motor 900 in a wye or a delta configuration.

In an embodiment, to facilitate connection of the stator windings 922 to the circuit board 906, circuit board 906 is peripherally provided with a series of outer notches 1002 positioned to receive the winding terminals 979 of the stator segments 924. Conductive routings of the circuit board 906 extend into the notches 1002 to facilitate the desired winding connection between the winding terminals 979. Axial retainers 980 of the stator segments 924, which may be, for example, elastic snaps, hooks, etc., snap onto the counterpart slots 1004 on outer periphery of the circuit board 906 to securely position and retain the circuit board 906 at the end of the stator 920.

FIGS. 71 and 72 depict partial cross-sectional views of motor assembly 900 during the assembly process, immediately prior to and after the insertion of the stator 920 and circuit board 906 into the heat sink 902, respectively, according to an embodiment. FIG. 73 depicts a partial perspective view of the motor assembly 900 immediately prior to the insertion of the stator 920 and circuit board 906 into the heat sink 902.

As shown in these figures, in an embodiment, during the assembly process, the stator segments 924 are initially held

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together and inserted partially into the heat sink 902, with winding terminals 979 and axial retainers 980 protruding outside the heat sink 902 body. In an embodiment, circuit board 906 is then mounted on the end of the stator 920 with winding terminals 979 disposed within openings of the notches 1002, and axial retainers 980 making snap connections with the slots 1004.

In an embodiment, legs 975 of end insulator 977 are positioned at an angle (e.g., approximately 5-15 degrees) from an insertion axis of the stator 920 to ensure that the winding terminals 979 are received into the openings of the notches 1002 with ease and without impeding the insertion process. In an embodiment, legs 975 include outer projections 1006 provided radially outwardly from the winding terminals 979. In an embodiment, after the circuit board 906 is mounted on the end of the stator 920, the stator 920 is further slid into the heat sink 902. During this step, the inner portion of the end surface of the heat sink 902 encounters the outer projections 1006, applying a radially inward force on the outer projections 1006 that causes the winding terminals 979 to be forced into contact and make an electrical connection with the inner surfaces of the notches 1002. This arrangement ensures that the winding terminals 979 do not block or interfere with the circuit board 906 during the assembly process, but make a secure electrical connection with the circuit board 906 at the completion of the process.

FIGS. 74 and 75 depict power output improvement of an exemplary segmented motor according to the above-described embodiments in comparison to a similarly-sized motor having a conventional stator design, according to an embodiment. In this example, both motors include 6-pole rotors and 9-slot stators. Both stators are 51 mm in outer diameter (i.e., inner diameter of the heat sink) and 25 mm in length (i.e., length of the segment core). In both simulations, the motor is powered by a 20V max battery pack. As shown, at different conduction band and angle advance commutations, the exemplary segmented motor outputs maximum power greater than 1800 W at 120/30 degrees conduction band/angle advance, and over 2000 W at 150/45 conduction band/angle advance. This power output represents an increase of 23% to 33% over the motor having a conventional stator design.

FIG. 76 depicts a power/torque diagram for another exemplary segmented motor according to the above-described embodiments. In this example, the motor includes a stator that is 30 mm in outer diameter and 25 mm in length, and produces a maximum power output of over 360 W when powered by a 20V max battery pack.

It was further found that an exemplary motor according to the above-described embodiments having a stator that is 61 mm in outer diameter and 45 mm in length produces a maximum power output of approximately 32000 W when powered by a 60V max battery pack.

Bobbin-Wound Motor with External Heat Sink

The above-described embodiment relates to a stator assembly in which the windings are wound directly onto the segment poles. In such a construction, each segment pole is placed into a winding machine and the magnet wires are wound around the segment pole. In an alternative embodiment, as described in this section, the stator assembly may be made of bobbin-wound segments. Bobbin windings refer to coils wound independently of the stator core. The bobbin coils are initially wound onto a bobbin carrier or a mandrel, and the wound coils are thereafter relocated and assembled onto the stator core poles.

According to an embodiment, any of the above-described embodiments, i.e., motor partially enveloped fully envel-

oped by an external heat sink, may be provided with a bobbin-wound motor as described herein.

FIG. 77 depicts a partially exploded view of a stator assembly 1100 received within a fully-enveloping heat sink 1102. FIG. 78 depicts a partially exploded view of the stator assembly 1100 alone. FIG. 79 depicts another partially exploded view of the stator assembly 1100. It is reiterated that heat sink 1102 may be constructed according to any of the above-described embodiments, and disposed as an outer-cooling mechanism for the stator assembly 1100 in any power tool construction previously described. The full motor construction, including details of the rotor assembly and other components, is not described herein to avoid repetition.

In an embodiment, stator assembly 1100 includes an inner stator core 1110 having a central annular body sized to receive a rotor assembly (not shown) with a small air gap therebetween, and a series of stator poles 1112 extending radially outwardly from the stator core 1110. A series of stator pieces 1120 are mounted on the stator poles 1112. Stator windings 1122 are wound on the stator pieces 1120, independently from the rest of the stator assembly 1100. The windings 1122 may be formed with bondable magnet wire. In an embodiment, the windings 1122 may be wound on a bobbin carrier or a mandrel and later pressed on the stator pieces 1120. Two winding terminals 1124 are disposed on each stator piece 1120 for attachment to a circuit board (not shown) as previously described. Each stator piece 1120 includes an elongate opening 1126 through which the stator pole 1112 extends after the stator piece 1120 is mounted on the stator pole 1112.

In an embodiment, an outer ring 1130 may be provided to constrain the stator pieces 1120 on the stator core 1110. In an embodiment, the outer ring 1130 includes a series of axial inner slots 1132 arranged to received pole tips 1114 of the stator poles 1112 therein when the stator core 1110 and the stator pieces 1120 are inserted inside the inner diameter of the outer ring 1130. In an embodiment, the outer ring 1130 may be pressed onto the stator pieces 1120 and the stator core 1110, forming the back iron for the stator assembly 1100 for the magnetic path between the stator poles 1112. The outer surface of the outer ring 1130 may be provided with appropriate geometry to be received within the heat sink 1102 for external cooling of the stator assembly 1100.

With the outer ring 1130, the heat sink 1102 may be provided as a partially-enveloping construction instead of a fully-enveloping construction as shown. It is noted that, in an embodiment, when heat sink 1102 has a fully-enveloping construction as shown in FIG. 77, the stator core 1110 and the stator pieces 1120 may be received inside the heat sink 1102 directly without the outer ring 1130. The heat sink 1102 in this embodiment may include axial inner slots similarly to slots 1132 of the outer ring 1130, and may be sized to provide mechanical radial (and even axial) constraint for the stator pieces 1120 on the stator core 1110. The heat sink in this embodiment may also form the back iron for the stator assembly 1100 for the magnetic path between the stator poles 1112.

FIGS. 80-83 depict various views of the stator core 1110 and the outer ring 1130, according to an embodiment.

FIGS. 84 and 85 depict perspective and cross-sectional views of a stator piece 1120, according to an embodiment. FIGS. 86 and 87 depict perspective and side views of a stator piece 1120, according to an embodiment.

According to an embodiment, in order to improve thermal conductivity between the stator windings 1122 and the heat sink 1102, the stator pieces 1120 are provided with thermal

filler material. In an embodiment, radial walls 1127 of each stator piece 1120, along the sides of the elongate opening 1126, are provided with side openings 1128, as shown in FIG. 86. In an embodiment, the stator pieces 1120 are coated with thermal filler material 1129, fully or partially within the side openings 1128. Thermal filler material may be made of thermally conductive but electrically insulating material, such as silicone caulk, grease, etc. The volume of the thermal filler material may be provided in excess of the spaces within the side openings 1128 to ensure proper bonding between stator windings 1122 and the stator poles 1112.

Outer-Cooled Motor with a Heatshield

Another aspect of the invention is described herein with reference to FIGS. 88-101, according to an embodiment.

Outer-cooled motor designs with external heat sinks, as described above in embodiments of this disclosure, dissipate heat generated by the motor to the outside environment. In many power tool applications, such designs result in heat sink outer surface temperatures tolerable for users to grip and operate. It has been found by the inventors, however, that in some power tool applications, particularly high voltage/high power applications, and/or power tool applications where the motor is required to run for extended periods of time, the outer surface of the heat sink may reach temperature levels intolerable or dangerous to users to touch and operate.

As previously discussed, one solution to alleviate this problem is to apply a coating of thermally non-conductive material selectively to outer extremities of the external heat sink. In an alternative embodiment, described herein, a sleeve (also referred to as a heatshield) is provided outside the heat sink with an airgap therebetween. The heatshield shields the user from high temperature of the heat sink, while providing an air gap for proper external cooling of the heat sink, as described herein in detail.

FIG. 88 depicts a side view of an exemplary power tool 1200 having a motor assembly 1300 provided within an outer heatshield 1220. While power tool 1200 in a grinder by way of example, it will be appreciated that the teachings of this disclosure is merely exemplary, and the power tool of this invention could be any type of cordless or corded power tool, including but not limited to, a drill, impact driver, hammer, hammer drill, circular saw, reciprocating saw, multi-tool, or any similar power tool constructed in accordance with the teachings of this disclosure.

In an embodiment, power tool 1200 may include a handle 1202 disposed on one side of the motor assembly 1300 (and heatshield 1220), and a gear case 1204 disposed on another side of the motor assembly 1300 (and heatshield 1220) opposite the handle 1202. In an embodiment, a battery pack 1206 is removably mounted to the end of the handle 1202. A trigger mechanism 1208 is also mounted on the handle 1202. In an embodiment, the handle 1202 houses a controller and other electronics (not shown) for driving the motor assembly 1300. In an embodiment, the motor assembly 1300 rotatably drives a motor spindle (not shown) within the gear case 1204, which in turns causes rotation of output spindle 1210.

FIGS. 89 and 90 depict front and rear perspective views of an exemplary motor assembly 1300 provided within a heatshield 1220, according to an embodiment. FIGS. 91 and 92 depict front and rear perspective exploded views of the motor assembly 1300 with stator 1320 and heat sink 1302 provided within the heatshield 1220, according to an embodiment.

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According to an embodiment, motor assembly **1300** includes mostly the same features as motor assembly **900** of FIGS. **49-52**. In summary, in an embodiment, motor assembly **1300** includes a rotor **1310** having a rotor shaft **1312** and a rotor lamination stack **1314** housing a series of permanent magnets, front and rear beings **1316a** and **1316b**, and a sense magnet ring **1352**. The motor assembly **1300** also includes a stator **1320** having segmented stator windings **1322** received within an outer heat sink **1302**. In an embodiment, circuit board **1306** includes positional sensors (not shown) for sensing the rotary position of the sense magnet ring **1352**, as well as conductive routings for connecting the stator windings **1322**. In an embodiment, a rotary fan **1304** having blades **1350** is mounted on one end of the rotor shaft **1312** to rotate with the rotor shaft **1312**. Front and rear end caps **1340** and **1330**, and a baffle **1356**, are also provided, as described later in detail.

Referring to FIGS. **93-95**, an exemplary arrangement of heat sink **1302** and heatshield **1220** is described herein. FIG. **93** depicts a perspective view of heatshield **1220** and heat sink **1302** housing the stator **1320** and circuit board **1306**. FIG. **94** depicts an axial view of heatshield **1220** alone. FIG. **95** depicts an axial view of heat sink **1302** disposed within heatshield **1220**.

In an embodiment, heat sink **1302** is provided for external cooling of the stator **1320**, as described in previous embodiments. In an embodiment, heat sink **1302** includes a generally cylindrical body surrounding the stator **1320**. Heat sink **1302** includes a cylindrical opening that is sized to fittingly receive the stator **1320** therein, with the inner surface of the heat sink **1302** being in thermal and physical contact with the outer surface of the stator **1320**. Since heat sink **1302** covers substantially the entire periphery of the stator **1320**, it provides for optimal heat transfer from the stator **1320**. Heat sink **1302** may be constructed according to the embodiments described above, particularly a fully-enveloping heat sink include a series of outer longitudinal fins **1360** forming longitudinal air channels **1362** therebetween on its outer surface.

In an embodiment, air channels **1362** are arranged to guide air flow generated by fan **1304** along the outer surface of heat sink **1302** parallel to the longitudinal axis of the motor **1300**. Fins **1360** increase the outer surface area of the heat sink **1302**, while passage of cooling air through the air channels **1362** provides for improved heat dissipation from the heat sink **1302**.

In an embodiment, heatshield **1220** is sized to receive the heat sink **1302** therein with a small airgap in between. The airgap allows cooling air to travel along the outer surface of heat sink **1302**, including inside the air channels **1362** in the direction of the fan **1304**, while the heatshield **1220** shields the user from direct contact with the heat sink **1302**.

In an embodiment, heatshield **1220** may be made of thermally insulating material such as plastic. Alternatively, and preferably, heatshield **1220** may be made of thermally conductive material such as metal to provide some heat transfer through the heatshield **1220**.

In an embodiment, heatshield **1220** may include an annular body with a fully circular inner and outer surface. Alternatively, in an embodiment, heatshield **1220** may include longitudinal extruded channels **1222** projecting inwardly from its main body **1224**. Extruded channels **1222** of heatshield **1220** are sized to project inwardly into air channels **1362** of the heat sink **1302**. This arrangement provides a uniform air gap (i.e., of substantially consistent width) between the heat sink **1302** and heatshield **1220**. In an embodiment, heatshield **1220** may include non-extruded

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portions **1226** in addition to extruded channels **1222** as needed to improve the power tool **1200** ergonomics. The airgap between non-extruded portions **1226** and the heat sink **1302** is not uniform. In an embodiment, the airgap between the heat sink **1302** and heatshield **1220** results in a temperature differential of at least 10 degrees Celsius between the heat sink **1302** and heatshield **1220**, making the heatshield **1220** more tolerable for a user to touch.

In an embodiment, heatshield **1220** is sized with outer geometry to provide a substantially uniform profile on the power tool **1200** between the handle **1202** and the gear case **1204**.

FIG. **96** depicts an axial view of stator **1320** and heat sink **1302** disposed within heatshield **1220**. FIG. **97** depicts a zoomed-in axial view of stator **1320** and heat sink **1302** disposed within heatshield **1220**. As shown herein, in an embodiment, stator **1320** is provided with segmented design as described earlier in this disclosure, and includes a series of stator windings **1322** wound over discrete stator segments **1324**. Stator segments **1324** are radially constrained and held together by the heat sink **1302**, while the air gap between the heat sink **1302** and heatshield **1220** allows for air flow to cool the stator **1320** externally outside the heat sink **1302**.

It should be noted that, while in this exemplary embodiment, the stator **1320** is segmented and fully enveloped by the heat sink **1302**, the teachings of this embodiment may be applied to other stator/heatsink designs. For example, the stator may be a self-contained assembly having a core, and the heat sink may partially or fully envelope the stator. Furthermore, the heat sink may be provided separately from the stator, or integrally with the stator core. In an embodiment, the heat sink may include outer fins, or may be provided with other geometry. The heatshield **1220** may be provided as an outer sleeve to any such stator/heatsink designs to provide an air channel outside the periphery of the heat sink for cooling the stator.

Referring to the partially exploded view of FIG. **98**, and with continued reference to FIGS. **89-92**, in an embodiment, rear end cap **1330** is provided on one end of the motor assembly **1300** for mounting the motor assembly **1300** to the power tool handle **1202**. Rear end cap **1330** is sized to mate with an end of the heat sink **1302**. Rear end cap **1330** includes peripheral receptacles **1332** that receive a series of fasteners **1333** for fastening the rear end cap **1330** to the end of the heat sink **1302**. The fasteners **1333** may be fastened into corresponding threaded receptacles **1335** in the heat sink **1302**. Alternatively, the fasteners **1333** may be fastened into the heat sink air channels **1362**. Front end cap **1330** further includes a center bearing pocket **1334** that receives the rear bearing **1316a** of the rotor shaft **1312**. In an embodiment, a rear portion of the rear end cap **1330** is provided with a pair of passthrough bosses **1338** for attachment to the tool handle **1202**.

In an embodiment, as shown in FIG. **98** and the zoomed-in perspective view of FIG. **99**, an outer periphery **1336** of the rear end cap **1330** is sized to be fittingly received inside the heatshield **1220** in contact with extruded channels **1222**. In an embodiment, outer periphery **1336** of the rear end cap **1330** is provided with peripheral teeth **1337** arranged to come into contact with inner radial surfaces of the of the extruded channels **1222**. The diameter of the outer periphery **1336** of the rear end cap **1330**, or at the very least the diameter of the peripheral teeth **1337** of the rear end cap **1330**, is larger than the diameter of the heat sink **1302**. As the rear end cap **1330** is fastened to the heat sink **1302** by fasteners **1333**, the rear end cap **1330** maintains a radial

small airgap between the heat sink **1302** and the heatshield **1220** on its end. This airgap, in an embodiment, may be in the range of 0.5 to 10 mm depending on the size of the motor and its thermal requirements. In an embodiment, openings **1339** formed between adjacent peripheral teeth **1337** and extruded channels **1222** form air inlets for airflow generated by the motor fan **1304** to enter the radial airgap between the heat sink **1302** and the heatshield **1220**.

In an embodiment, openings **1339** may be aligned to receive air from within the handle **1202** of the power tool **1200** for improved cooling of the electronic components within the handle **1202**. Alternatively, openings **1339** may be provided to suction in air from the outside environment.

FIG. **100** depicts a rear zoomed-in axial view of the front end cap **1340**, according to an embodiment. FIG. **101** depicts a front axial view of the front end cap **1340**, according to an embodiment. As shown in these figures, and with continued reference to FIGS. **91** and **92**, in an embodiment, front end cap **1340** is provided on an end of the motor assembly **1300** opposite the rear end cap **1330**. Front end cap **1340** includes a main annular body **1341** provided with outer receptacles **1342** for fastening the front end cap **1340** to the gear case **1204** of the power tool **1200** via fasteners (not shown). Front end cap **1340** also includes a disc-shaped inner body **1343** arranged around a through-hole **1344** of the front end cap **1340**. Inner body **1343** is provided with inner receptacles **1345** for fastening the front end cap **1340** to the heat sink **1302** via fasteners **1347**. Fasteners **1347** may be fastened into corresponding threaded receptacles in the heat sink **1302**, or within the air channels **1362**. In an embodiment, through-hole **1344** is provided as a bearing support for front bearing **1316b** of the rotor shaft **1312**. Alternatively, through-hole **1344** may be a pass-through opening through which the rotor shaft **1312** freely extends and is piloted in the gear case **1204** via a driver **1317**.

In an embodiment, front end cap **1340** may also be provided with a series of peripheral openings forming air inlets **1346a** between the main annular body **1341** and the inner body **1343**. Air inlets **1346a** are aligned with the radial airgap between the heat sink **1302** and the heatshield **1220** to receive air traversing through the radial airgap. Air inlets **1346a** on the rear side of the front end cap **1340** are guided through the front air cap **1340** to air conduits **1346b** on the front side of the front end cap **1340**. A baffle **1356** direct air from the air conduits **1346b** towards a middle portion of the fan **1304**. Fan **1304** includes blades **1355** that generate air flow that enters through the openings **1399** (between the heatshield **1220** and the rear end cap **1330**), passes axially through the radial air gap (between the heatshield **1220** and the heat sink **1302**), and exists the radial air gap through the air inlets **1346a**, and are guided from the air conduits **1346b** in the direction of the fan **1304**. In an embodiment, the air flow generated by the fan **1304** may be expelled out of air exhaust ports (not shown) in the gear case **1204**.

In an embodiment, a rear face of the inner body **1343** may be provided with a series of recesses **1348** around the through-hole **1344**. Recesses **1348** are positioned where the ends of the stator windings **1322** facing the inner body **1343**, to account with tolerances associated with the stator windings **1322**.

In an embodiment, on the rear face of the front end cap **1340** facing the heatshield **1220**, annular area **1357** between the inner body **1343** and the main annular body **1341** is axially recessed to receive the end of the heatshield **1220** therein. This annular area **1357**, which is formed by an outer periphery **1358** of the inner body **1343** and an inner rim **1364** of the main annular body **1341**, is sized to form-fittingly

receive the end of the heatshield **1220** therein. The view of FIG. **100** depicts a cut-out portion of the end of the heatshield **1220** received within the annular area **1357** of the front end cap **1340**.

In an embodiment, outer periphery **1358** of the inner body **1343** includes peripheral teeth **1366** arranged to come into contact with inner radial surfaces of the of the extruded channels **1222** of the heatshield **1220**. The diameter of the outer periphery **1358** of the inner body **1343**, or at the very least the diameter of the peripheral teeth **1366** of the inner body **1343**, is larger than the diameter of the heat sink **1302**. As the front end cap **1340** is fastened to the heat sink **1302** by fasteners **1347**, the front end cap **1340**, similarly to the rear end cap **1330**, maintains the radial small airgap between the heat sink **1302** and the heatshield **1220** on its end.

In an embodiment, the heat sink **1302** and rear and front end caps **1330** and **1340** together substantially encapsulate and seal the stator **1320** and rotor **1310** from the outside environment, preventing or at the very least minimizing air flow, and in particular contaminated air flow, from entering the magnetic areas of the motor assembly **1300**. In an embodiment, additional sealants, such as for example, gaskets, adhesive, etc. may be applied as needed to the rear and front end caps **1330** and **1340** to form a water-sealed containment around the magnetic areas of the motor assembly **1300**.

By form-fitting insertion of the stator segments **1324** within the heat sink **1302**, via for example press-fitting, the heat sink **1302** provides radial as well as axial constraint to hold the stator segments **1324** together around the rotor **1310** with a high level of precision. Moreover, securing the rear and front end caps **1330** and **1340** to the ends of the heat sink **1302** provides axial and radial retention between the rotor **1310** and the stator **1320** within the heat sink **1302**, while maintaining a small radial gap between the rotor **1310** and the stator **1320**. Disposition of the rear end cap **1340** and the inner body **1343** of the front end cap **1340** within the two ends of the heatshield **1220** also provides axial and radial retention between the heat sink **1302** and the heatshield **1220**, while maintaining the radial gap therebetween.

The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms "a," "an," and "the" may be intended to include the plural

forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

LIST OF NUMERICAL REFERENCES

Power Tool **100**
 Tool Housing **102**
 Clam Shells **102a, 102b**
 Motor housing **104**
 Handle portion **106**
 Transmission assembly **108**
 Integrated switch module **110**
 Chuck **112**
 Trigger switch **114**
 Battery terminal **116**
 Battery receptacle **118**
 Heat Sink **120a, 120b**
 Fan **130**
 Air intakes **140**
 Side openings **142a, 142b**
 End cap **150**
 Motor assembly **200**
 BLDC motor **202**
 Rotor **210**
 Rotor shaft **212**
 Rotor lamination stack **214**
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The invention claimed is:

1. A power tool comprising:

a motor;
 an output shaft driven by the motor along a longitudinal axis;
 a fan rotationally coupled to the motor to generate an airflow;
 a housing comprising two housing halves mated together to support the motor and forming a rear wall behind the motor along the longitudinal axis, wherein the rear wall includes a plurality of air intakes;
 an end cap attached to the housing adjacent the rear wall with an air gap therebetween, wherein at least a portion of the end cap does not overlap an outer circumferential surface of the housing so as to form a circumferential opening around the air gap, wherein the airflow generated by the fan suctions ambient air from an outside environment directly through the circumferential opening into the air gap in a substantially transverse direction relative to the longitudinal axis, and the airflow then passes through the plurality of air intakes to enter the housing in thermal contact with the motor.

2. The power tool of claim 1, wherein the end cap is oriented substantially along a plane that is perpendicular to the longitudinal axis.

3. The power tool of claim 1, wherein the fan is disposed between the motor and the rear wall of the housing.

4. The power tool of claim 1, wherein the plurality of air intakes is oriented radially around the longitudinal axis.

5. The power tool of claim 4, wherein the fan is located adjacent the rear wall of the housing and includes a plurality of fan blades facing the rear wall, wherein a distance between the plurality of air intakes and the longitudinal axis is less than a distance between the plurality of fan blades and the longitudinal axis, to allow the fan to repel the airflow entering through the plurality of air intakes radially outwardly.

6. The power tool of claim 1, wherein the end cap comprises two side caps extending axially to overlap opposite side portions of the housing.

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7. The power tool of claim 1, wherein the motor comprises:

a stator including a stator core and a plurality of stator windings connected in phases;

5 a rotor rotatably disposed within the stator to magnetically interact with the plurality of stator windings; and
 at least one heat sink in contact with the stator core and supported by the housing.

8. The power tool of claim 7, wherein the heat sink comprises a plurality of fins forming a plurality of air channels therebetween extending longitudinally along an outer surface thereof, wherein the plurality of air channels is in fluid communication with the airflow.

9. The power tool of claim 8, wherein the end cap comprises two side caps extending axially to overlap opposite side portions of the heat sink to direct the airflow into the plurality of air channels.

10. The power tool of claim 1, wherein the end cap does not include any air intakes in its rear surface to allow entry of airflow.

11. A power tool comprising:

a tool housing including a motor housing defining a longitudinal axis and having a rear wall formed at a rear end thereof, a handle portion extending from the motor housing in a direction transversal to the longitudinal axis, and a battery receptacle formed at an end of the handle portion opposite the motor housing;

a motor disposed within the motor housing;

an output assembly mounted forward of the motor housing and driven by the motor along the longitudinal axis;

30 a fan rotationally coupled to the motor to generate an airflow;

an end cap attached to the motor housing adjacent the rear wall with an air gap therebetween, wherein at least a portion of the end cap includes a diameter that is at most substantially equal to a diameter of the tool housing so as to form a circumferential opening around the air gap, wherein the airflow generated by the fan enters through the circumferential opening into the air gap in a substantially radial direction, then passes through the plurality of air intakes to enter the housing in thermal contact with the motor.

12. The power tool of claim 11, wherein the end cap is oriented substantially along a plane that is perpendicular to the longitudinal axis.

13. The power tool of claim 11, wherein the fan is disposed between the motor and the rear wall of the housing.

14. The power tool of claim 11, wherein the plurality of air intakes is oriented radially around the longitudinal axis.

15. The power tool of claim 14, wherein the fan is located adjacent the rear wall of the housing and includes a plurality of fan blades facing the rear wall, wherein a distance between the plurality of air intakes and the longitudinal axis is less than a distance between the plurality of fan blades and the longitudinal axis, to allow the fan to repel the airflow entering through the plurality of air intakes radially outwardly.

16. The power tool of claim 11, wherein the end cap comprises two side caps extending axially to overlap opposite side portions of the housing.

17. The power tool of claim 11, wherein the motor comprises:

a stator including a stator core and a plurality of stator windings connected in phases;

a rotor rotatably disposed within the stator to magnetically interact with the plurality of stator windings; and
 at least one heat sink in contact with the stator core and supported by the housing.

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18. The power tool of claim 17, wherein the heat sink comprises a plurality of fins forming a plurality of air channels therebetween extending longitudinally along an outer surface thereof, wherein the plurality of air channels is in fluid communication with the airflow.

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19. The power tool of claim 18, wherein the end cap comprises two side caps extending axially to overlap opposite side portions of the heat sink to direct the airflow into the plurality of air channels.

20. The power tool of claim 11, wherein the end cap does not include any air intakes in its rear surface to allow entry of airflow.

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